

Universidade de Lisboa
Faculdade de Ciências
Departamento de Física



Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics

Henrique Gomes da Cruz Martins Farinha

Mestrado em Engenharia Física
Versão Provisória

Dissertação orientada por:
Dr. João Nuno Ramalho Gonçalves Pires
Dr. Jorge Miguel de Brito Almeida Sampaio

Agradecimientos

Resumo

A crescente precisão alcançada pelas experiências realizadas em colisores de partículas exige previsões teóricas com um grau de rigor comparável. Uma parte substancial destes processos é governada pela interação forte, descrita pela Cromodinâmica Quântica (QCD). A energias elevadas, esta teoria pode ser tratada perturbativamente, organizando as previsões físicas como uma expansão em potências da constante de acoplamento forte, α_s . Contudo, aumentar a precisão de um cálculo não corresponde apenas a acrescentar diagramas de Feynman: exige controlar uma estrutura complexa de contribuições de emissão real e virtual, bem como as divergências que surgem em cada uma delas.

As divergências infravermelhas constituem uma das principais dificuldades dos cálculos perturbativos em QCD. As contribuições de emissão real tornam-se singulares quando um partão é emitido com energia muito baixa, o limite mole, ou quando dois partões se tornam colineares. Paralelamente, as amplitudes com laços contêm polos infravermelhos associados às mesmas regiões cinemáticas. Embora estas divergências se cancelem em observáveis físicos suficientemente inclusivos, as componentes individuais do cálculo não são finitas. É, por isso, necessário reorganizar a previsão teórica de modo a isolar as singularidades e a efetuar o seu cancelamento de forma analítica e controlada.

O formalismo de subtração por antenas fornece um enquadramento adequado para esse objetivo. Neste método, a radiação não resolvida é descrita entre dois partões duros ligados por cor, designados por radiadores. As funções de antena reproduzem os limites moles e colineares relevantes de forma local no espaço de fase. Quando são subtraídas às contribuições reais, eliminam as singularidades antes da integração numérica. Depois de integradas analiticamente sobre o espaço de fase da radiação não resolvida, originam polos em regularização dimensional que cancelam os polos das contribuições virtuais. O cancelamento destas singularidades permite que cada contribuição seja integrada numericamente em quatro dimensões.

Esta dissertação apresenta o desenvolvimento de *AntCalc*, um pacote simbólico para a construção e integração automatizadas de funções de antena em QCD. O projeto procura automatizar etapas que requerem implementações específicas para cada antena, reunindo num mesmo fluxo as convenções de normalização, construção e integração. O trabalho centra-se nas antenas quark–antiquark sem massa, na configuração cinemática final–final, até à ordem next-to-next-to-leading order (NNLO). Para além deste objetivo principal, é estudada uma extensão exploratória ao regime massivo através da implementação da antena A_3^0 .

A base teórica da dissertação inclui uma introdução à QCD perturbativa, aos campos de quarks e glúons e à construção de amplitudes a partir de diagramas de Feynman. São discutidas as divergências ultravioleta, tratadas por renormalização, e as divergências infravermelhas, que exigem a combinação entre várias contribuições perturbativas. Utiliza-se regularização dimensional, na qual o número de dimensões do espaço-tempo é tomado como $d = 4 - 2\epsilon$. Neste enquadramento, as divergências manifestam-se como polos em ϵ , e as quantidades integradas são expressas por séries de Laurent. O cancelamento dos polos pode, assim, ser acompanhado explicitamente ao combinar as diferentes contribuições para uma

quantidade física.

É também apresentada a estrutura das funções de antena e a sua decomposição em cor. As antenas são obtidas a partir de amplitudes físicas normalizadas e organizadas de acordo com os radiadores e as partículas emitidas. A fatorização do espaço de fase é essencial neste processo, pois permite separar o espaço de fase reduzido, associado aos radiadores após a emissão não resolvida ser removida, do espaço de fase correspondente à própria emissão. A integração pode então ser efetuada apenas sobre o espaço de fase da radiação não resolvida, independentemente do processo duro em que a antena venha a ser inserida. A dissertação introduz ainda os objetos $\mathcal{T}$, que combinam antenas integradas, fatores de cor e normalizações de forma apropriada para a construção de observáveis inclusivos.

A integração das funções de antena é uma das etapas mais exigentes do cálculo. As integrais de espaço de fase são reescritas como integrais de laço através da técnica de reverse unitarity. Em seguida, são reduzidas a um conjunto finito de integrais mestres usando identidades de integração por partes (integration by parts, IBP) e o algoritmo de Laporta. Em casos adequados, recorre-se também à redução de Passarino–Veltman. Uma vez identificados os integrais mestres e substituídas as respetivas expressões conhecidas, obtém-se a forma integrada da antena como uma expansão em ϵ . O resultado da redução fica, deste modo, expresso num conjunto limitado de integrais mestres, cujas expansões em ϵ determinam as partes singular e finita da antena integrada.

O pacote *AntCalc* foi implementado no ambiente *Wolfram Mathematica* e integra ferramentas especializadas para as diferentes fases do cálculo. *FeynArts* é utilizado para gerar os diagramas de Feynman relevantes, enquanto *FeynCalc* e *FeynHelpers* apoiam a construção e a manipulação das amplitudes. A redução IBP é realizada com *LiteRed2*. A arquitetura do pacote é orientada por perfis, isto é, por descrições estruturadas que especificam o processo, as amplitudes, as famílias de integrais, os fatores de normalização e os integrais mestres necessários. A informação específica de cada antena fica, assim, concentrada no respetivo perfil, enquanto as rotinas de construção e integração permanecem comuns aos diferentes casos.

O fluxo de trabalho divide-se em duas fases principais: construção e integração. Na fase de construção, o utilizador fornece a descrição do processo e o pacote gera os diagramas e amplitudes correspondentes. São depois efetuadas as manipulações algébricas necessárias para extrair a função de antena não integrada, ou as suas componentes de cor quando aplicável. Na fase de integração, essa expressão é convertida para uma forma apropriada à redução, associada à família de integrais correta e reduzida aos integrais mestres. Finalmente, a substituição dos resultados dos integrais mestres e a expansão em torno de $\epsilon = 0$ produzem a antena integrada. Ao longo deste fluxo são efetuados testes de consistência e apresentados diagnósticos do cálculo.

Como exemplo detalhado, é apresentada a derivação da antena A_3^0 , que descreve a emissão de um glúon entre um quark e um antiquark sem massa ao nível de árvore. Este caso ilustra todo o percurso, desde a geração da amplitude até à expressão integrada. A validação local da antena é efetuada através dos seus limites infravermelhos. No limite mole, verifica-se que a expressão reproduz o fator eikonal esperado; nos limites colineares, confirma-se a recuperação da função de separação de Altarelli–Parisi apropriada. A concordância nestes limites constitui uma verificação local da expressão obtida, independente da comparação posterior da antena integrada com resultados da literatura.

Para além de A_3^0 , *AntCalc* é aplicado a um conjunto de antenas sem massa até NNLO. São obtidas contribuições de dois, três e quatro partões, incluindo os conjuntos A_2^2 , A_3^1 e A_4^0 , bem como antenas com outras estruturas de cor, tais como B_4^0 e C_4^0 . Os resultados não integrados e integrados são comparados com resultados estabelecidos na literatura, verificando-se concordância. A concordância com os resultados publicados verifica, em simultâneo, as convenções de regularização dimensional e normalização, bem

como o tratamento da estrutura de cor e do espaço de fase adotado na implementação.

A validação global é realizada através do cálculo do rácio hadrónico sem massa, ou R -ratio. Este observável relaciona a secção eficaz de produção de hádrons em aniquilação elétron–positrão com a secção eficaz de produção de pares de múons. Por ser inclusivo, constitui um teste rigoroso da consistência do formalismo: os polos infravermelhos provenientes das contribuições reais e virtuais devem cancelar exatamente. A montagem do R -ratio a partir dos objetos integrados produzidos por *AntCalc* confirma esse cancelamento e reproduz o resultado finito conhecido até NNLO. A reprodução do R -ratio complementa, deste modo, os testes locais das antenas, verificando o cancelamento dos polos quando as diferentes contribuições são combinadas num observável físico.

A dissertação explora ainda a extensão ao caso massivo pela construção e integração da antena massiva A_3^0 . A presença de uma massa altera a cinemática, a estrutura do espaço de fase e os limites infravermelhos, eliminando, em particular, algumas singularidades colineares existentes no caso sem massa. Este cálculo fornece uma primeira implementação do fluxo massivo em *AntCalc* e identifica as alterações necessárias à construção e integração de futuras antenas massivas.

A principal limitação da versão atual do pacote é o custo computacional da execução inteiramente em Mathematica. Embora este ambiente seja particularmente útil para desenvolvimento simbólico, alguns cálculos tornam-se lentos e exigentes em memória. O trabalho futuro deverá, por isso, avançar para uma arquitetura modular e multilinguagem. Prevê-se a utilização de Python como orquestrador, FORM para manipulação algébrica eficiente e Kira para reduções IBP mais robustas, mantendo as ferramentas atuais nas tarefas em que são mais adequadas. Entre as extensões previstas encontram-se outras antenas massivas, antenas com radiadores quark–gluão e gluão–gluão, configurações inicial–final e inicial–inicial e, a mais longo prazo, cálculos a ordens perturbativas superiores, incluindo N³LO.

Em síntese, o trabalho desenvolvido mostra que as etapas de construção, redução e integração das funções de antena consideradas podem ser reunidas num único procedimento automatizado. O pacote reproduz resultados conhecidos para antenas sem massa até NNLO, valida os limites infravermelhos locais e recupera corretamente o R -ratio hadrónico. A implementação obtida pode servir de base à inclusão de novas classes de antenas sem exigir que todo o procedimento de construção e integração seja novamente implementado de raiz.

Palavras chave: QCD; Subtração por Antenas; NNLO; Singularidades Infravermelhas; *Mathematica*

Abstract

High-precision collider measurements require theoretical predictions that are accurate to their same level. In perturbative Quantum Chromodynamics (QCD), these predictions are organised as an expansion of the strong coupling constant, α_s . At higher orders, the calculation is complicated by infrared singularities associated with soft and collinear parton radiation. Although these singularities cancel in physical observables, the real-emission and virtual contributions are separately divergent and therefore cannot be handled numerically without first isolating and cancelling them.

Antenna subtraction provides a systematic way to do this. It uses universal antenna functions to describe unresolved radiation between colour-connected hard partons. These functions are subtracted locally from the real-emission contribution and then integrated analytically in dimensional regularisation, making the cancellation of infrared poles against the virtual terms explicit.

This thesis develops *AntCalc*, a symbolic framework in Wolfram Mathematica for constructing and integrating QCD antenna functions. The package combines *FeynArts*, *FeynCalc*, *FeynHelpers*, and *LiteRed2*, while keeping process-dependent information separate from the general calculation workflow. It generates the required amplitudes, constructs unintegrated antennae or colour components, maps the corresponding phase-space integrals onto integral families, and reduces them to master integrals through integration-by-parts identities.

The implementation reproduces the massless final-final quark-antiquark antennae through next-to-next-to-leading order. The tree-level three-parton antenna A_3^0 is used as a detailed example and is checked in its soft and collinear limits. The integrated results agree with the literature, and the full NNLO massless hadronic R -ratio is reconstructed as an additional validation. A preliminary massive A_3^0 implementation shows how the same workflow can be extended beyond the massless case. Future work will extend *AntCalc* to further massive antennae, additional partonic configurations, and a more modular multi-language architecture.

Keywords: QCD; Antenna Subtraction; NNLO; Infrared Singularities; *Mathematica*

Contents

Agradecimentos	1
Resumo e Abstract	5
List of Abbreviations and Symbols	12
1 Introduction	14
2 Theoretical Foundations	16
2.1 QCD: Fields, Interactions and Perturbative Amplitudes	16
2.1.1 Quark and Gluon Fields	16
2.1.2 From the QCD Lagrangian to Feynman Diagrams	17
2.2 Divergences in Perturbative QCD	18
2.2.1 Infrared Divergences	18
2.2.2 Ultraviolet Divergences	20
2.3 Dimensional Regularisation	21
2.4 The Antenna Subtraction Formalism	22
2.4.1 Subtraction Terms	23
2.4.2 Antenna Functions	27
2.4.3 Colour Algebra and the Antenna Colour Decomposition	28
2.4.4 Antenna Construction	30
2.4.5 Phase-Space Factorisation	32
2.5 Phase-Space Integration and IBP Reduction	33
2.5.1 IBP Identities and the Laporta Algorithm	35
2.5.2 Master Integrals	38
3 Package Framework	40
3.1 Design Philosophy and Scope	40
3.2 Profile-Driven Architecture	41
3.3 The Build Stage	43
3.3.1 Tree-Level Routes	45
3.3.2 One-Loop Routes	45
3.3.3 Two-Loop Routes	46
3.4 From the Build Stage to Integration	46
3.5 The Integration Stage	47
3.5.1 Integration Profiles and Selection	49
3.5.2 Integration Routes	50

3.6	Higher-Level Functions	51
3.7	Validation and Development Diagnostics	51
3.7.1	Amplitude-Level and Release-Acceptance Validation	52
3.7.2	Runtime and Memory Characteristics	52
4	Antenna Construction and Integration: Selected Case Studies	54
4.1	Master Integrals Used in <i>AntCalc</i>	54
4.1.1	PaVe Scalar Functions	55
4.1.2	IBP Master Integrals	56
4.1.3	Exploratory Massive Master Integrals	64
4.2	NLO Antenna Results	64
4.2.1	The Tree-Level Three-Parton Antenna A_3^0	65
4.2.2	Local Validation of A_3^0	71
4.2.3	The Virtual A_2^1 Contribution	73
4.3	NNLO Antenna Results	73
4.3.1	Two-Parton Contributions: the A_2^2 Set	73
4.3.2	Three-Parton Contributions: the A_3^1 Set	74
4.3.3	Four-Parton Contributions	75
4.4	Summary of the Massless Antenna Set	76
4.5	Exploratory Massive Extension: A_3^0	77
4.5.1	Build and Integrate Results	78
4.6	Overview of Implemented Routes	78
5	Global Validation: The Massless Hadronic R-Ratio	80
5.1	R-Ratio Definition	80
5.2	Global Validation with <i>AntCalc</i>	80
5.2.1	Assembly from Integrated Antennae	81
6	Future Work	83
7	Conclusion	85
A	Complete Antenna Expressions	86
A.1	LO Antenna	86
A.1.1	A_2^0	86
A.2	NLO Antennae	86
A.2.1	A_2^1	86
A.2.2	A_3^0	87
A.3	NNLO Antennae	87
A.3.1	A_2^2	87
A.3.2	$\tilde{A}_2^2$	87
A.3.3	$\hat{A}_2^2$	88
A.3.4	$\check{A}_2^2$	88
A.3.5	A_3^1	88
A.3.6	$\tilde{A}_3^1$	89
A.3.7	$\hat{A}_3^1$	90

A.3.8	A_4^0	90
A.3.9	$\tilde{A}_4^0$	94
A.3.10	B_4^0	102
A.3.11	C_4^0	103
A.4	Exploratory Massive A_3^0	105
B	Supporting Derivations	106
B.1	Feynman Rules	106
B.1.1	Propagators	106
B.1.2	Vertices	107
B.2	PaVe Reduction	107
B.2.1	Rank-One Bubble Example	108
B.3	IBP Reduction Example	109
B.4	Three-Particle Phase-Space to Cut Propagators	109
B.5	Laporta Algorithm Implementation Example	110
B.6	The Massive Extension	110
B.6.1	Massive Kinematics and Infrared Structure	110
B.6.2	Massive Phase-Space Integration and Reduction Methods	111

List of Figures

2.1	Tree-level diagrams for the process $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$	18
2.2	The colour connections in between parent and daughter partons before and after an emission, accompanied also by a schematic Feynman diagram using the momenta labels.	24
2.3	The colour connection of the partons for the double unresolved antenna.	26
2.4	The colour connection of the partons for two adjacent single unresolved antennae.	26
2.5	The colour connections of the partons for two disconnected single unresolved antennae.	26
3.1	Profile-driven architecture of <i>AntCalc</i> . Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The <code>AntennaObject</code> provides the integration-ready handoff between the two stages.	41
3.2	Profile-driven architecture of <i>AntCalc</i> 's Build stage. The principal build-specific options to the <code>BuildAntenna</code> function are displayed on the red card.	44
3.3	Diagram showing the build-side workflow leading to the generation of the A_2^2 antenna set. The two subsets are separated by the interferences used in their generation.	46
3.4	Diagram of the build-only and build-and-integrate routes: the combined route is shown above, the build-only route below. The dashed lines mark the options to separate the result into single components (build-only route) or merge them into one ordered list (combined route). The A_3^1 antenna set is used as an example.	47
3.5	Profile-driven architecture of <i>AntCalc</i> 's Integration stage. The principal integration-only options are shown inside the red card.	48
3.6	Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the A_2^2 antenna set.	49
3.7	PaVe reduction for the A_2^1 antenna integration.	50
3.8	IBP reduction workflow for the remaining antennae integrations. The MI computation is external to <i>AntCalc</i> 's internal operation, and hence is represented by an external incoming arrow.	51
4.1	Feynman diagrams depicting the two topologies described under the A_3^0 antenna, corresponding to the process $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$	65
6.1	Diagram showing the current and planned workflow steps relative to their programming languages.	83

List of Tables

2.1	Infrared limits of the real-radiation matrix elements for $\gamma^* \rightarrow q\bar{q}$ through NNLO, listed by final-state parton configuration. Each configuration is colour-ordered, with the hard quark and antiquark first and last; p_i is the momentum of parton i , $\parallel$ denotes a collinear pair, and $s_{i\dots k} = (p_i + \dots + p_k)^2$. Limits related by $q \leftrightarrow \bar{q}$ or by gluon relabelling are not listed separately. The subtraction counterterm of Eq. 2.13 must reproduce the matrix element in each of these limits [4].	20
2.2	NNLO unresolved configurations and their matching subterm, part of the $d\sigma_{\text{NNLO}}^S$ and $d\sigma_{\text{NNLO}}^T$ subtraction counterterms.	25
2.3	Quark-antiquark antenna families used in this work, including the $q\bar{q}$ form factors [4]. Final states are ordered with the two hard radiator listed first and last, following that paper’s convention.	27
3.1	Overview of the antenna functions currently implemented in <i>AntCalc</i> , by perturbative order and validation source.	42
3.2	Overview of the settings for the A_2^1 route used as a representative example of the route system used in <i>AntCalc</i>	43
3.3	Overview of the tree-level production and extraction modes.	45
3.4	Overview of the one-loop production and extraction modes.	46
3.5	Overview of the two-loop contributions and extraction modes.	46
3.6	Overview of the integration-side route-profiles.	49
3.7	Representative resource measurements for the implemented public antenna routes. The build time measures <code>BuildAntenna[... , IntegrableForm -> True]</code> , followed by <code>IntegrateAntenna[...]</code> for the integration time. Each route was evaluated in a fresh Wolfram kernel. Peak memory is the maximum kernel-memory increase relative to the initial baseline.	53
4.1	Overview of the master integrals used in the build- and integration-stage by each antenna type [4, 11].	54
4.2	Construction and integration overview of every antenna route implemented in <i>AntCalc</i> . .	79
4.3	Validation evidence and status of every antenna route implemented in <i>AntCalc</i> . “Not applicable” marks routes with no real unresolved parton to take a soft/collinear limit of (A_2^0, A_2^1, A_2^2) ; “Not performed” marks routes where such a limit exists but was not checked in this work — see Section 4.4 for why A_3^0 was chosen to demonstrate the method rather than repeating it on every route. The massive A_3^0 route is deliberately marked as less mature than the massless set: it agrees with the literature locally, but has not yet been checked against any global observable.	79

B.1	Overview of the propagators found throughout the thesis, alongside their mathematical and graphical representations.	106
B.2	Overview of the vertices encountered throughout this thesis, alongside their mathematical and graphical representations.	107

List of Abbreviations and Symbols

Abbreviations and Acronyms

Term	Meaning
QCD	Quantum Chromodynamics
LO / NLO / NNLO	Leading / Next-to-Leading / Next-to-Next-to-Leading Order
IR / UV	Infrared / Ultraviolet
MI	Master Integral(s)
RR / RV / VV	double-Real, Real-Virtual and double-Virtual NNLO contributions
AP	Altarelli–Parisi (splitting function)
CDR	Conventional Dimensional Regularisation
KLN	Kinoshita–Lee–Nauenberg (theorem)
SM	Standard Model of Particle Physics
EFT	Effective Field Theory
SUSY	Supersymmetry
MC	Monte Carlo
IBP	Integration by Parts

Antenna and Regularisation Notation

Symbol	Meaning
X_n^l	Generic antenna symbol: n is the final-state multiplicity, l the loop order
A, B, C	Quark-antiquark antenna families (implemented)
D, E	Quark-gluon antenna families (not implemented in the present work)
F, G, H	Gluon-gluon antenna families (not implemented in the present work)
X (undecorated)	Leading-colour component (from the raw “Lead” interference term)
$\tilde{X}$	Subleading-colour component (raw “Sublead” interference term)
$\hat{X}$	Quark-loop component (raw “QL” interference term)

Symbol	Meaning
$\check{X}$	Double-fermion-loop self-interference component (NNLO A_2^2 only)
S_ϵ	$(4\pi)^{-\epsilon} e^{\epsilon\gamma_E}$, the $\overline{\text{MS}}$ normalisation factor
Λ_l	$(8\pi^2)^l$, loop-order normalisation
G_k	$(8\pi^2)^k$, perturbative-order scale normalisation
$C(\epsilon, k)$	$G_k S_\epsilon$, combined normalisation factor
$\mathcal{N}_X$	Build-stage colour and coupling normalisation
$\mathcal{N}(\epsilon, k)$	Full integration normalisation, $C(\epsilon, k)/\Phi_2$
$\mathcal{T}$ -objects	Process-wide colour- and kinematics-conscious objects assembled from antennae, used in the R -ratio construction
Φ_n	d -dimensional n -body phase-space volume

General Kinematics

Symbol	Meaning
α_s	Strong coupling constant
$\epsilon, d = 4 - 2\epsilon$	Dimensional-regularisation parameter; spacetime dimension
s_{ij}	Mandelstam invariant, $2p_i \cdot p_j$
μ	Renormalisation/regularisation scale
q^2	Hard-process invariant mass squared
N	Number of colours (= 3)
N_f	Number of active quark flavours
C_F, C_A	Quadratic Casimir invariants (fundamental / adjoint representation)
g_s	Strong coupling, $g_s = \sqrt{4\pi\alpha_s}$

Chapter 1

Introduction

High-energy collider experiments, such as the proton-proton collisions studied at the Large Hadron Collider (LHC) at CERN, have, in recent years, provided ever-increasing precision in measurements of processes governed by the strong interaction. To be able to use these measurements fully, theoretical calculations must achieve similar levels of precision. At high energies, Quantum Chromodynamics (QCD) permits such predictions to be organised as a perturbative expansion in powers of the strong coupling constant α_s .

Obtaining higher-order perturbative predictions is not simply a matter of calculating additional Feynman diagrams. Real-emission and virtual-loop contributions develop infrared singularities in kinematic configurations involving soft (low energy) or collinear partons. These individual contributions are divergent, and their infrared singularities must be extracted and cancelled before numerical integration can be performed. For infrared-safe observables, such as the hadronic R -ratio, the singularities cancel between the real and virtual contributions, leaving a finite physical result.

The antenna subtraction formalism provides a framework for extracting and cancelling these infrared singularities, rendering the individual contributions suitable for numerical integration. The formalism describes unresolved radiation between pairs of colour-connected hard partons through universal antenna functions. These functions reproduce the relevant infrared limits locally and, after analytic integration over the unresolved phase space in dimensional regularisation, $d = 4 - 2\epsilon$, yield Laurent series whose ϵ -pole structure cancels that of the virtual contributions. Their systematic construction and integration are therefore central to higher-order QCD calculations within the antenna subtraction framework.

This thesis develops and validates *AntCalc*, a symbolic framework for the automated construction and integration of QCD antenna functions. The work focuses on massless final-final quark-antiquark antennae through next-to-next-to-leading order (NNLO), combining symbolic amplitude generation with reduction methods for phase-space integrals. The resulting implementation is tested globally through the calculation of the massless hadronic R -ratio, a well-known inclusive observable in e^+e^- annihilation that provides a benchmark for the complete implementation, up to NNLO. Locally, individual antenna functions are tested, where appropriate, by verifying that they reproduce the expected eikonal factors and Altarelli-Parisi (AP) splitting functions in the relevant unresolved limits.

The remainder of the thesis is organised as follows. Chapter 2 introduces the theoretical foundations required throughout this work, including perturbative QCD, infrared singularities, dimensional regularisation, and the antenna subtraction formalism. It defines the massless antenna functions and discusses their colour structure, phase-space factorisation, and role in fixed-order calculations. The chapter also introduces the normalisation conventions used throughout. Chapter 3 describes *AntCalc*'s framework, including its architecture, antenna construction and integration workflows, reduction procedures, and the

conventions and validation tools used to obtain the results. Chapter 4 presents the antenna functions obtained using *AntCalc*, beginning with the master integrals required for their integration. The A_3^0 antenna is discussed in detail, from its unintegrated construction to its integrated form and the validation of its soft and collinear limits. Selected NLO and NNLO antenna results are then presented more concisely. The chapter concludes with an exploratory extension of the A_3^0 antenna to the massive case. Chapter 5 provides a global validation of the framework through the calculation of the massless hadronic R -ratio through NNLO. The relevant antenna contributions are combined, demonstrating the cancellation of infrared poles and agreement with the known finite result. Chapter 6 discusses future directions for *AntCalc*, including more efficient symbolic and reduction backends, completion of the massive antenna sector, general model support, and extensions to higher perturbative orders, particularly N³LO. Finally, Chapter 7 summarises the main conclusions of the thesis.

Chapter 2

Theoretical Foundations

2.1 QCD: Fields, Interactions and Perturbative Amplitudes

QCD is the quantum field theory of the strong interaction, whose elementary particles are the quarks and the gluons. Quarks are fermions that carry colour and interact through the exchange of gluons. Owing to the non-Abelian gauge structure of QCD, gluons themselves also carry colour charge and, therefore, undergo self-interaction.

At sufficiently high energy scales, the strong coupling, α_s , becomes sufficiently small for physical observables to be calculated perturbatively, as an expansion in powers of α_s . In this framework, the interaction structure of QCD determines the amplitudes for quark and gluon radiation that are studied throughout this thesis.

2.1.1 Quark and Gluon Fields

Quarks are spin-1/2 fields carrying flavour and a three-valued colour charge. They are QCD's matter fields. Gluons are the eight gauge fields associated with local $SU(3)_c$ colour symmetry. The QCD Lagrangian describes their propagation and interaction. It is written as follows,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f) \psi_f. \quad (2.1)$$

In this expression, $G_{\mu\nu}^a$ is the non-Abelian field-strength tensor, D_μ is the covariant derivative, ψ_f is the quark field and m_f is the quark mass. The flavour label f is summed over the quark species included in the theory, while the adjoint colour index a runs from one to eight. This Lagrangian is cleanly divided into two pieces, the quark and the gluon pieces.

Defining the covariant derivative as,

$$D_\mu = \partial_\mu - ig_s T^a A_\mu^a \quad (2.2)$$

where T^a represents the generators of the fundamental representation of $SU(3)_c$ and A_μ^a is the gluon gauge field, the quark Lagrangian term may be expanded to,

$$\mathcal{L}_q = \sum_f [\bar{\psi}_f (i\gamma^\mu \partial_\mu - m_f) \psi_f + g_s \bar{\psi}_f \gamma^\mu T^a \psi_f A_\mu^a]. \quad (2.3)$$

Here g_s is the strong coupling. In the expanded quark term, the first piece describes free quark propagation

and the second quark-gluon interaction. This second piece is also responsible for generating the $q\bar{q}g$ vertex.

We now also define the totally antisymmetric structure constants of the $SU(3)_c$ Lie algebra f^{abc} as,

$$[T^a, T^b] = if^{abc}T^c. \quad (2.4)$$

They may be used to define the non-Abelian field-strength tensor $G_{\mu\nu}^a$ as,

$$G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c. \quad (2.5)$$

The first two terms of the field-strength tensor are Abelian-like and, in fact, they can be rewritten using the Abelian-like linear part of the non-Abelian field-strength tensor $F_{\mu\nu}^a$, as,

$$G_{\mu\nu}^a = F_{\mu\nu}^a + g_s f^{abc} A_\mu^b A_\nu^c, \quad F_{\mu\nu}^a \equiv \partial_\mu A_\nu^a - \partial_\nu A_\mu^a. \quad (2.6)$$

The second term arises because, as shown in Eq. 2.4, the $SU(3)_c$ generators do not commute. The non-Abelian field-strength tensor is used to write the gluon Lagrangian term. After expanding it becomes,

$$\mathcal{L}_g = -\frac{1}{4} \left(F_{\mu\nu}^a \right)^2 - \frac{g_s}{2} f^{abc} F^{a\mu\nu} A_\mu^b A_\nu^c - \frac{g_s^2}{4} f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (2.7)$$

Here, much like in $\mathcal{L}_q$, the gluon Lagrangian is also divided into three terms. The first term is quadratic in the gluon field and, after gauge-fixing, describes free gluon propagation as [1],

$$\Delta_{ab,\mu\nu}(k) = \delta_{ab} \frac{i}{k^2} \left(-g_{\mu\nu} + (1-\lambda) \frac{k_\mu k_\nu}{k^2} \right), \quad \mathcal{L}_{\text{gf}} = -\frac{1}{2\lambda} \left(\partial^\mu A_\mu^a \right)^2 \quad (2.8)$$

where $\mathcal{L}_{\text{gf}}$ is the gauge-fixing Lagrangian, δ_{ab} is the Kronecker delta, and λ is the gauge parameter. Throughout this text the 't Hooft-Feynman convention is used, which makes $\lambda = 1$, following the default convention from *FeynArts* [2]. It is important to note, however, that physical observables are independent of the gauge choice. Unlike the remaining terms, the first contains no non-Abelian self-interaction. The cubic and quartic terms in the gluon fields generate, respectively, the three- and four-gluon vertices, ggg and $gggg$.

Using a covariant gauge-fixing term in a non-Abelian theory like QCD requires it to be supplemented by a ghost Lagrangian, described as [1],

$$\mathcal{L}_{\text{ghost}} = \partial_\mu \eta^{a\dagger} \left(D_{ab}^\mu \eta^b \right). \quad (2.9)$$

This is the default ghost sector as used in *FeynArts* [2] and, as for the λ convention, is the one used for the duration of the present text and in *AntCalc*. These ghost fields cancel the unphysical timelike and longitudinal gluon polarisations that would, otherwise, propagate in covariant-gauge loop diagrams. This ensures that only the two physical transverse polarisation states contribute to physical amplitudes [1].

Together, the two pieces, $\mathcal{L}_q$ and $\mathcal{L}_g$, constitute the gauge-invariant core of QCD. They generate the three elementary interaction vertices among the quark and gluon fields: $q\bar{q}g$, ggg and $gggg$ [1, 3].

2.1.2 From the QCD Lagrangian to Feynman Diagrams

The interaction terms in $\mathcal{L}_q$ and $\mathcal{L}_g$ determine the interaction vertices of QCD, which, together with the corresponding propagators, form the Feynman rules we use to compute scattering amplitudes

perturbatively. Feynman diagrams are graphical representations of the individual contributions to the perturbative expansion of an amplitude. Each process may require more than one diagram to be fully represented; in that case, the complete amplitude is given as the sum of the amplitudes associated with all contributing diagrams. Fig. 2.1 depicts the two Feynman diagrams contributing to the process $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$, where a virtual photon splits into a quark-antiquark pair, with the gluon emitted from either the quark or antiquark line.

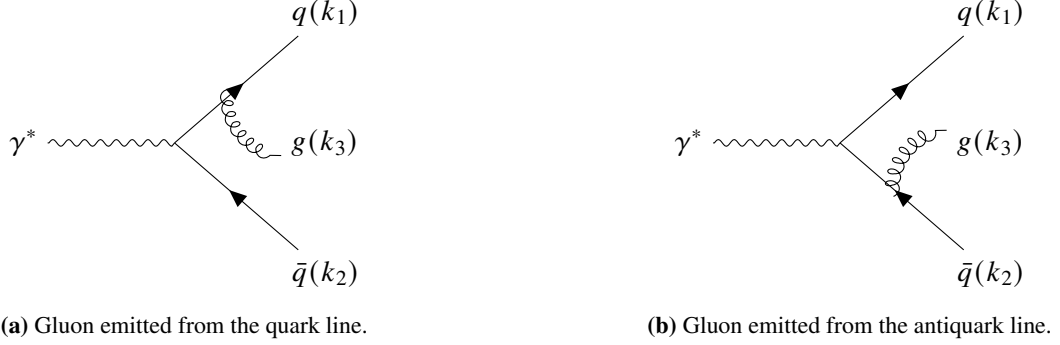


Figure 2.1: Tree-level diagrams for the process $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$.

The final amplitude in this case would then follow,

$$\mathcal{M}_{\gamma^* \rightarrow qg\bar{q}}^{(0)} = \mathcal{M}_{q\text{-emission}}^{(0)} + \mathcal{M}_{\bar{q}\text{-emission}}^{(0)}. \quad (2.10)$$

In the diagrams in Fig. 2.1, straight, curly, and wavy lines denote fermions (quarks), gluons and photons, respectively. These diagrams represent the reduced hadronic-current process with γ^* as an off-shell and colourless source. In the full process, $e^+e^- \rightarrow \gamma^* \rightarrow qg\bar{q}$, the photon is internal between the lepton (electron-positron) and hadron (quark-antiquark) currents. All other lines are external, meaning that they represent outgoing asymptotic partons, the quarks and gluons. Each of the diagrams features two vertices, where the lines meet. The vertices represent the interactions between the particles, which are determined by their respective Lagrangians. Each diagram contains an electromagnetic $\gamma^* q\bar{q}$ current insertion and a QCD $q\bar{q}g$ interaction vertex. The former follows from the QED sector, whereas the latter is generated by $\mathcal{L}_{\text{QCD}}$. A rate or cross section often depends on the squared amplitude $|\mathcal{M}|^2$, given as,

$$|\mathcal{M}|^2 = \sum_{D,D'} \mathcal{M}_D \mathcal{M}_{D'}^*, \quad (2.11)$$

where D and D' index the Feynman diagrams contributing to the amplitude, so that the sum includes both the diagonal $D = D'$ terms and the interference between distinct diagrams.

An amplitude $\mathcal{M}$ is computed from a Feynman diagram using Feynman rules. This ruleset is given in Section B.1.

2.2 Divergences in Perturbative QCD

2.2.1 Infrared Divergences

Beyond leading order in perturbation theory, QCD predictions receive contributions from both real and virtual corrections. Real corrections correspond to processes with additional parton emission in the final state, while virtual corrections arise from loop contributions to the amplitude. Both are required

at a given perturbative order and can develop infrared singularities, which must ultimately cancel in infrared-safe physical observables.

The two QCD infrared limits, soft and collinear, can be derived explicitly as follows. We start by defining the physical system under study: consider two massless final-state partons i and j with four-momenta p_i and p_j . By momentum conservation, the four-momentum of the internal virtual propagator is $P = p_i + p_j$. The matrix element contains this propagator and in the square amplitude it appears as a $1/P^2$ factor. Expanding it and applying the massless condition ($p^2 = 0$), we get $P^2 = 2p_i \cdot p_j = 2E_i E_j (1 - \cos \theta_{ij})$ [1]. Importantly, this quantity defines the Mandelstam invariants as $s_{ij} = 2p_i \cdot p_j = P^2$, in the massless regime¹, which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when θ_{ij} does. In both cases,

$$\lim_{E_i \rightarrow 0} s_{ij} = 0, \quad \lim_{\theta_{ij} \rightarrow 0} s_{ij} = 0. \quad (2.12)$$

A soft limit exists only where removing the unresolved parton leaves a physical lower-multiplicity process. For the $\gamma^* \rightarrow q(k_1) g(k_3) \bar{q}(k_2)$ process, this permits only a soft gluon because, since $\gamma^* \rightarrow g\bar{q}$ does not exist, the soft quark limit is not possible. That means that a quark can only become unresolved collinearly. At NNLO, however, a quark-antiquark pair can become soft as a unit, when being emitted by an intermediate gluon. In that topology, apart from the soft gluon limit, only the collinear gq and $\bar{q}$ limits exist, with the collinear $q\bar{q}$ limit not existing at this perturbative order. These and all other limits considered throughout this thesis are shown in Table 2.1.

Unresolved limit	Kinematic condition	Vanishing invariant(s)
$(1_q, 3_g, 2_{\bar{q}})$ — NLO real radiation		
Soft gluon	$p_3 \rightarrow 0$	s_{13}, s_{23}
Collinear	$1_q \parallel 3_g$	s_{13}
$(1_q, 3_g, 4_g, 2_{\bar{q}})$ — NNLO real radiation		
Soft gluon	$p_3 \rightarrow 0$	s_{13}, s_{34}
Collinear (qg)	$1_q \parallel 3_g$	s_{13}
Collinear (gg)	$3_g \parallel 4_g$	s_{34}
Double soft	$p_3, p_4 \rightarrow 0$	s_{13}, s_{34}, s_{24}
Soft-collinear	$p_3 \rightarrow 0, 4_g \parallel 2_{\bar{q}}$	s_{13}, s_{34}, s_{24}
Triple collinear	$1_q \parallel 3_g \parallel 4_g$	s_{134}
Double single collinear	$1_q \parallel 3_g, 2_{\bar{q}} \parallel 4_g$	s_{13}, s_{24}
$(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}})$ — NNLO real radiation		
Soft $q'\bar{q}'$ pair	$p_3, p_4 \rightarrow 0$	s_{34} (double pole)
Collinear ($q'\bar{q}'$)	$3_{q'} \parallel 4_{\bar{q}'}$	s_{34}
Triple collinear	$1_q \parallel 3_{q'} \parallel 4_{\bar{q}'}$	s_{134}

¹The general definition of the Mandelstam invariant is $s_{ij} = (p_i + p_j)^2$, but as $p_k^2 = 0$ for any k , in this massless regime, then the massless Mandelstam invariant is defined as $s_{ij} = 2p_i \cdot p_j$.

Unresolved limit	Kinematic condition	Vanishing invariant(s)
$(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})$ — NNLO real radiation, identical flavour		
Triple collinear	$2_{\bar{q}} \parallel 3_q \parallel 4_{\bar{q}}$	s_{234}

Table 2.1: Infrared limits of the real-radiation matrix elements for $\gamma^* \rightarrow q\bar{q}$ through NNLO, listed by final-state parton configuration. Each configuration is colour-ordered, with the hard quark and antiquark first and last; p_i is the momentum of parton i , $\parallel$ denotes a collinear pair, and $s_{i\dots k} = (p_i + \dots + p_k)^2$. Limits related by $q \leftrightarrow \bar{q}$ or by gluon relabelling are not listed separately. The subtraction counterterm of Eq. 2.13 must reproduce the matrix element in each of these limits [4].

From an experimental point of view, a real emission that becomes soft or collinear may be unresolved, meaning that it cannot be distinguished from a configuration in which that emission is absent. For an infrared-safe observable, the two configurations yield the same value of the observable in the corresponding limit.

The virtual corrections contain infrared singularities related to those arising from unresolved real radiation. According to the Kinoshita–Lee–Nauenberg (KLN) theorem, these singularities cancel once all degenerate contributions to an infrared-safe physical observable are appropriately combined [5, 6]. The resulting observable is therefore finite, despite the presence of infrared divergences in the individual real and virtual contributions.

In practice, however, this cancellation cannot be performed directly at the integrand level, since the real and virtual contributions are separately divergent and are defined on phase spaces with different particle multiplicities. At NLO, the real contribution contains one additional final-state parton and is integrated over the $(n + 1)$ -particle phase space, whereas the virtual contribution is integrated over the n -particle phase space. A subtraction method resolves this mismatch by introducing a local counterterm $d\sigma^S$ that reproduces the singular behaviour of the real contribution $d\sigma^R$ in all unresolved limits. The counterterm can then be subtracted from the real contribution, rendering their difference finite and suitable for numerical integration. The same counterterm is analytically integrated over the unresolved one-particle phase space and added to the virtual contribution, where its explicit infrared poles cancel those of the virtual correction. Schematically, at NLO, this is expressed as,

$$\sigma = \int_{n+1} (d\sigma^R - d\sigma^S) + \int_n \left(d\sigma^V + \int_1 d\sigma^S \right) \quad (2.13)$$

where $d\sigma^R$ represents the differential real radiation cross-section, $d\sigma^V$ the differential virtual corrections cross-section and $d\sigma^S$ the differential subtraction counterterm [7]. Importantly, $d\sigma^S$ must reproduce $d\sigma^R$'s exact singular behaviour at every point in the phase space, such that the divergences cancel locally.

2.2.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. In loop diagrams, the loop momentum l is integrated over the whole of momentum space. Regions in which l becomes soft or collinear to an external massless momentum can give rise to infrared singularities, which contribute to the singular structure of $d\sigma^V$. In addition, loop integrals may become divergent when the magnitude of the loop momentum becomes arbitrarily large, $|l| \rightarrow \infty$. These are referred to as ultraviolet (UV) divergences.

The UV behaviour of a loop integral can be understood through simple power counting. In d

dimensions, the loop integration measure can be written in spherical coordinates as,

$$d^d l = l^{d-1} dl d\Omega_{d-1}. \quad (2.14)$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the l^{d-1} measure term. Since each propagator contributes $\sim 1/l^2$ to the integrand, schematically, a diagram with n_p propagators in the loop and no extra powers of l in the numerator becomes,

$$\int^\infty l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^\infty l^{d-1-2n_p} dl \quad (2.15)$$

which is UV divergent whenever $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$ [3, 8].

The presence of UV divergences means that loop corrections expressed in terms of the bare coupling α_0 , implicit in the covariant derivative of the QCD Lagrangian in Eq. 2.1, are not finite beyond tree level, since their higher-order coefficients diverge [1]. This property makes α_0 unsuitable as the order-by-order perturbative series variable we require. These divergences are removed through renormalisation, in which the bare coupling is traded for a finite renormalised coupling, α_s , defined at a renormalisation scale μ [8]. This renormalisation is made explicit for the one-loop antenna sets in Subsection 3.3.2.

2.3 Dimensional Regularisation

The subtraction scheme from Eq. 2.13 separates the NLO cross-section into contributions that can ultimately be evaluated numerically. The difference $d\sigma^R - d\sigma^S$ is finite in the unresolved regions and can therefore be integrated numerically in four dimensions. The subtraction term must, however, still be integrated analytically over the unresolved one-parton phase space,

$$\int_1 d\sigma^S. \quad (2.16)$$

Since $d\sigma^S$ is designed explicitly to carry over all of $d\sigma^R$'s divergent structures, this integral blows up once integrated over four dimensions, as does the virtual contribution $d\sigma^V$, whose infrared divergences originate from the loop integration, as explored in Subsection 2.2.2. To regulate these divergent contributions we continue the number of space-time dimensions away from four, via,

$$d = 4 - 2\epsilon \quad (2.17)$$

so that infrared divergences appear as explicit analytic poles in ϵ . After analytic integration over the unresolved phase space, this leaves us with two ϵ -dependent expressions, for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left(d\sigma^V + d\sigma_1^S \right). \quad (2.18)$$

From the KLN theorem, as described in Subsection 2.2.1, we know that these poles must cancel. This combination is therefore finite as $\epsilon \rightarrow 0$, allowing the remaining phase-space integration to be performed numerically in four dimensions.

Though the underlying principle is always the same, dimensional regularisation can be implemented using different schemes. The most common are 't Hooft-Veltman [9], Four-Dimensional Helicity [10] and Conventional Dimensional Regularisation (CDR) [1]. In CDR, all momenta and polarisation states are

treated in $d = 4 - 2\epsilon$ dimensions. This is the scheme used throughout the present work, since *AntCalc*'s primary objective is the simplification and democratisation of the antenna framework established in [4], which uses CDR throughout.

While using CDR, the integrated and unintegrated antennae considered in this work are normalised using the following convention. First, the antennae are expressed in the $\overline{\text{MS}}$ (modified minimal subtraction)-convention using $S_\epsilon = (4\pi)^{-\epsilon} e^{\epsilon\gamma_E}$ [4]. Second, they are also normalised in relation to the strong coupling constant expansion using,

$$G_k = \left(8\pi^2\right)^{n-2} \Lambda_l = \left(8\pi^2\right)^{n-2} \left(8\pi^2\right)^l \quad (2.19)$$

defining also $k = n - 2 + l$ as the perturbative-order index, which is zero for LO, one for NLO and two for NNLO, for n final-state particles and l loops. These two normalisations are then bundled into a general convention used throughout, which is written as,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}, \quad C(\epsilon, k) = G_k S_\epsilon \quad (2.20)$$

using the d -dimensional two-particle phase-space measure Φ_2 . This is the general representation of the work's normalisation convention.

Furthermore, the square matrix element amplitudes used to compute the antenna functions are normalised according to their number of final-state particles n , using,

$$N(n) = 4\pi\alpha \sum_q e_q^2 \left(g^2\right)^{n-2} \left(N^2 - 1\right) |\mathcal{M}_2^0|^2. \quad (2.21)$$

As described in Subsection 2.2.2, we need to trade the bare coupling constant α_0 for a suitable order by order perturbative series variable. That is achieved as follows [4],

$$\begin{aligned} \alpha_0 \mu_0^{2\epsilon} S_\epsilon &= \alpha_s \mu^{2\epsilon} \left[1 - \frac{\beta_0}{\epsilon} \frac{\alpha_s}{2\pi} + \left(\frac{\beta_0^2}{\epsilon^2} - \frac{\beta_1}{2\epsilon} \right) \left(\frac{\alpha_s}{2\pi} \right)^2 + \mathcal{O}(\alpha_s^3) \right], \\ \beta_0 &= \frac{11N - 2N_f}{6}, \quad \beta_1 = \frac{34N^3 - 13N^2 N_f + 3N_f}{12N} \end{aligned} \quad (2.22)$$

where for the massless case, the renormalisation scale is set to $\mu^2 = q^2$. Here, μ_0 is the bare mass parameter keeping α_0 dimensionless and β_0 is the one-loop coefficient of the QCD beta function, with $N = 3$ the number of colours and N_f the number of light quark flavours [1, 4].

2.4 The Antenna Subtraction Formalism

In perturbative QCD, computing the perturbative corrections to a certain jet observable $J_m^{(n)}$, requires summing over all partonic multiplicity channels up to the desired order. As the orders increase in complexity, so does the number of channels: there is one at LO, two at NLO, five at NNLO, and so on. Each channel contains ultraviolet and infrared singularities. The former are removed via renormalisation, as shown in Eq. 2.22, and the latter cancel among each other when all channels are summed over, as per the KLN theorem [4].

At LO, the approximation to the m -jet cross section is computed as the integral over the appropriate

phase space,

$$d\sigma_{\text{LO}} = \int_{d\Phi_m} d\sigma^B, \quad d\sigma^B = \sum_n d\Phi_n(p_1, \dots, p_n; q) \frac{1}{S_n} |\mathcal{M}_n(p_1, \dots, p_n)|^2 J_m^{(n)}(p_1, \dots, p_n) \quad (2.23)$$

where $d\sigma^B$ is the tree-level n -parton contribution to the m -jet cross section. S_n is a symmetry factor for identical partons in the final state, and Φ_n is the n -parton final-state phase space in d dimensions [4],

$$d\Phi_n(p_1, \dots, p_n; q) = \frac{d^{d-1}p_1}{2E_1(2\pi)^{d-1}} \cdots \frac{d^{d-1}p_n}{2E_n(2\pi)^{d-1}} (2\pi)^d \delta^{(d)}(q - p_1 - \dots - p_n). \quad (2.24)$$

At NLO, however, the m -jet cross section is now written using the two contributions, a real emission channel characterised by the $\gamma^* \rightarrow q\bar{q}g$ process, and a virtual channel, where a gluon is exchanged by the hard quarks, $\gamma^* \rightarrow q\bar{q}$. These contributions are summed to reach,

$$d\sigma_{\text{NLO}} = \int_{d\Phi_{m+1}} d\sigma_{\text{NLO}}^R + \int_{d\Phi_m} d\sigma_{\text{NLO}}^V. \quad (2.25)$$

At NNLO, the m -jet cross section is written using the three contributions that come from five different channels: RR has three, $\gamma^* \rightarrow q\bar{q}gg$, $q\bar{q}q\bar{q}$, $q\bar{q}q'\bar{q}'$, and RV and VV only have one each, $\gamma^* \rightarrow q\bar{q}g$, with a gluon loop, and $\gamma^* \rightarrow q\bar{q}$ with two gluon loops,

$$d\sigma_{\text{NNLO}} = \int_{d\Phi_{m+2}} d\sigma_{\text{NNLO}}^{RR} + \int_{d\Phi_{m+1}} d\sigma_{\text{NNLO}}^{RV} + \int_{d\Phi_m} d\sigma_{\text{NNLO}}^{VV}. \quad (2.26)$$

2.4.1 Subtraction Terms

The contributions introduced for the NLO and NNLO m -jet cross sections are finite when summed, but not individually. As such, when the naive integrals are numerically computed, they do not converge. Our solution to this problem is the introduction of subtraction terms for each contribution, that follow their respective contributions' infrared-singularity structure for the whole phase space. After subtracting the singularities, the subtraction terms are added to the last term, integrated over $d\Phi_m$.

NLO

At NLO, there is need of one subtraction term, $d\sigma_{\text{NLO}}^S$, which cancels $d\sigma_{\text{NLO}}^R$'s singularities. Using it, the m -jet cross section becomes,

$$d\sigma_{\text{NLO}} = \int_{d\Phi_{m+1}} \left(d\sigma_{\text{NLO}}^R - d\sigma_{\text{NLO}}^S \right) + \int_{d\Phi_m} \left(d\sigma_{\text{NLO}}^V + \int_{d\Phi_1} d\sigma_{\text{NLO}}^S \right) = \int_{d\Phi_m} \left(d\sigma_{\text{NLO}}^V + \int_1 d\sigma_{\text{NLO}}^S \right). \quad (2.27)$$

The real-emission contribution is essentially equal to the Born-level cross-section, as written in Eq. 2.23, but for $m+1$ particles. Its subtraction counterterm has its same unintegrated behaviour for every singular limit and must be analytically integrable over the required phase space. It must also be local and not introduce spurious singularities elsewhere. From the KLN theorem, the sum of the virtual contribution and the subtraction counterterm is infrared-finite, that is, after analytic integration, the integrated real-emission subtraction cross-section carries explicit ϵ -poles which cancel virtual-correction's singularity

structure pole-by-pole. It is obtained as a sum of antenna functions, as follows,

$$d\sigma_{\text{NLO}}^S = \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \sum_j X_{ijk}^0 |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}). \quad (2.28)$$

The subtraction term is defined in terms of the redefined on-shell momenta $p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}$, where $\tilde{p}_I$ and $\tilde{p}_K$ are linear combinations of p_i, p_j, p_k and denote the momenta of the parent partons, in our case the $q\bar{q}$ hard pair, with i, j, k denoting the daughter partons after the gluon emission, that is $qg\bar{q}$. Parent partons I and K are colour connected. Daughter partons i and j , and j and k are also connected, but i and k are not. A diagram of these relationships is shown in Fig. 2.2.

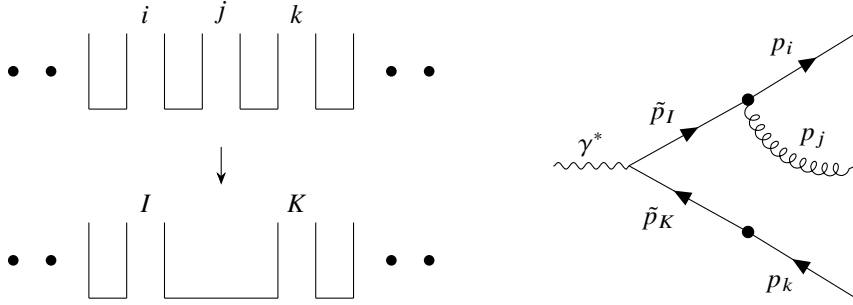


Figure 2.2: The colour connections in between parent and daughter partons before and after an emission, accompanied also by a schematic Feynman diagram using the momenta labels.

The jet observable considered throughout this work is the massless hadronic R -ratio. This quantity is fully inclusive in the phase space, which means that $J_m^{(n)} = 1$ everywhere, independent of the order. Without the jet function in the expression, the phase space can be factorised as,

$$d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) = d\Phi_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) d\Phi_{X_{ijk}}(p_i, p_j, p_k; q = \tilde{p}_I + \tilde{p}_K) \quad (2.29)$$

which here equals,

$$d\Phi_3 = P_2 d\Phi_{X_{ijk}}, \quad P_2 = \int d\Phi_2. \quad (2.30)$$

NNLO

Analogously, at NNLO, there is need of two subtraction counterterms, $d\sigma_{\text{NNLO}}^S$ and $d\sigma_{\text{NNLO}}^T$, subtracting respectively $d\sigma_{\text{NNLO}}^{RR}$ and $d\sigma_{\text{NNLO}}^{RV}$. Using them, the m -jet cross section becomes,

$$\begin{aligned} d\sigma_{\text{NNLO}} &= \int_{d\Phi_{m+2}} \left(d\sigma_{\text{NNLO}}^{RR} - d\sigma_{\text{NNLO}}^S \right) + \int_{d\Phi_{m+1}} \left(d\sigma_{\text{NNLO}}^{RV} - d\sigma_{\text{NNLO}}^T \right) + \int_{d\Phi_m} d\sigma_{\text{NNLO}}^{VV} \\ &= \int_{d\Phi_m} \left(d\sigma_{\text{NNLO}}^{VV} + \int_{d\Phi_2} d\sigma_{\text{NNLO}}^S + \int_{d\Phi_2} d\sigma_{\text{NNLO}}^T \right). \end{aligned} \quad (2.31)$$

At this order, the complexity rises. The subtraction counterterms must take into account the multiple possible configurations. The subtraction counterterms are written as,

$$\begin{aligned} d\sigma_{\text{NNLO}}^S &= d\sigma^{S,a} + d\sigma^{S,b} + d\sigma^{S,c} + d\sigma^{S,d} \\ d\sigma_{\text{NNLO}}^T &= d\sigma^{T,a} + d\sigma^{T,b} + d\sigma^{T,c}. \end{aligned} \quad (2.32)$$

The subterms of each of the counterterms match the real-real and real-virtual configurations present at this order. This matching is shown in Table 2.2.

Configuration	SubTerm
RR subtraction counterterm – $d\sigma_{\text{NNLO}}^S$	
Single unresolved parton	$d\sigma^{S,a}$
Two colour-connected unresolved partons	$d\sigma^{S,b}$
Two almost colour-unconnected unresolved partons	$d\sigma^{S,c}$
Two fully colour-unconnected unresolved partons	$d\sigma^{S,d}$
RV subtraction counterterm – $d\sigma_{\text{NNLO}}^T$	
Explicit infrared poles from the one-loop matrix elements	$d\sigma^{T,a}$
One of the partons becomes soft or collinear	$d\sigma^{T,b}$
Spurious poles introduced by the other two subterms	$d\sigma^{T,c}$

Table 2.2: NNLO unresolved configurations and their matching subterm, part of the $d\sigma_{\text{NNLO}}^S$ and $d\sigma_{\text{NNLO}}^T$ subtraction counterterms.

The subterms for both counterterms are given in terms of the antenna functions as follows, noting that at this order, four-parton and three-parton one-loop antenna functions are necessary, written as X_{ijkl}^0 and X_{ijk}^1 . First for the real-real subtraction counterterm,

$$\begin{aligned}
d\sigma^{S,a} &= \sum_{m+2} d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) \frac{1}{S_{m+2}} \\
&\quad \sum_j X_{ijk}^0 |\mathcal{M}_{m+1}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+2})|^2 J_m^{(m+1)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+2}), \\
d\sigma^{S,b} &= \sum_{m+2} d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) \frac{1}{S_{m+2}} \\
&\quad \sum_{j,k} \left(X_{ijkl}^0 - X_{ijk}^0 X_{IKl}^0 - X_{jkl}^0 X_{iJL}^0 \right) |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2})|^2 \\
&\quad J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2}),
\end{aligned} \tag{2.33}$$

And then the real-virtual counterterm,

$$\begin{aligned}
d\sigma^{T,a} &= \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \\
&\quad \sum_{i,k} -X_{ijk}^0(s_{ik}) |\mathcal{M}_{m+1}(p_1, \dots, p_i, p_k, \dots, p_{m+1})|^2 J_m^{(m+1)}(p_1, \dots, p_i, p_k, \dots, p_{m+1}), \\
&\quad \text{with } X_{ijk}^0 = C(\epsilon, 1) \int d\Phi_{X_{ijk}} X_{ijk}^0, \\
d\sigma^{T,b} &= \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \\
&\quad \sum_j X_{ijk}^0 |\mathcal{M}_m^1(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) \\
&\quad + X_{ijk}^1 |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}), \\
&\quad \text{with } X_{ijk}^1 \rightarrow X_{ijk}^1 + \frac{\beta_0 (4\pi)^\epsilon e^{-\epsilon\gamma_E}}{8\pi^2} X_{ijk}^0 (s_{ijk}^{-\epsilon} - (\mu^2)^{-\epsilon}),
\end{aligned} \tag{2.34}$$

$d\sigma^{S,c}$, $d\sigma^{S,d}$, and $d\sigma^{T,c}$ are only required for processes with three or more hard partons, beyond those considered in the current text. Their complete form is written in [4].

The colour-connection notation is,

$$\begin{aligned}
(i, j, k) &\rightarrow (\tilde{I}, \tilde{K}) \\
(\tilde{K}, l, m) &\rightarrow (\tilde{K}, \tilde{M})
\end{aligned} \tag{2.35}$$

and the three characteristic colour-connection configurations are shown in Figs. 2.3, 2.4, and 2.5.

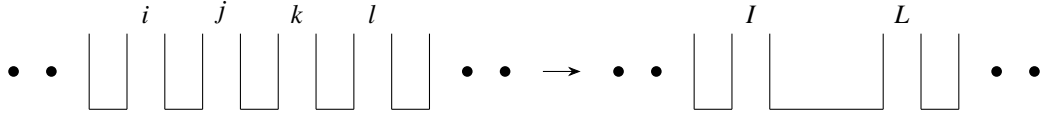


Figure 2.3: The colour connection of the partons for the double unresolved antenna.

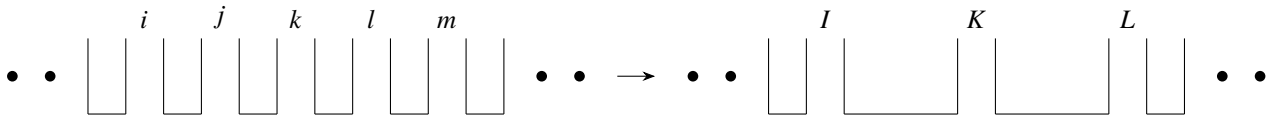


Figure 2.4: The colour connection of the partons for two adjacent single unresolved antennae.

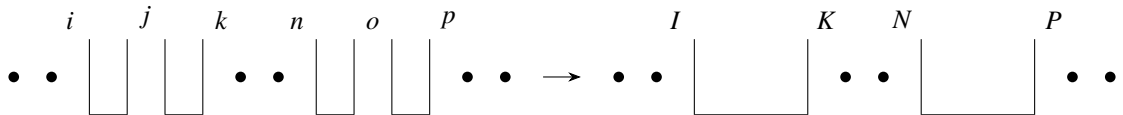


Figure 2.5: The colour connections of the partons for two disconnected single unresolved antennae.

Once again, as for NLO, the massless hadronic R -ratio is the jet observable considered, and hence $J_m^{(n)} = 1$. The phase space factorises as,

$$d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) = d\Phi_m(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2}; q) d\Phi_{X_{ijkl}}(p_i, p_j, p_k, p_l; q = \tilde{p}_I + \tilde{p}_L) \tag{2.36}$$

and hence,

$$d\Phi_4 = P_2 d\Phi_{X_{ijkl}}. \quad (2.37)$$

2.4.2 Antenna Functions

In this thesis all quark-antiquark final-final antenna functions are considered and will be computed. All considered antennae have the same underlying hard parton pair, and might have as many as one unresolved parton at NLO and two at NNLO. We also consider three antenna-like structures, one per order, which are quark-antiquark form factors. Since their structure is very similar to other antenna functions, as a shorthand, they will also be called that throughout this text.

Each antenna is generally determined by its external state and the hard parton pair it collapses to. In this thesis, the hard parton pair is always $q\bar{q}$. Up to NNLO, quark-antiquark final-final antennae are organised into three families, which differ in the type of particle the hard parton pair particles emit. *A*-type antennae are characterised by gluon emissions, leading to final states like $qgg\bar{q}$ at NNLO or $qg\bar{q}$ at NLO; *C*-type antennae are characterised by a final-state quark-antiquark pair with the original hard pair's flavour, as $q\bar{q}q\bar{q}$; and *B*-type by a quark-antiquark pair of a different flavour, notated as $q\bar{q}q'\bar{q}'$. Following QCD vertex rules, described in Section B.1, *A*-type antennae may exist at any order, while *B*- and *C*-type antennae can only exist at NNLO.

Throughout the rest of the present text, a general *X*-type antenna will be written as X_n^l , for n final-state particles and l loops. An antenna may have up to three non-zero terms (or components): leading, written plainly as X_n^l ; subleading, written as $\tilde{X}_n^l$; and quark-loop, as $\hat{X}_n^l$. An antenna topology that has more than one term is called an antenna set and is likewise called the X_n^l set. All antenna functions are part of larger sets, but only some have non-zero subleading or quark-loop terms. The A_2^2 antenna set will appear with four terms, with the extra one written as $\check{A}_2^2$. This is not an exception to the rule, however, but rather a shorthand notation, bundling two distinct perturbative contributions with the same final-state particles in a single set. The antenna functions (and $q\bar{q}$ form factors) considered are organised by order, loop number, and final state in Table 2.3.

Final state	Tree level	One loop	Two loop
$q\bar{q}$	A_2^0	A_2^1	$A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$
$qg\bar{q}$	A_3^0	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$	—
$qgg\bar{q}$	$A_4^0, \tilde{A}_4^0$	—	—
$qq'\bar{q}'\bar{q}$	B_4^0	—	—
$qq\bar{q}\bar{q}$	C_4^0	—	—

Table 2.3: Quark-antiquark antenna families used in this work, including the $q\bar{q}$ form factors [4]. Final states are ordered with the two hard radiator listed first and last, following that paper's convention.

Antenna functions are derived from physical matrix elements following the structure of the subtraction terms they are used in. These matrix elements are written following the antenna functions' own notation. Tree-level ($l = 0$) antennae are derived as [4],

$$\begin{aligned}
X_{ijk}^0 &= S_{ijk,IK} \frac{|\mathcal{M}_{ijk}^0|^2}{|\mathcal{M}_{IK}^0|^2} \\
X_{ijkl}^0 &= S_{ijkl,IL} \frac{|\mathcal{M}_{ijkl}^0|^2}{|\mathcal{M}_{IL}^0|^2}
\end{aligned} \quad (2.38)$$

where for $n = 3$ the three final-state partons take the notation ijk , and for $n = 4$ take $ijkl$, for both antenna function and matrix elements. One-loop antennae are derived using the colour-ordered renormalised one-loop three-parton matrix element as [4],

$$\begin{aligned} X_{ij}^1 &= S_{ij,IJ} \frac{|\mathcal{M}_{ij}^1|^2}{|\mathcal{M}_{IJ}^0|^2} \\ X_{ijk}^1 &= S_{ijk,IK} \frac{|\mathcal{M}_{ijk}^1|^2}{|\mathcal{M}_{IK}^0|^2} - X_{ijk}^0 \frac{|\mathcal{M}_{IK}^1|^2}{|\mathcal{M}_{IK}^0|^2}. \end{aligned} \quad (2.39)$$

The two-loop two-parton antenna, which here is only used for the A_2^2 antenna set, is defined as,

$$X_{ij}^2 = S_{ij,IJ} \frac{|\mathcal{M}_{ij}^2|^2}{|\mathcal{M}_{IJ}^0|^2}. \quad (2.40)$$

Loop antennae are additionally coupling-renormalised using Eq. 2.22 wherever the reduced process carries a strong coupling. This procedure is described in further detail in Subsection 2.4.4.

To compute the considered antenna functions, we require the squared matrix element at each loop order, implicitly normalised with $N(n)$ as described in Eq. 2.21. For the tree-level, one-loop, and two-loop cases these are given as,

$$\begin{aligned} |\mathcal{M}_n^0|^2 &= \mathcal{M}_n^{0\dagger} \mathcal{M}_n^0 \\ |\mathcal{M}_n^1|^2 &= 2 \operatorname{Re} \left(\mathcal{M}_n^{0\dagger} \mathcal{M}_n^1 \right) \\ |\mathcal{M}_n^2|^2 &= 2 \operatorname{Re} \left(\mathcal{M}_n^{0\dagger} \mathcal{M}_n^2 \right) + \mathcal{M}_n^{1\dagger} \mathcal{M}_n^1. \end{aligned} \quad (2.41)$$

Lastly, the whole system is normalised by the $|\mathcal{M}_2^0|^2$ squared matrix element, setting the Born-level antenna to one. It is also useful to define the notation for a full-process normalised squared matrix element amplitude for more complex cases, particularly at NNLO. It is defined as,

$$M_{p_1 p_2 \dots p_n}^l = \frac{1}{|\mathcal{M}_2^0|^2} \sum_{r=0}^l \mathcal{M}_n^{r\dagger}(p_1, \dots, p_n) \mathcal{M}_n^{l-r}(p_1, \dots, p_n). \quad (2.42)$$

where, for final states admitting more than one colour ordering, flavour assignment, or coupling configuration, the product of the amplitudes is understood to be a complete process-level contraction, including the corresponding squared contributions and interference terms. The symmetry factors are also all equal to one, for all $q\bar{q}$ topologies.

The colour-ordered matrix elements carry colour structure and the results they produce have to be decomposed into the three components for each set. The methods and tools used to complete that decomposition are discussed in the next subsection [4].

2.4.3 Colour Algebra and the Antenna Colour Decomposition

Quarks carry a fundamental colour index $i \in \{1, 2, 3\}$ while gluons carry an adjoint colour index $a \in \{1, \dots, 8\}$. The interactions between them are governed by the $SU(3)_c$ generators T^a . Those generators are normalised by the trace relation $\operatorname{Tr}(T^a T^b) = T_R \delta^{ab}$, where by standard convention $T_R = 1/2$ [1, 3]. This is particularly useful since when the amplitude is squared, the two colour generators must be contracted with one another, that is T^a and $(T^a)^\dagger$, and since the generators are

Hermitian, $(T^a)^\dagger = T^a$. Hence, summing over all colour we get,

$$\sum_{i_1, i_2, a} (T^a)_{i_1 i_2} (T^a)_{i_2 i_1} = \sum_a \text{Tr} (T^a T^a) = \sum_a T_R \delta^{aa}. \quad (2.43)$$

For a fixed a , the trace equals T_R . Summing this result over all $N^2 - 1$ adjoint indices we get,

$$\sum_a T_R \delta^{aa} = \frac{1}{2} (N^2 - 1). \quad (2.44)$$

Eq. 2.43 may also be generalised to four quark colour indices instead of only two. The generalised expression is the *colour Fierz identity*, or $SU(N)$ completeness relation and is particularly useful for the decomposition of the B_4^0 and C_4^0 antenna functions [1, 3]. It is written as,

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2} \left(\delta_{i_1 i_4} \delta_{i_2 i_3} - \frac{1}{N} \delta_{i_1 i_2} \delta_{i_3 i_4} \right). \quad (2.45)$$

We now also introduce the Casimir invariants: the fundamental and adjoint Casimir invariants, C_F and C_A . These represent, respectively, the total colour charge squared of the quark and of the gluon. They are defined as,

$$C_F = \frac{N^2 - 1}{2N} \\ C_A = N. \quad (2.46)$$

The invariants are particularly useful to normalise the two hard quarks' colour from the antenna functions, by taking out a factor of $2C_F$. This factor equals,

$$2C_F = N - \frac{1}{N} \quad (2.47)$$

which will be a normalisation used throughout the present work.

In some processes, colour can flow in different ways depending on the ways emitted particles couple with the hard partons. These different topologies may be understood as planar, non-planar, and quark-loop topologies, each yielding a different antenna-set component. These components, after decomposition become:

- The leading colour component (N): these come from planar topologies (where virtual gluons are exchanged without crossing over each other). This component is represented by the base antenna in standard notation, e.g. A_3^1 . An NNLO leading antenna-set component is proportional to the diagonal pieces, $X_4^0 \sim |\mathcal{M}_4^0(1, 3, 4, 2)|^2$ or $X_4^0 \sim |\mathcal{M}_4^0(1, 4, 3, 2)|^2$.
- The subleading colour component ($1/N$): these come from non-planar topologies (where gluon lines do cross each other). This component is represented by the tilde antenna, e.g. $\tilde{A}_3^1$. An NNLO subleading antenna-set component is proportional to the interference pieces, $\tilde{X}_4^0 \sim 2 \text{Re} \left(\mathcal{M}_4^{0\dagger}(1, 3, 4, 2) \mathcal{M}_4^0(1, 4, 3, 2) \right)$.
- The fermion loop (N_f): Instead of a virtual gluon, the loop can have a virtual quark-antiquark pair. This contribution depends strictly on the number of active flavours, N_f . This component is represented by the hat antenna, e.g. $\hat{A}_3^1$. An NNLO quark-loop one-loop-antenna-set component is proportional to the pieces with quark-loop terms, $\hat{X}_n^1 \sim 2 \text{Re} \left(\mathcal{M}_n^{0\dagger} \mathcal{M}_{n,q\text{-loop}}^1 \right)$ [4].

2.4.4 Antenna Construction

Each antenna set considered in Table 2.3 is now built explicitly in this subsection. As mentioned, every antenna is normalised by the Born-level squared amplitude given as,

$$|\mathcal{M}_2^0|^2 = 4(1 - \epsilon)q^2. \quad (2.48)$$

The LO antenna then becomes,

$$A_2^0(1_q, 2_{\bar{q}}) = \frac{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2} = 1. \quad (2.49)$$

Advancing to the NLO tree-level antenna, there are only two possible similar processes, both considered within the same squared amplitude. The antenna becomes,

$$A_3^0(1_q, 3_g, 2_{\bar{q}}) = \frac{|\mathcal{M}_3^0(1_q, 3_g, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \quad (2.50)$$

The NLO one-loop antenna introduces the squared one-loop two-particle amplitude as,

$$A_2^1(1_q, 2_{\bar{q}}) = \frac{2 \operatorname{Re}(\mathcal{M}_2^{0\dagger}(1_q, 2_{\bar{q}})\mathcal{M}_2^1(1_q, 2_{\bar{q}}))}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \quad (2.51)$$

The NNLO A-type tree-level antenna set is the first one of this list with multiple components, due to the way in which its two unresolved partons may couple with different hard partons and may also interfere. The complete set linear combination is given,

$$\begin{aligned} M_{qg\bar{q}}^0 &= \frac{1}{|\mathcal{M}_2^0|^2} \left(N \left(|\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}})|^2 + |\mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})|^2 \right) \right. \\ &\quad \left. - \frac{1}{N} |\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) + \mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})|^2 \right) \\ &= \left(NA_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) - \frac{1}{N} \tilde{A}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \right) \end{aligned} \quad (2.52)$$

reaching the two components that make up the set $\{A_4^0, \tilde{A}_4^0\}$.

The NNLO one-loop antenna set is given following the methods shown for the one-loop NLO and the tree-level NNLO antennae, appearing with three bare components,

$$\begin{aligned} M_{qg\bar{q}}^1 &= \frac{2 \operatorname{Re}(\mathcal{M}_3^{0\dagger}(1_q, 3_g, 2_{\bar{q}})\mathcal{M}_3^1(1_q, 3_g, 2_{\bar{q}}))}{|\mathcal{M}_2^0|^2} \\ &= \left(NA_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}) - \frac{1}{N} \tilde{A}_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}) + N_f \hat{A}_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}) \right). \end{aligned} \quad (2.53)$$

As mentioned, this set needs to be renormalised according to Eq. 2.22 as,

$$M_{qg\bar{q}}^1 = \left(N \left(A_{3,\text{ren}}^1 + \frac{11}{6\epsilon} A_3^0 \right) - \frac{1}{N} \tilde{A}_{3,\text{ren}}^1 + N_f \left(\hat{A}_{3,\text{ren}}^1 - \frac{1}{3\epsilon} A_3^0 \right) \right). \quad (2.54)$$

Finally, the renormalised source still contains the one-loop correction to the underlying $q\bar{q}$ pair. This correction term has to also be subtracted and is as follows,

$$M_{q\bar{q}}^1 = \left(N \left(A_3^1 + A_3^0 \left(\frac{11}{6\epsilon} + A_2^1 \right) \right) - \frac{1}{N} \left(\tilde{A}_3^1 + A_2^1 A_3^0 \right) + N_f \left(\hat{A}_3^1 - \frac{1}{3\epsilon} A_3^0 \right) \right) \quad (2.55)$$

leaving the final antenna set $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$.

The NNLO two-loop antenna-set is made up of two different configurations which generate four components. These need to be UV renormalised but do not require subtracting further correction terms. The set linear combination is given as,

$$\begin{aligned} M_{q\bar{q}}^2 &= \left(\frac{2 \operatorname{Re} \left(\mathcal{M}_2^{0\dagger}(1_q, 2_{\bar{q}}) \mathcal{M}_2^2(1_q, 2_{\bar{q}}) \right) + \mathcal{M}_2^{1\dagger}(1_q, 2_{\bar{q}}) \mathcal{M}_2^1(1_q, 2_{\bar{q}})}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2} \right) = \frac{|\mathcal{M}_2^{2,[2\times 0]}|^2 + |\mathcal{M}_2^{2,[1\times 1]}|^2}{|\mathcal{M}_2^0|^2} \\ &= \left(N A_{2,\text{bare}}^2 + \frac{1}{N} \tilde{A}_{2,\text{bare}}^2 + N_f \hat{A}_{2,\text{bare}}^2 + \left(N - \frac{1}{N} \right) \check{A}_{2,\text{bare}}^2 \right) \\ &= \left(N \left(A_2^2 + \frac{11}{6\epsilon} A_2^1 \right) + \frac{1}{N} \tilde{A}_2^2 + N_f \left(\hat{A}_2^2 - \frac{1}{3\epsilon} A_2^1 \right) + \left(N - \frac{1}{N} \right) \check{A}_2^2 \right) \end{aligned} \quad (2.56)$$

reaching the four components that make up the set $\{A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2\}$.

The remaining two NNLO tree-level antenna functions considered throughout this thesis come from the same process, $\gamma^* \rightarrow q\bar{q}q\bar{q}$. This process produces the two relevant antennae, B_4^0 and C_4^0 , and a third antenna $\hat{B}_4^0$, which exchanges which quark pair couples to the electroweak current. This is out of the current scope of this work and will not be derived past this mention. The three antennae appear as a set-like linear combination, as,

$$\begin{aligned} M_{4q}^0 &= \frac{1}{|\mathcal{M}_2^0|^2} \left(|\mathcal{M}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}})|^2 + |\mathcal{M}_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})|^2 \right) \\ &= \left[N_f B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) + N_{f,\gamma} \hat{B}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) \right. \\ &\quad \left. - \frac{1}{N} \left(C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) + C_4^0(2_{\bar{q}}, 4_{\bar{q}}, 3_q, 1_q) \right) \right] \end{aligned} \quad (2.57)$$

where $N_{f,\gamma}$ is the charge weighted sum of the quark flavours, given as,

$$N_{f,\gamma} = \frac{\left(\sum_q e_q \right)^2}{\sum_q e_q^2}. \quad (2.58)$$

We can decouple the two relevant antennae from the general expression and write proper antenna-specific expressions. For that, we begin by defining $q\bar{q}\text{-}\gamma$ coupling amplitude using the Fierz identity,

$$\mathcal{M}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) = i e_q g^2 \delta_{q_1 q_2} \delta_{q_3 q_4} \left(\delta_{i_1 i_4} \delta_{i_3 i_2} - \frac{1}{N} \delta_{i_1 i_2} \delta_{i_3 i_4} \right) \mathcal{M}_{q\bar{q}\text{-}\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) + (1 \leftrightarrow 3, 2 \leftrightarrow 4). \quad (2.59)$$

Using this amplitude, and comparing its structure with Eq. 2.57's, we are able to write,

$$\begin{aligned} B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) &= \frac{|\mathcal{M}_{q\bar{q}-\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0|^2} \\ C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) &= -\frac{\text{Re}\left(\mathcal{M}_{q\bar{q}-\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})\mathcal{M}_{q\bar{q}-\gamma}^\dagger(1_q, 4_{\bar{q}}, 3_q, 2_{\bar{q}})\right)}{|\mathcal{M}_2^0|^2}. \end{aligned} \quad (2.60)$$

2.4.5 Phase-Space Factorisation

The previous Subsections focused on the antenna functions as objects: how they are built, combined, and normalised. As ingredients of the subtraction term $d\sigma^S$, the antenna functions must then be integrated over the unresolved-parton phase space. Schematically, as described in Eq. 2.13, this is the integral,

$$\int_1 d\sigma^S \quad (2.61)$$

where $\int_1$ denotes the integral over the unresolved parton phase-space. This integral does not only denote the integral over the single unresolved parton, alone, on two counts: first the fact that the integral must cover all unresolved partons which, for three-final-particle antennae is only one, but for four-final-particle antennae it becomes two. Second, the hard partons and their interactions with the unresolved partons must also be integrated over. The integral becomes over the n -final-state-particle antenna phase space. This makes,

$$\int_1 d\sigma^S = \sum_{\text{unresolved pairs}} \left(\int d\Phi_{ijkl\dots} X_n^l \right) d\sigma^B. \quad (2.62)$$

The n -final-state-particle antenna phase space $\Phi_{ijkl\dots}$ ² is not the standard n -particle phase space, as in Φ_3 or Φ_4 . It is, rather, the antenna's own phase space: the unresolved-parton phase space associated with the unresolved-parton emission between the two hard radiators. It has no independent parametrisation and is only accessed by factorising the known n -particle phase space [4, 11]. Since, through the recursive factorisation of the Lorentz-invariant phase space property [7], the phase space factorises, we are able to denote that the n -particle phase-space equals the $(n - n_u)$ -particle, where n_u is the number of unresolved partons, phase-space times the antenna phase-space for n final-state particles. For our case, two useful identities are produced for this formula [4],

$$\begin{aligned} d\Phi_3 &= d\Phi_2 d\Phi_{ijk} \\ d\Phi_4 &= d\Phi_2 d\Phi_{ijkl}. \end{aligned} \quad (2.63)$$

Since $d\Phi_n$ is perfectly factorised [7], and the X_n^l antenna functions are explicitly built to be functions of the antenna's internal invariants alone and to have no dependence on how the reduced system is oriented [4], the two phase spaces are integrated separately, with the two-particle phase-space integral producing Φ_2 directly. The antenna's own phase-space integral is then rewritten as [4],

$$\int d\Phi_{ijkl\dots} X_n^l = \frac{1}{\Phi_2} \int d\Phi_n X_n^l \quad (2.64)$$

which is how the phase-space prefactor appears in $\mathcal{N}(\epsilon, k)$, as described in Section 2.3.

²This phase space is notated by Φ_{ijk} for three-final-state-particle antennae like A_3^1 and Φ_{ijkl} for four-parton-final-particle antennae like A_4^0 .

Hence, finally, the integrated antenna $\mathcal{X}_n^l$ is given generally³ as, taking the aforementioned overall $\mathcal{N}$ normalisation factor [4, 11],

$$\mathcal{X}_n^l = \mathcal{N}(\epsilon, k) \int d\Phi_n X_n^l. \quad (2.65)$$

Two antenna types are exempt from this rule: A_2^1 and A_2^2 , as they do not present with unresolved partons, but rather only with loops, and thus are completely virtual. These antennae are only integrated over the loop momenta rather than over the phase space, thus not requiring normalisation using the two-particle phase-space [12]. A_2^0 also has no unresolved parton, but is excluded here for a different reason: it is the Born-level antenna itself, not an integrated subtraction ingredient.

2.5 Phase-Space Integration and IBP Reduction

In the previous sections, antenna functions were introduced together with their role in the local subtraction of infrared singularities. Eq. 2.65 defined their integration over the unresolved phase space, leading from an unintegrated antenna X_n^l to its integrated counterpart $\mathcal{X}_n^l$. For loop-level antennae ($l \geq 1$), the loop momenta must additionally be integrated.

Both the unintegrated and integrated forms are required in a subtraction calculation, but they play different roles. The unintegrated antenna functions provide local counterterms for real-radiation matrix elements and are therefore required in numerical Monte Carlo phase-space integration. They must retain the dependence on the local kinematics of each phase-space point, which can be expressed in terms of Mandelstam invariants s_{ij} . This allows the subtraction term to reproduce the singular behaviour of the real-emission matrix element point by point in the unresolved limits. For loop-level antennae, the term *unintegrated* refers here to the unresolved phase space: the loop-momentum integration is first performed analytically, leaving an antenna function that depends on the external kinematic invariants and can be evaluated locally within the phase-space integration. The subsequent analytic integration over the unresolved phase space then produces the corresponding integrated antenna function, whose explicit infrared poles enter the cancellation against virtual contributions.

From Eq. 2.64, we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space Φ_X . If required, they are also integrated over their loop momenta ℓ_r , $r = 1, \dots, l$. The propagator or Mandelstam-invariant denominators entering these integrals may generally be defined as,

$$D_j = \left(\sum_{r=1}^l c_{jr} \ell_r + r_j \right)^2 - m_j^2 + i0 \quad (2.66)$$

where r_j is built from momenta held fixed during the loop integration, including the mapped hard partons and unresolved antenna momenta. For the massless, on-shell external partons considered throughout this work, r_j reduces to a combination of Mandelstam invariants s_{ij} .

An algebraic expansion of the integrand of an antenna function can generate multiple integrals with the same kinematic structure but different numerator and denominator powers, since any numerator built from scalar products can itself be absorbed into the same propagator basis $\{D_j\}$. These integrals may then be organised into integral families, which fix the denominators and integration variables, and may

³ This definition is general for antenna functions that are already cleanly and completely given in terms of Mandelstam invariants and ϵ . This is only automatically true for antenna where $l = 0$. When $l \geq 1$, which happens for all loop antennae by definition, a prior loop-momentum integration step is necessary. This step is explored further in Section 2.5.

then be systematically reduced. These families are denoted as the complete collection [11],

$$\mathcal{F}[\{D_j\}; \{\ell_r\}, d\Phi_X] = \{I(a_1, \dots, a_N)\}, \quad a_j \in \mathbb{Z} \quad (2.67)$$

with,

$$I(\vec{a}) = \int \left[\prod_{r=1}^l d^d \ell_r \right] d\Phi_X \frac{1}{D_1^{a_1} \dots D_N^{a_N}}, \quad \vec{a} = (a_1, \dots, a_N). \quad (2.68)$$

The integrals within each family are not independent. Through integral reduction, they can be expressed in terms of a finite and substantially smaller set of master integrals,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \quad (2.69)$$

where M_k are the master integrals and c_k are coefficients that depend on the integral indices, the dimensional regulator ϵ , and the relevant kinematic invariants. The original integration problem is therefore separated into two tasks: determining the reduction coefficients c_k and evaluating the much smaller set of master integrals M_k . Once these are known, every required integral in the family can be reconstructed from the same master-integral basis. In $\vec{a}$, positive a_j indices denote propagator powers, while non-positive indices account for absent denominators or irreducible scalar products in the numerator.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants s_{ij} , since they must evaluate the subtractions at a sampled physical external phase-space point p . The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino-Veltman (PaVe) reduction mechanism, described in detail in Section B.2, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by $A_0, B_0, \dots$, while keeping the phase-space variables and integration untouched, as [13],

$$\int d^d \ell \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{PaVe}} \sum_k c_k(\{s_{ij}\}) \{A_0, B_0, C_0, D_0, \dots\}_k. \quad (2.70)$$

To finally reach the integrated antenna functions $\mathcal{X}_n^l$ we do require integrating over the phase-space. For this integral we also use a reduction method: Integration By Parts (IBP) reduction, described in detail in Subsection 2.5.1. This method generates relations among scalar integrals in a common family through vanishing total derivatives, as [14],

$$0 = \int d^d \ell \frac{\partial}{\partial \ell^\mu} \left(v^\mu \frac{1}{\prod_j D_j^{a_j}} \right). \quad (2.71)$$

As mentioned in previous sections, emitted final-state particles are real and on shell. That condition is enforced in the phase-space through the use of Dirac-delta functions like $\delta^+(p_i^2)$. Note that these are not ordinary Feynman propagators. To bring the phase-space integration into the same algebraic framework as the loop integrals, we introduce the method of reverse unitarity to represent these Dirac-delta functions

as cut propagators, through the discontinuity relation as [11],

$$2\pi i \delta^+(p_i^2) = \left[\frac{1}{p_i^2 - i0} - \frac{1}{p_i^2 + i0} \right]_+ . \quad (2.72)$$

where both $+$ signs, in δ^+ and $]_+$, represent the positive-energy prescription.

The family now contains ordinary propagators from virtual loops and cut propagators representing the unresolved phase-space. The IBP identities can then reduce the resulting combined cut-integral family to a finite set of MI.

After UV renormalisation and evaluation of the MI, the result is expanded as a Laurent series in ϵ , making the remaining explicit IR poles visible order by order [4, 9].

2.5.1 IBP Identities and the Laporta Algorithm

IBP Identities

The integral family has already been defined in Eq. 2.68. This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X \mathcal{I}(\vec{a}; \{D_1, \dots, D_N\}), \quad \mathcal{I}(\vec{a}; \{D_1, \dots, D_N\}) = \int \prod_{r=1}^l d^d \ell_r \frac{1}{D_1^{a_1} \dots D_N^{a_N}}. \quad (2.73)$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators D_j were already defined generally in Eq. 2.66. Take one loop momentum ℓ_s , for $s = 1, \dots, l$, and one vector v^μ from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside $\mathcal{I}$, as [11, 14],

$$0 = \int \prod_{r=1}^l d^d \ell_r \frac{\partial}{\partial \ell_s^\mu} \left[v^\mu \frac{1}{D_1^{a_1} \dots D_N^{a_N}} \right] \quad (2.74)$$

as previously introduced in Eq. 2.71. In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals⁴.

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^l d^d \ell_r \left[\frac{\partial_{\ell_s} \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^N a_j \frac{v^\mu \partial_{\ell_s^\mu} D_j}{D_1^{a_1} \dots D_j^{a_j+1} \dots D_N^{a_N}} \right]. \quad (2.75)$$

For these quadratic denominators D_j , differentiation with respect to ℓ_s^μ is linear in momenta,

$$\partial_{\ell_s^\mu} D_j = 2c_{js} \left(\sum_r c_{jr} \ell_r + r_j \right)_\mu \quad (2.76)$$

and therefore, $v^\mu \partial_{\ell_s^\mu} D_j$ is a scalar product. Scalar products may be expressed in terms of chosen denominators and irreducible scalar products. The latter are represented by negative family indices a_j ,

⁴This method is explored further and completely in Subsubsection 4.2.1.

while zero indices denote absent denominators. We are then able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}(\vec{a}, \epsilon, \{s_{ij}\}) \mathcal{I}(\vec{b}) = 0 \quad (2.77)$$

where each vector $\vec{b}$ labels a family member obtained from $\vec{a}$ through index shifts in one or more denominator powers. $\rho_{\vec{b}}$ is the corresponding coefficient in the IBP relation.

An example of the application of this method is written in Section B.3.

Reverse Unitarity and Cut Integrals

In the previous subsection, we have derived the IBP identities relative to $\mathcal{I}$. As described in Eq. 2.73, the entire integrated antenna is not merely $\mathcal{I}$, but rather,

$$I(\vec{a}) = \int d\Phi_X \mathcal{I}(\vec{a}). \quad (2.78)$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before $I(\vec{a})$ is reducible via the present method.

As described in Subsection 2.4.5, phase-space factorisation isolates the antenna measure through $d\Phi_{m+r} = d\Phi_m d\Phi_X$, or in the two-hard-radiator final-final case we are considering throughout the present work, $d\Phi_X = d\Phi_n/d\Phi_2$, for n final-state particles. That n -particle phase space is given generally as,

$$d\Phi_n(q : p_1, \dots, p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^n p_i\right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+(p_i^2 - m_i^2) \quad (2.79)$$

where the $\delta^{(d)}$ -function enforces overall momentum conservation, and each δ^+ -function imposes the mass-shell condition for the i -th final-state particle, with the superscript + selecting its positive-energy solution. This may be written as,

$$\delta^+(p_i^2 - m_i^2) = \theta(p_i^0) \delta(p_i^2 - m_i^2) \quad (2.80)$$

This δ^+ -function may be reexpressed as a discontinuity as follows [11, 15],

$$2\pi i \delta^+(p_i^2 - m_i^2) = \left[\frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0} \right]_+ \quad (2.81)$$

where the $\pm i0$ terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [11]. Using the discontinuity relation to refactor the δ -functions, we are able to rewrite the integrated antenna

expression schematically as,

$$\chi_n^l \sim \int \left[\prod_{r=1}^l d^d \ell_r \right] \left[\prod_u d^d p_u \right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \quad (2.82)$$

where $D_{v,\alpha}$ are ordinary virtual propagators, $D_{c,\beta}$ are cut propagators, $\tilde{p}_{\text{hard}}$ is the hard-radiator combined momenta and p_u are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsubsection 2.5.1 may now be taken with respect to any loop momentum ℓ_r^μ or integrated unresolved momentum p_u^μ . This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [11, 15].

An example of the method used to rewrite $d\Phi_3$ in terms of cut propagators is available in Section B.4.

Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic-elimination method that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [16].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \leq 0, \end{cases} \quad \vec{\theta} = (\theta_1, \dots, \theta_N). \quad (2.83)$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present t , the excess denominator power r , and the numerator complexity encoded through negative indices s^5 . These quantities are defined using each integral's a -index, taken from the family's $\vec{a}$ vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^N \theta(a_j), \quad r = \sum_{j=1}^N \max(a_j - 1, 0), \quad s = \sum_{j=1}^N \max(-a_j, 0). \quad (2.84)$$

Here, $\theta(a_j)$ records whether denominator D_j is present. The hierarchy is defined by comparing t first, then r and finally s . The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first.

An example of the application of the Laporta algorithm is written by hand in Section B.5.

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. These seed integrals are a part of a finite, bounded set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for $I(1, 1, 1)$, the highest-ranked member, and the resulting expression is substituted recursively into the remaining system [16].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially

⁵The present t , r and s metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

relevant sectors. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [11].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

2.5.2 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \quad (2.85)$$

where c_k are the reduction coefficients and represent what comes from the reduction algebra and M_k are the MI. Our remaining task is evaluating the finite set M_k [16].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [16].

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[\prod_r d^d \ell_r \right] \left[\prod_u d^d p_u \right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \quad (2.86)$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite; in fact, in dimensional regularisation they commonly contain ϵ poles [11].

For a massless fully inclusive antenna system, q^2 is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left(q^2 \right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon), \quad f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)} \quad (2.87)$$

where λ_k is fixed by mass dimension and f_k is dimensionless, depending only on ϵ [11].

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as $x = m^2/q^2$, the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to

reduce to the master basis, as [17],

$$\frac{d}{dx}\vec{M}(x, \epsilon) = A(x, \epsilon)\vec{M}(x, \epsilon). \quad (2.88)$$

where $\vec{M} = (M_1, \dots, M_{N_{\text{MI}}})^T$ is the vector of masters and A is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. The MI used through this work and the methods used to compute them are discussed in further detail in Section 4.1.

Once the MI are evaluated, we expand them around $\epsilon = 0$, as a Laurent series [11],

$$M_k = \sum_{n=n_{\min}}^{\infty} m_{k,n} \epsilon^n \quad (2.89)$$

to expose their pole and finite structure. The lowest power $n_{\min}$ can be, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + \mathcal{O}(\epsilon). \quad (2.90)$$

The required expansion depth is determined by the reduction coefficient. If $c_k \sim \epsilon^{-p}$, the M_k must be expanded through to ϵ^p to obtain the finite contribution to $c_k M_k$.

The reduction techniques introduced in this chapter provide the computational tools required for the analytic integration of the antenna functions considered in this work. In particular, the IBP identities are systematically solved through the Laporta algorithm, reducing the phase-space and loop integrals to a finite set of master integrals. As described in Chapter 3, this reduction procedure is implemented within the *AntCalc* workflow and is subsequently applied in Chapter 4 to obtain the integrated antenna functions in dimensional regularisation. The resulting Laurent expansions in ϵ provide the explicit infrared pole structure required for their use in higher-order subtraction at NLO and NNLO.

Chapter 3

Package Framework

A central contribution of this thesis is the development of *AntCalc*¹, a *Wolfram Language* package created from scratch as part of this thesis work to automate the symbolic construction and integration of QCD antenna functions. The package provides direct access to both unintegrated and integrated antenna functions and implements the computational framework described in the previous chapter. The results presented in this thesis were produced with *AntCalc*, version 0.3.0 β 2, corresponding to commit 4ceb991.

AntCalc makes use of *FeynCalc* [18, 19, 20, 21] for symbolic amplitude manipulation and interfaces with *LiteRed2* [22] for IBP reduction. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports massless and massive antenna routes. All routes were validated against the established literature, with all massless routes being validated against [4] except for the two-loop antenna set, which was validated against [23]. The single massive route results were validated against [24]. All routes are public: none is held back as a partial or experimental implementation, and each produces stable and reproducible results notwithstanding the current beta version. The release-acceptance procedure that ensures this is described in Section 3.7.

AntCalc's inner working structure can be described using the diagram in Fig. 3.1.

3.1 Design Philosophy and Scope

AntCalc was built to combine the symbolic amplitude capabilities of the *FeynCalc* environment with the IBP-reduction capabilities of *LiteRed2*, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return antenna functions in the conventional forms required for higher-order QCD calculations using the antenna subtraction method. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its two central functions are designed to reflect this objective. For example, the unintegrated one-loop three-parton antennae $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$ returned by `BuildAntenna[...]` are expressed after Passarino-Veltman reduction, in terms of scalar one-loop functions and the external kinematic invariants, following

¹Source code available at <https://github.com/HenFar/AntCalc/tree/release/0.3.x>.

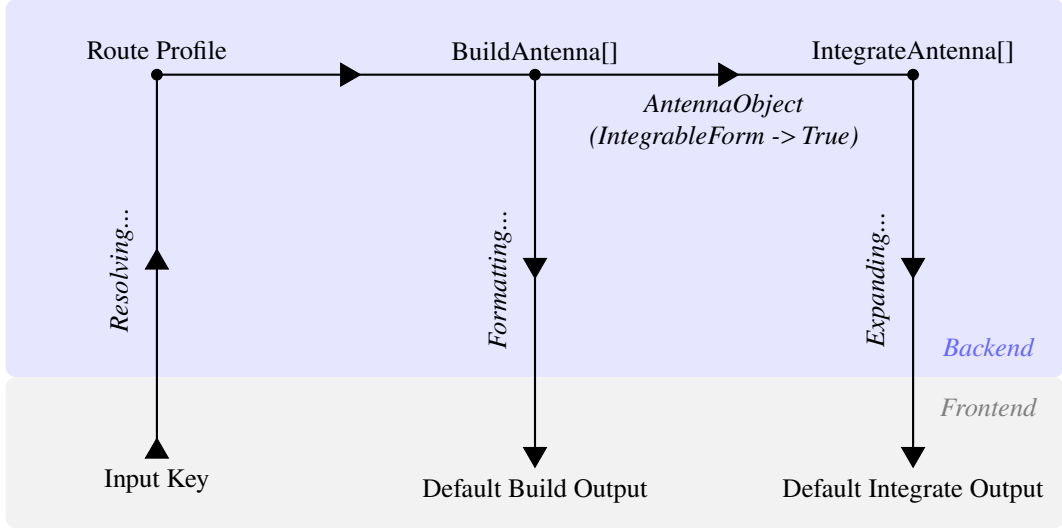


Figure 3.1: Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The *AntennaObject* provides the integration-ready handoff between the two stages.

the conventional representation used in the literature [4]. By default, `IntegrateAntenna[...]` returns the integrated antennae as Laurent series in ϵ , with the truncation order chosen to match the conventions used in the antenna literature: the NLO antennae $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$ are expanded through $\mathcal{O}(\epsilon^2)$, while the NNLO antennae $\mathcal{A}_4^0$, $\mathcal{A}_3^1$, $\mathcal{A}_2^2$, $\mathcal{B}_4^0$, and $\mathcal{C}_4^0$ are returned through $\mathcal{O}(\epsilon^0)$ [4, 23]. Furthermore, the massive NLO tree-level antenna $\mathcal{A}_3^{0,(m_q)}$ is given up to $\mathcal{O}(\epsilon)$ [24].

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module that communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, facilitates the addition of further modules while reducing duplicated logic and direct dependencies. These new modules need not be compatible with previous implemented functions through function-specific couplings. They need only conform with the profile system, being selected and configured by route profiles, and accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised by mass scale, into massless and massive routes. All massless quark-antiquark final-final antenna functions through NNLO have been made public. Alongside these, a single massive route, $\mathcal{A}_3^0$, is finalised, validated, and likewise public. This is the first massive antenna of what is expected to be a complete expansion of *AntCalc*'s capabilities into the massive regime for quark-antiquark final-final antennae. The antenna functions currently implemented are summarised in Table 3.1.

3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

A public antenna route key is a three-element list which describes a canonical route whose execution may be refined by options. The route key takes the form,

$$\{\text{Antenna Type, Final-State Multiplicity, Loop Order}\}. \quad (3.1)$$

Perturbative Order	Antenna	Validated Against
LO	A_2^0	Born normalisation
NLO	A_2^1, A_3^0	[4, 23]
NNLO	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1, A_4^0, \tilde{A}_4^0, B_4^0, C_4^0$	[4, 23]
NNLO	$A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$	[23]
NLO, massive	$A_3^{0,(m_q)}$	[24]

Table 3.1: Overview of the antenna functions currently implemented in *AntCalc*, by perturbative order and validation source.

As an example, the route key for the one-loop three-parton antenna set associated with the full-process amplitude $M_{qg\bar{q}}^1$ is $\{A, 3, 1\}$. This key represents the complete set of colour contributions, comprising the leading-colour antenna A_3^1 , the subleading-colour antenna $\tilde{A}_3^1$, and the quark-loop contribution $\hat{A}_3^1$. The default public output is therefore the ordered list $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$.

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system consists of a main profile and two further profiles: one build-side, handling reduction, and one integration-side. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the Passarino-Veltman-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the $\{A, 2, 1\}$ and $\{A, 3, 1\}$ loop routes. `AntennaIntegrationProfile[key]` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective Sections 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As a representative example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A, 2, 1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
  <|
    "Key" -> {A, 2, 1},
    "Name" -> "A21",
    "AntennaType" -> A,
    "NumFinalParticles" -> 2,
    "LoopOrder" -> 1,
    "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
    "BornInterference" -> BornInterference[],
    "Extraction" -> "LoopScalar",
    "ColourNorm" -> colourNorm,
    "ExcludedParticles" ->
      {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
    "DropSumOver" -> True,
    "ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
```

```
"ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
|>
```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The settings used in this route are explained in Table 3.2.

Item	Use
TreeAmplitude	Selects the tree-level amplitude. In this case, it is $\mathcal{M}_2^0$.
BornInterference	Generates the general normalisation as $ \mathcal{M}_2^0 ^2$.
Extraction	Selects how the components are extracted.
ColourNorm	Colour normalisation: $N - 1/N$.
ExcludedParticles	Restricts <i>FeynArts</i> internal particle insertions.
DropSumOver	Replaces every <code>SumOver[]</code> statement with 1.
ReductionProfile	Points to the reduction profile.
ConventionProfile	Points to the normalisation-convention ruleset profile.

Table 3.2: Overview of the settings for the A_2^1 route used as a representative example of the route system used in *AntCalc*.

3.3 The Build Stage

The build stage constructs the unintegrated antenna functions from their corresponding matrix elements. Its main public interface is `BuildAntenna[type, multiplicity, loop order]`. The workflow begins with a route key specified through a public call, for example `BuildAntenna[A, 3, 1]`. From this key, *AntCalc* identifies the appropriate construction workflow and the ingredients required for the corresponding antenna set. The relevant amplitudes are then constructed and combined according to the route definition, after which the resulting expressions are processed into the required unintegrated antenna functions. Internally, the intermediate results are stored in an association containing the different components of the calculation. The final step reformats this internal representation into the antenna expression, or ordered set of antenna components, returned to the user. The build workflow can therefore be summarised as follows,

```
BuildAntenna[type, multiplicity, loop order]
-> route key;
-> loop-order dispatcher;
-> route-specific build workflow;
```

```

-> build-data association / route context;
-> public-output boundary;
-> output: expression, component list, AntennaObject, diagnostics,
    or run record.

```

The loop-order dispatcher is a central element of this workflow. Using the route key, it directs each antenna to the appropriate computational path, since different loop orders require different computational treatments, most notably the tensor reduction required for the loop-momentum dependence of one- and two-loop antennae.

Functionally, the route-context for this stage usually contains the initial profile itself (the declarative *AntennaProfile*), the amplitude generated from it, the sectors, computed interferences, components after separation, diagnostics, and its normalised interference. Depending on the route, the context might also retain prototype components, full colour components, reference square components or loop-specific amplitudes.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the build stage is shown below, in Fig. 3.2.

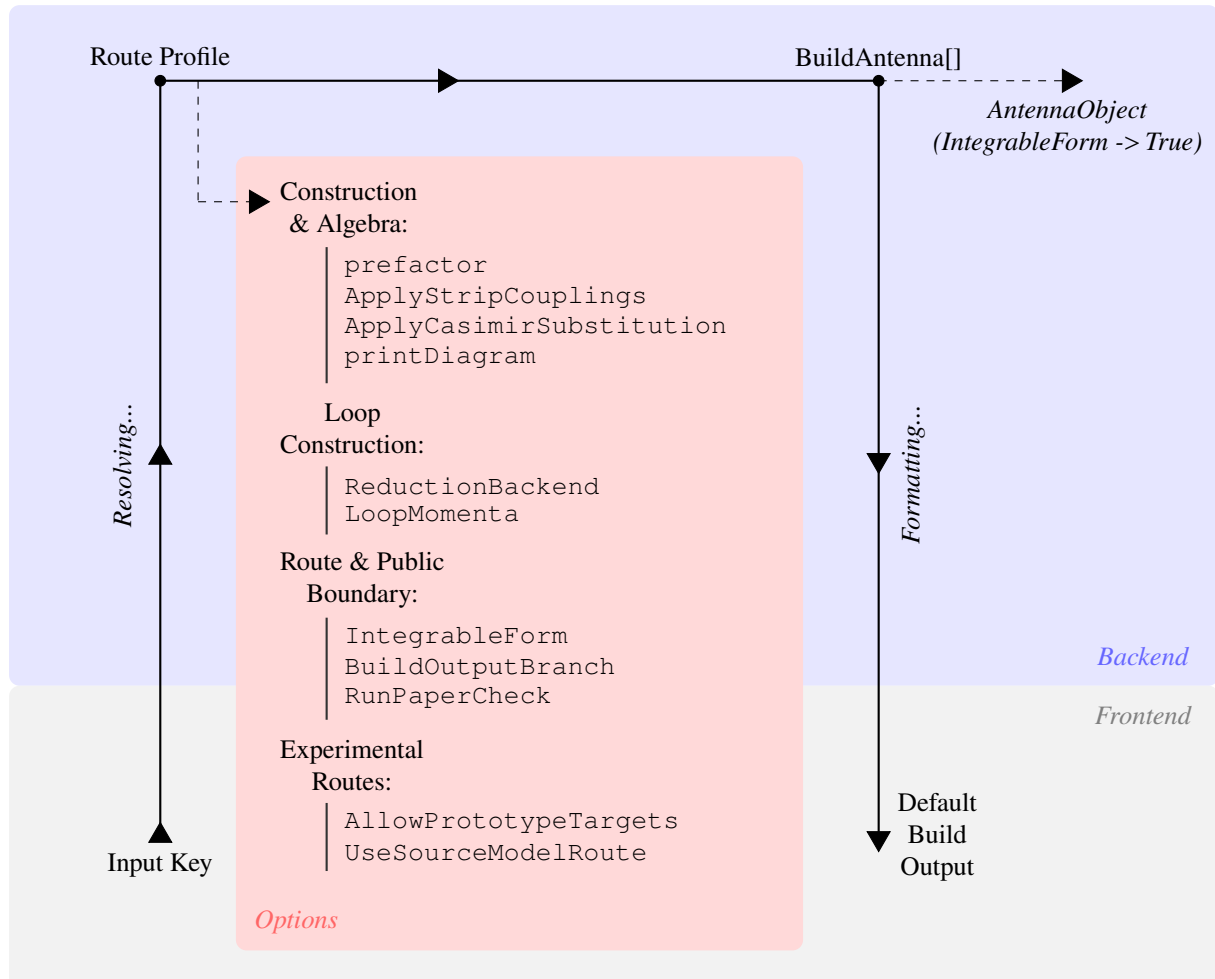


Figure 3.2: Profile-driven architecture of *AntCalc*'s Build stage. The principal build-specific options to the *BuildAntenna* function are displayed on the red card.

3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes like those relative to the A_3^0 antenna, start by retrieving the profile and computing the relevant process amplitude. It uses the profile's `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq. 2.47, and the `Extraction` mode also introduces the Born normalisation. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.3.

Production Mode	Extraction Mode	Current Route
SelfInterference	BornScalar	A_2^0
SelfInterference	TreeColourCoefficients	A_3^0
ColourOrderedAntenna	TreeColourCoefficients	$A_4^0, \tilde{A}_4^0$
SectorSelfInterference	ColourNormScalar	B_4^0
SectorSymmetrisedInterference	MinusTwoColourNormOverSUNN	C_4^0
SelfInterference	TreeColourCoefficients	$A_3^{0,(m_q)}$

Table 3.3: Overview of the tree-level production and extraction modes.

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna's multiplicity. The different production methods are shown in completeness in Subsection 2.4.4, when each of the elements of the *Current Route* column is discussed. The extraction paths are also implicitly discussed in that subsection. A_2^0 is set to 1 as discussed. Both A_3^0 and the A_4^0 -set antennae are directly extracted from the usual structure of $NX_n^l + 1/N\tilde{X}_n^l$. Lastly we extract the B_4^0 and C_4^0 antennae from the more complex inner structure of the M_{4q}^0 full-process amplitude and hence require a more complex extraction process.

3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is Passarino-Veltman reduction, as described in Section B.2.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, as described in Eqs. 2.19 and 2.47. This normalised interference is then reduced and its components extracted. The production and extraction modes implemented for the supported one-loop routes are summarised in Table 3.4, below.

The two one-loop routes make use of a hybrid production mode that combines the `TreeAmplitude` and `MAmpOneLoop` objects. As shown in Table 3.2, the former is $\mathcal{M}_n^0$, while the latter is its one-loop counterparts $\mathcal{M}_n^1$. These two amplitudes are used as described in Subsection 2.4.4. Their extraction modes are the one-loop analogous to the methods used for A_2^0 and A_3^0 , as discussed in Subsection 3.3.1.

Production Mode	Extraction Mode	Current Route
TreeAmplitude + MAmpOneLoop	LoopScalar	A_2^1
TreeAmplitude + MAmpOneLoop	LoopColourCoefficients	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$

Table 3.4: Overview of the one-loop production and extraction modes.

3.3.3 Two-Loop Routes

Loop-order-two routes are dispatched to `BuildTwoLoopAntennaData`. There is only, formally, one set of this kind up to NNLO, A_2^2 . It is remarkably different from the rest, however. The four-component set is not generated like the A_4^0 and A_3^1 sets were, rather its four components come from two distinct source contributions, as discussed in Subsection 2.4.4. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.5, their sources differ,

Production Mode	Extraction Mode	Current Route
TwoLoopInterf	TwoLoopTTermComponents	$A_2^2, \tilde{A}_2^2, \hat{A}_2^2$
OneLoopSelf	TwoLoopTTermComponents	$\check{A}_2^2$

Table 3.5: Overview of the two-loop contributions and extraction modes.

One of the ways in which this route is particular inside *AntCalc*'s architecture is illustrated by their different production, but equal extraction modes. This route may be schematically represented as shown in Fig. 3.3, below.

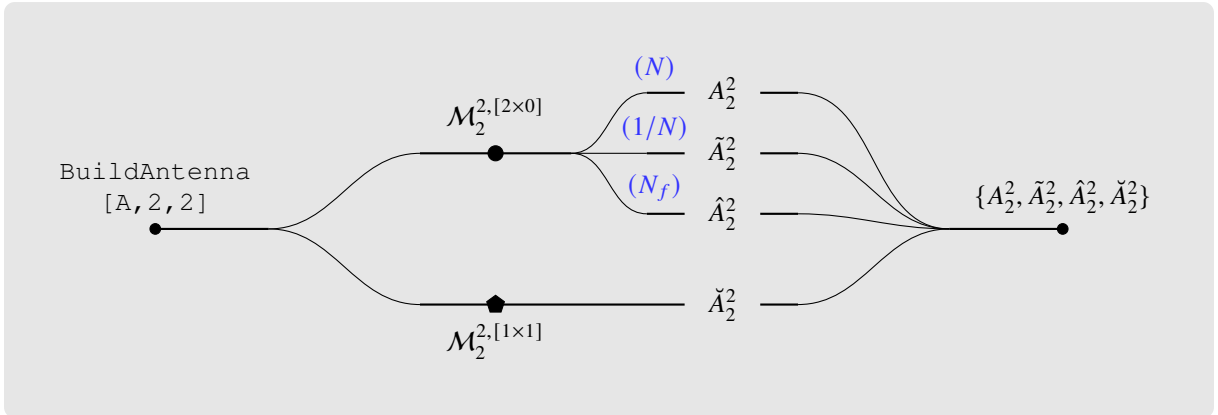


Figure 3.3: Diagram showing the build-side workflow leading to the generation of the A_2^2 antenna set. The two subsets are separated by the interferences used in their generation.

The programme reads the profile and generates the two necessary amplitudes, for the two separate loops and for the one-loop self-interference case, $M_2^{2,[2 \times 0]}$ and $M_2^{2,[1 \times 1]}$. The former amplitude generates three antenna-set components, while the latter produces only one, but all four are extracted using the same usual method used for tree-level and one-loop antenna functions, applied to two loops.

3.4 From the Build Stage to Integration

The build stage described above produces the unintegrated antenna functions in terms of the external kinematic invariants s_{ij} . This is the natural public representation of the antenna and is suitable for its

use as a local subtraction term in a numerical phase-space integration, as discussed in Chapter 2. A standard `BuildAntenna[...]` call therefore returns either a single antenna expression or, where several antenna-set components are present, an ordered list made up of those components.

For the analytic calculation of the integrated antennae, however, the final expression in terms of Mandelstam invariants is not by itself sufficient. The integration stage requires additional information from the construction of the antenna, including the route definition and the integral structures needed for the subsequent reduction and phase-space integration. *AntCalc* therefore provides the option `IntegrableForm -> True`. With this option, `BuildAntenna[...]` returns an `AntennaObject` that retains both the constructed antenna and the additional information required by `IntegrateAntenna[...]`. This object provides the interface between the build and integration stages.

At this point, *AntCalc* can therefore provide the unintegrated antennae in the kinematic form required for local subtraction. The next stage addresses the complementary part of the antenna subtraction procedure: their analytic integration over the unresolved phase space in dimensional regularisation. These integrated antennae are required to combine the subtraction terms with the virtual contributions and achieve the cancellation of explicit infrared poles. The implementation of this integration stage is described in the following section.

3.5 The Integration Stage

The integration stage performs the analytic integration of the antenna functions constructed in the build stage over the corresponding unresolved phase space in $d = 4 - 2\epsilon$ dimensions, as in Eq. 2.65. The integrated antennae are denoted with calligraphic letters.

Within *AntCalc*, this stage is initiated by passing the `AntennaObject` produced by `BuildAntenna[...]` to `IntegrateAntenna[...]`. A diagram representing this architecture and how it splits from an unchanged `BuildAntenna[...]` call is shown in Figure 3.4.

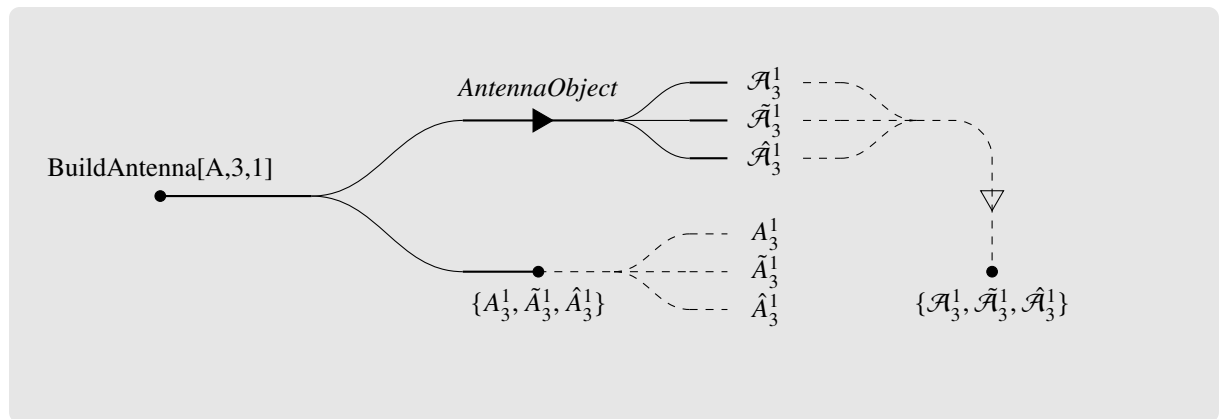


Figure 3.4: Diagram of the build-only and build-and-integrate routes: the combined route is shown above, the build-only route below. The dashed lines mark the options to separate the result into single components (build-only route) or merge them into one ordered list (combined route). The A_3^1 antenna set is used as an example.

The route key stored in this object determines the appropriate integration workflow, including the required reduction procedure, master integrals, and normalisation conventions. After these steps have been applied, `IntegrateAntenna[...]` returns the integrated antenna, or the corresponding set of antenna components, expanded as a Laurent series in ϵ . The integration workflow is structured as

follows,

```
IntegrateAntenna[AntennaObject]
-> recover route key and resolve profile;
-> choose integration backend;
-> complete IBP/Passarino-Veltman reduction;
-> substitute master integrals when applicable;
-> extract public integrated antennae;
-> output: selected component, list, diagnostics, run record, or
    master-combination view.
```

It is relevant to note that `IntegrateAntenna` requires the profile to complete its function like `BuildAntenna`, since it also encodes the relevant routing information for the reduction and integration process.

Much like for `BuildAntenna`, the route state is not held in a single context object but spread across several objects, accessed as required: the key and profile, the input (unintegrated) antenna and selected component, the IBP bases and master integrals, the series result, and the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.5.

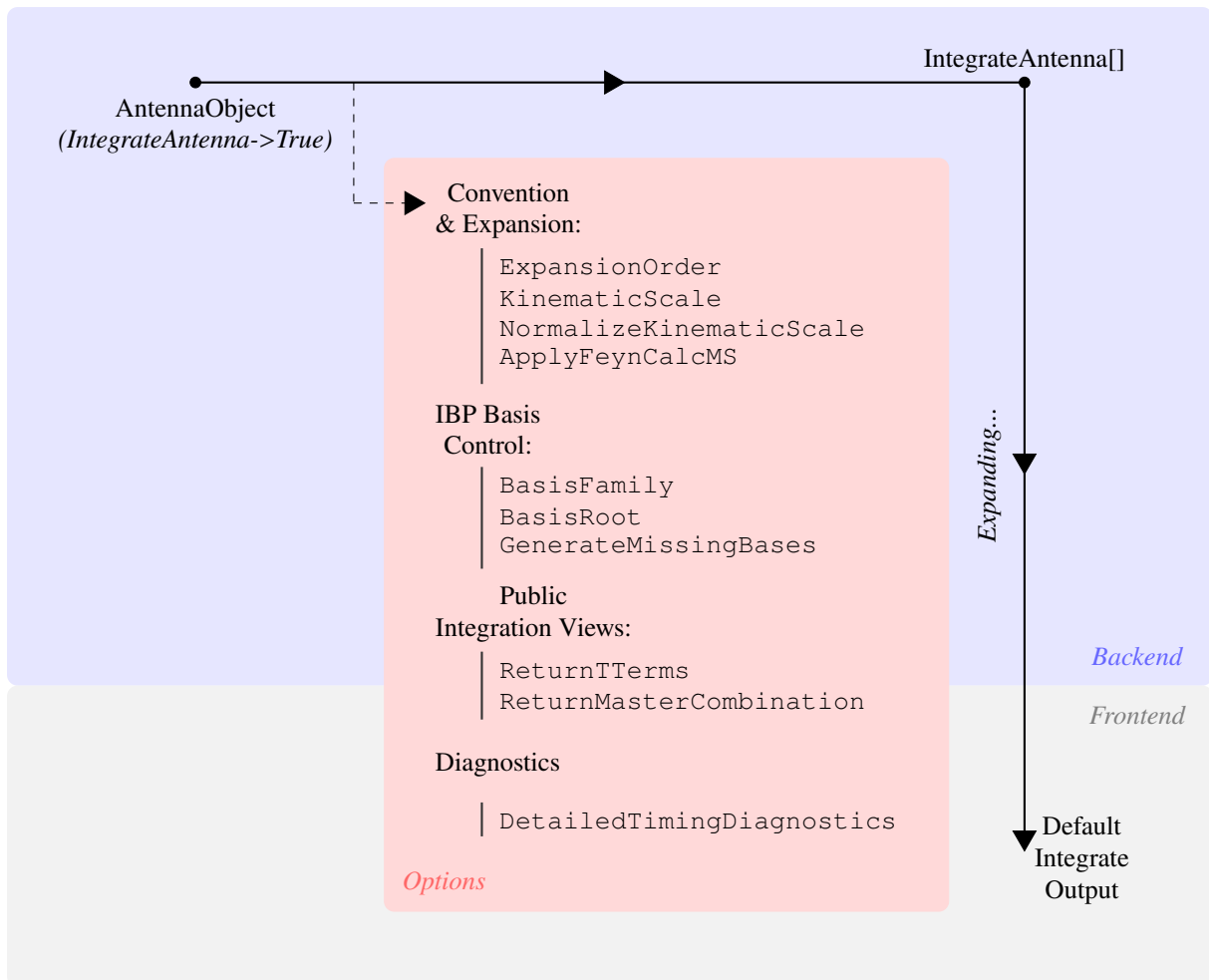


Figure 3.5: Profile-driven architecture of `AntCalc`'s Integration stage. The principal integration-only options are shown inside the red card.

3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. Since IBP reduction acts on the kinematic integral structure and is independent of the colour coefficients, the components of a set are reduced separately. *AntCalc* can therefore integrate the complete set or a single selected component, avoiding unnecessary reductions. The exception to this rule is the A_2^2 set, in which different sets are used depending on configuration. A diagram of this route's multi-configuration structure is shown in Figure 3.6; comparing it with the upper path of Figure 3.4 shows how the A_2^2 route differs from the single-family routes used for A_3^1 and most others.

The A_2^1 antenna is also an exception to the use of integral families: its loop integrals are reduced directly to scalar one-loop integrals using Passarino-Veltman reduction, whose known analytic expressions are then substituted and expanded as Laurent series in ϵ . An overview of these is shown in the following Table 3.6.

Default Backend	Bases-Family Set	Current Route
Passarino-Veltman / <i>FeynHelpers</i>	massless two-parton vertex	$\mathcal{A}_2^1$
IBP / <i>LiteRed2</i>	X30	$\mathcal{A}_3^0$
IBP / <i>LiteRed2</i>	X40	$\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0, \mathcal{B}_4^0, \mathcal{C}_4^0$
IBP / <i>LiteRed2</i>	A31	$\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1$
IBP / <i>LiteRed2</i>	A22TwoLoopTree	$\mathcal{A}_2^2, \tilde{\mathcal{A}}_2^2, \hat{\mathcal{A}}_2^2$
IBP / <i>LiteRed2</i>	A22OneLoopSelf	$\check{\mathcal{A}}_2^2$
IBP / <i>LiteRed2</i>	MX30	$\mathcal{A}_3^{0,(m_q)}$

Table 3.6: Overview of the integration-side route-profiles.

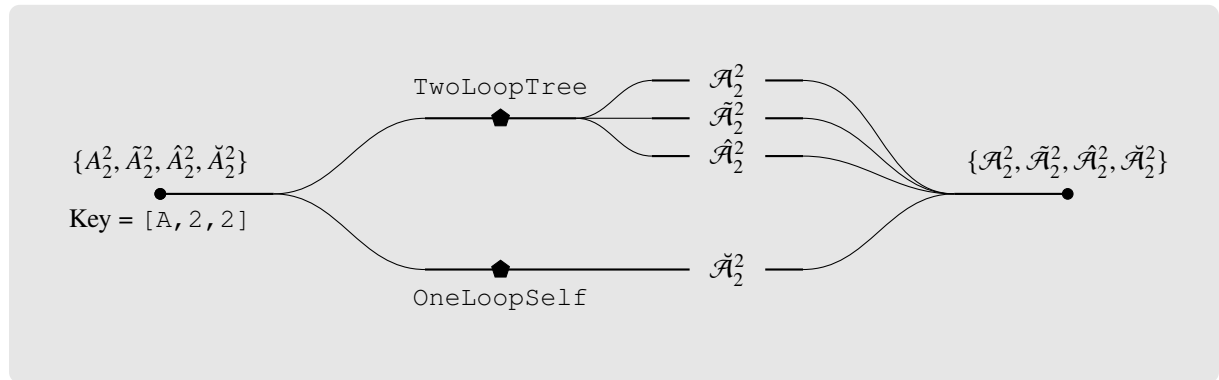


Figure 3.6: Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the A_2^2 antenna set.

As on the build side, the integration profile does not itself perform any calculation; rather, it specifies the information required to execute the corresponding integration workflow. Consider, for example, the integration of the B_4^0 antenna. From the route key stored in the *AntennaObject*, *AntCalc* identifies the appropriate integration profile and the integral family associated with the X_4^0 phase-space topology. The unintegrated antenna is then mapped onto this integral family and reduced by IBP, using *LiteRed2*, to

the corresponding set of master integrals. Their known analytic expressions are subsequently substituted to obtain the integrated antenna.

The master-integral basis used in these reductions is discussed in Chapter 4, where selected master integrals are evaluated explicitly and their role in the analytic integration of antenna functions is illustrated through representative examples.

3.5.2 Integration Routes

Passarino-Veltman Route

The A_2^1 antenna constitutes a special low-multiplicity virtual case and is the only integration route implemented without IBP reduction. As a two-parton one-loop form-factor contribution, its loop integrals can be treated directly using Passarino-Veltman reduction, which reduces the tensor loop integrals to scalar one-loop functions. *AntCalc* evaluates these scalar functions analytically using *Package-X* [25] through *FeynHelpers* [26]. The required branch and scale conventions are then applied, followed by the appropriate kinematic normalisation and Laurent expansion in ϵ , yielding the integrated $\mathcal{A}_2^1$ antenna. The workflow is summarised schematically in Fig. 3.7.

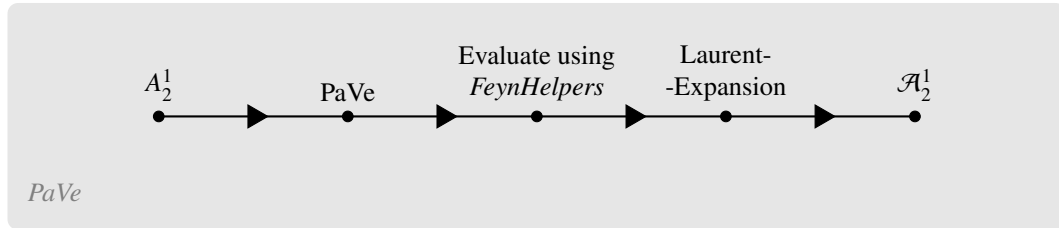


Figure 3.7: PaVe reduction for the A_2^1 antenna integration.

IBP Route

All remaining antenna functions are integrated using IBP reduction, implemented through *LiteRed2* [22], following the formalism described in Section 2.5. The antenna integrand is first expressed in terms of the momenta, invariants, and denominator structures defining the relevant integral families. The appropriate phase-space cut propagators are then introduced, allowing the phase-space integrals to be treated within the same integral-family framework as loop integrals. After expanding the antenna into individual integral terms, each term is assigned to the corresponding integral family and reduced by *LiteRed2* to a linear combination of master integrals. The reduced terms are subsequently recombined to obtain the complete integrated antenna in terms of the corresponding master-integral basis.

For the public routes, the master integrals (MIs) identified by the IBP reduction are matched to their corresponding analytic expressions known from the literature and evaluated explicitly in this work. Their expressions are then substituted by *AntCalc*. Finally, the required normalisation and dimensional-regularisation conventions are applied, and the result is expanded as a Laurent series in ϵ , yielding the integrated antenna function. Selected master integrals and explicit examples of their analytic evaluation are presented in Chapter 4, together with representative applications of this procedure to the integration of antenna functions. The complete IBP-based integration workflow is summarised schematically in Fig. 3.8.

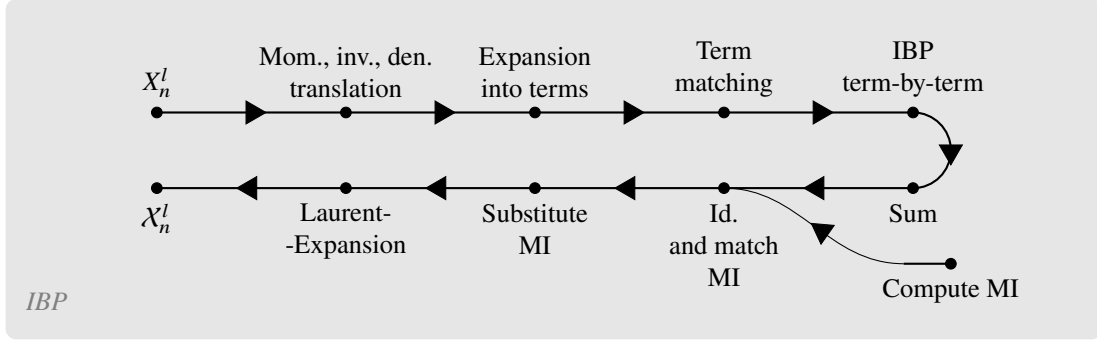


Figure 3.8: IBP reduction workflow for the remaining antennae integrations. The MI computation is external to *AntCalc*’s internal operation, and hence is represented by an external incoming arrow.

3.6 Higher-Level Functions

Besides the two canonical functions, `BuildAntenna` and `IntegrateAntenna`, *AntCalc* also supports higher-level functions. This layer does not introduce new antenna-construction or integration methodology; instead it orchestrates the two canonical functions to compute observables such as the R -ratio, to assemble the topology-grouped combinations used in it, and to streamline repeated build or integrate evaluation. The five higher-level functions are:

- `BuildAndIntegrateAntenna` — chains `BuildAntenna[... , IntegrableForm -> True]` into `IntegrateAntenna` in a single call, forwarding the component-selection and Laurent-expansion-order controls to their respective stages.
- `BuildAllAntennae` — builds every public antenna function in one call.
- `BuildAndIntegrateAllAntennae` — builds and integrates every public antenna function in one call. Both bulk wrappers can be limited to a chosen perturbative order.
- `TObject` — assembles integrated antennae into the topology-grouped combinations used in the R -ratio, defined in Subsection 5.2.1 of Chapter 5.
- `BuildRRatio` — assembles the massless hadronic R -ratio from the integrated antenna ingredients required at the selected perturbative order.

The R -ratio produced by `BuildRRatio` is the observable used for *AntCalc*’s global validation. Called as `BuildRRatio[SMQCD2, maxOrder -> ...]`, it generates the R -ratio through NNLO and, by default, returns the finite coefficient after application of the observable convention bridge. This result has been validated against a literature reference target, as discussed in Chapter 5.

3.7 Validation and Development Diagnostics

As outlined in the introduction, the implementation is validated at two levels. Locally, each integrated antenna function is compared against its published Laurent expansion [4, 23] and, where applicable, against the expected eikonal factors and Altarelli–Parisi splitting functions in the relevant unresolved limits (Chapter 4). Globally, the integrated antennae are assembled into the inclusive R -ratio and compared with the known finite result (Chapter 5).

²The label SMQCD is not standard in the literature. It is used here to denote the massless QCD sector of the Standard Model.

As discussed in Section 3.2, the route system can retain selected diagnostic information from an evaluation. That makes it easy to locate any inconsistency during a route calculation, which can then be inspected using the built-in diagnostics tool — for example, by retrieving the antenna ingredients of an R -ratio order and checking how their ϵ -poles cancel.

The `Ingredients` association is only one part of the diagnostic information. For a complete retained history of a calculation, *AntCalc* can instead return an `AntennaRunRecord`, which records the relevant route stages and metadata. This information was used to trace the source of discrepancies during development; the resulting validation comparisons are presented in Chapter 5.

3.7.1 Amplitude-Level and Release-Acceptance Validation

The checks described above validate an antenna function against an external reference: a published Laurent series, or a known finite observable. *AntCalc* also implements a complementary validation family that requires no such reference. The function `VerifyWardIdentity[type, numFinalParticles, loopOrder]` acts directly on the unsquared amplitude produced by the build stage, before spin summation, colour decomposition, or phase-space integration is performed. For a chosen external gluon leg, it replaces that gluon’s polarisation vector by its own momentum, $\epsilon^\mu(k) \rightarrow k^\mu$, while keeping every other external gluon polarisation explicitly transverse, and checks whether the resulting expression vanishes. This is the standard amplitude-level Ward identity, applied leg by leg to the raw generated amplitude rather than inferred from a property of the extracted or integrated antenna. It is currently applied to every public route with an external gluon leg, that is, to A_3^0 , A_4^0 and the A_3^1 set, and returns a structured pass, fail, or time-out report for each checked leg. Because it depends only on the internal gauge structure of the generated amplitude, and not on comparison with any external result, it certifies a property the literature-comparison checks above cannot: that the amplitude’s diagrams were assembled with the correct relative signs and couplings before extraction even begins. In principle, this makes it applicable to a route for which no literature result exists at all.

Separately from this route-level testing, *AntCalc*’s development also relies on a release-acceptance practice. Before a version is prepared, every public route is re-evaluated in a fresh kernel, with any stored or cached intermediate result disabled, and its build, integration, and one-shot outputs are compared against each other and against the literature target discussed above. Each run is recorded, together with its timing and the specific checks it satisfied, as dated evidence. This practice has already caught, and allowed the correction of, release-preparation issues that were not visible from the final Laurent-series outputs alone.

3.7.2 Runtime and Memory Characteristics

To characterise the computational cost and performance of the implemented public routes, each antenna was evaluated in a fresh *Wolfram Mathematica* kernel. Package loading was excluded from the measurements. Both the build and integration stages were timed using the `AbsoluteTiming` function. Memory usage was measured using `MaxMemoryUsed[]`, relative to the initial kernel baseline. These measurements were taken using a MacBook Pro (Mac16,8), with an Apple M4 Pro 12 core (8 performance and 4 efficiency) CPU, and 24 GB unified RAM, running macOS 26.6.2 (build 25G83), *Wolfram Mathematica* 15.0.0 for Mac OS X ARM (64-bit), and *AntCalc* 0.3.0 β 2.

The lower-order routes, up to NLO, complete both stages in approximately one second, with the cost increasing substantially for the NNLO routes, in particular A_2^2 , which surpasses 31 minutes. The A_4^0

route has the largest memory requirement, reaching 567 MB above the kernel benchmark. The complete results are shown in Table 3.7.

Route	Build time	Integration time	End-to-end time	Peak memory above baseline
A_2^0	0.09 s	0.49 s	0.58 s	22.7 MiB
A_3^0	0.30 s	0.85 s	1.15 s	24.8 MiB
A_2^1	0.57 s	0.40 s	0.98 s	19.3 MiB
A_3^1	74.67 s	15 min 27 s	16 min 42 s	195.3 MiB
A_2^2	21.79 s	31 min 06 s	31 min 27 s	227.7 MiB
A_4^0	3 min 03 s	13 min 51 s	16 min 53 s	567.3 MiB
B_4^0	8.65 s	17.27 s	25.92 s	143.2 MiB
C_4^0	5.56 s	5 min 26 s	5 min 32 s	174.5 MiB
$A_3^{0,(m_q)}$	2.81 s	0.25 s	3.09 s	12.07 MiB

Table 3.7: Representative resource measurements for the implemented public antenna routes. The build time measures `BuildAntenna[... , IntegrableForm -> True]`, followed by `IntegrateAntenna[...]` for the integration time. Each route was evaluated in a fresh Wolfram kernel. Peak memory is the maximum kernel-memory increase relative to the initial baseline.

Chapter 4

Antenna Construction and Integration: Selected Case Studies

4.1 Master Integrals Used in *AntCalc*

As discussed in Chapter 3, *AntCalc* makes use of two reduction methods, Passarino-Veltman reduction and integration-by-parts reduction. Both of those methods reduce a larger loop-momentum or phase-space integral into a linear combination of smaller and often less complex master integrals, as discussed in Subsection 2.5.2¹. The MI required by each route depend on the antenna being integrated, though some antennae share the same MI, especially those of the same type and in the same perturbative order, like all tree-level antenna functions at NNLO [11]. *AntCalc* also applies integral reduction at two different stages: loop antennae are reduced before producing the public build-stage output, but only over the loop momenta, and all antennae are reduced over the whole phase space during the integration stage. This means that loop antenna functions like A_3^1 , for example, may require more than one set of MI throughout a complete build-and-integrate run, and the ones required for one stage may not be required for the other. Table 4.1 shows the MI used in each stage for each antenna [4].

Antenna type(s)	Build-Side MI	Integration-Side MI
A_2^0	–	–
A_3^0	–	R_3
$A_4^0, \tilde{A}_4^0, B_4^0, C_4^0$	–	$R_4, R_6, R_{8,a}, R_{8,b}$
A_2^1	–	B_0, C_0
$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$	B_0, C_0, D_0	$V_{5,a}, V_{5,b}, V_8$
A_2^2	$A_{22,LO}, A_3, A_4, B_0, C_0$	$A_{22,LO}, A_3, A_4$
$\tilde{A}_2^2$	$A_{22,LO}, A_3, A_4, A_6$	$A_{22,LO}, A_3, A_4, A_6$
$\hat{A}_2^2$	A_4, B_0, C_0	A_4
$\check{A}_2^2$	$A_{22,LO}$	$A_{22,LO}$

Table 4.1: Overview of the master integrals used in the build- and integration-stage by each antenna type [4, 11].

Both PaVe and IBP MI are shown in the table, but they are quite different in structure and integration

¹The PaVe MI are most commonly referred to as *scalar functions*, instead of *master integrals*. They are, in effect, still master integrals, however, and the moniker will be kept throughout the rest of the present text [27].

processes and will be discussed separately. The PaVe are B_0 , C_0 , and D_0 . All others are IBP MI.

4.1.1 PaVe Scalar Functions

As mentioned, only three of the MI considered are PaVe scalar functions. Those are B_0 , C_0 and D_0 and are the PaVe two-, three- and four-point scalar integrals, respectively. The PaVe method also features the A_0 (one-point) scalar function, that does not feature in *AntCalc*. It also accounts for scalar integrals with more than four external legs, as those can be further reduced in four dimensions to combinations of D_0 functions [9, 13].

The B_0 Function

The B_0 function represents a scalar one-loop diagram with two external lines and two internal propagators. It is represented, in d dimensions as,

$$B_0(k^2, m_0, m_1) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{(p^2 - m_0^2 + i\varepsilon)((p+k)^2 - m_1^2 + i\varepsilon)} \quad (4.1)$$

where k is the external momentum, and m_0 and m_1 are the internal masses. μ is the t'Hooft normalisation scale.

It is also noteworthy that this integral, unlike higher-point ones, is UV divergent and generates the pole term Δ_B ,

$$\Delta_B = \frac{2}{4-d} - \gamma_E + \log(4\pi) \quad (4.2)$$

where γ_E is Euler's gamma constant. This function may also be written as an integral over the Feynman parameter x . That yields the explicit analytical form [13],

$$B_0(k^2, m_0, m_1) = \Delta_B - \int_0^1 dx \log \left[\frac{k^2 x^2 - x(k^2 - m_0^2 + m_1^2) + m_1^2 - i\varepsilon}{\mu^2} \right] + O(d-4). \quad (4.3)$$

The C_0 Function

The C_0 function represents a scalar loop with three external lines and three internal propagators. It is a common feature in the vertex corrections of electroweak physics. It is written in d dimensions as,

$$C_0(k_1, k_2, m_0, m_1, m_2) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{(p^2 - m_0^2 + i\varepsilon)((p+k_1)^2 - m_1^2 + i\varepsilon)((p+k_1+k_2)^2 - m_2^2 + i\varepsilon)}. \quad (4.4)$$

This integral is UV finite and may be rewritten as a double integral over the Feynman parameters x and y as,

$$C_0 \propto \int_0^1 dx \int_0^{1-x} dy [ax^2 + by^2 + cxy + dx + ey + f]^{-1} \quad (4.5)$$

where the coefficients a, b, c, d, e and f are linear combinations of the external momenta and internal mass parameters [9].

The D_0 Function

The D_0 function describes a scalar loop with four external lines and four internal propagators. It is often called the scalar box integral. It is written as,

$$D_0(k_1, k_2, k_3, m_0, m_1, m_2, m_3) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{D_0 D_1 D_2 D_3} \quad (4.6)$$

with denominator factors,

$$\begin{aligned} D_0 &= p^2 - m_0^2 + i\varepsilon \\ D_i &= \left(p + \sum_{a=1}^i k_a \right)^2 - m_i^2 + i\varepsilon. \end{aligned} \quad (4.7)$$

Evaluating D_0 is highly non-trivial as it features numerous physical cuts and branch points. The resulting analytical expression is expressed in terms of Spence functions (dilogarithms) [9].

4.1.2 IBP Master Integrals

The IBP MI employed in *AntCalc* can be divided into four groups, based on the antenna functions they are used with. This division goes as,

- NLO tree-level: R_3 ;
- NNLO tree-level: $R_4, R_6, R_{8,a}, R_{8,b}$;
- NNLO one-loop: $V_{5,a}, V_{5,b}, V_8$;
- NNLO two-loop: $A_{22,LO}, A_3, A_4, A_6$.

Phase-Space Measures and Normalisations

These MI are computed over different phase-space measures, depending on the number of final-state particles. These phase-space integrals are defined over all n -particle Mandelstam invariants for each specific phase-space. For that reason, we introduce the shorthands,

$$\begin{aligned} s_{12} &= s_{12}, & d\mathcal{S}_2 &= ds_{12} \\ s_{123} &= s_{12} + s_{13} + s_{23}, & d\mathcal{S}_3 &= ds_{12} ds_{13} ds_{23} \\ s_{1234} &= s_{12} + s_{13} + s_{14} + s_{23} + s_{24} + s_{34}, & d\mathcal{S}_4 &= ds_{12} ds_{13} ds_{14} ds_{23} ds_{24} ds_{34}. \end{aligned} \quad (4.8)$$

Using the shorthands, the differential phase-spaces are defined in d dimensions as [11, 15],

$$\begin{aligned} d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta(q^2 - s_{12}) \\ d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} \left(q^2\right)^{2-\frac{d}{2}} (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta(q^2 - s_{123}) \\ d\Phi_4 &= (2\pi)^{4-3d} \left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \\ &\quad d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} d\mathcal{S}_4 \delta(q^2 - s_{1234}). \end{aligned} \quad (4.9)$$

Here Δ_4 is the Gram determinant given by the Källén function λ [11, 15],

$$\Delta_4 = \lambda(s_{12}s_{34}, s_{13}s_{24}, s_{14}s_{23}), \quad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2(xy + xz + yz). \quad (4.10)$$

Also, Θ is the Heaviside step function and $d\Omega$ the differential solid hyperspherical angle element in d dimensions, given as,

$$\Omega_n = \int d\Omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}. \quad (4.11)$$

Some MI appear normalised using the S_Γ term, to make the infrared singularity cancellations explicit and absorb common dimensional factors. The normalising term is given as [23, 11],

$$S_\Gamma = \left(\frac{(4\pi)^\epsilon}{16\pi^2\Gamma(1-\epsilon)} \right)^2. \quad (4.12)$$

It is also common to normalise with

$$S_{\Gamma,2} = P_2 S_\Gamma, \quad P_2 = \int d\Phi_2 = \frac{2^{-3+2\epsilon}\pi^{-1+\epsilon}\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)} (q^2)^{-\epsilon}. \quad (4.13)$$

The R_3 Integral

This MI is used only on the integration of the massless A_3^0 antenna. It is the direct integral over the three-final-state-particle phase-space. It is written as [27],

$$\begin{aligned} R_3 &= \int d\Phi_3 \\ &= \frac{1}{4} \frac{\Omega_{d-1}\Omega_{d-2}}{(2\pi)^{2d-3}} \iiint dE_1 dE_2 d\theta_{12} (E_1 E_2 \sin \theta_{12})^{d-3} \delta(q^2 - S_3) \\ &= \frac{1}{128q^2\pi^3} \int dS_3 \delta(q^2 - S_3). \end{aligned} \quad (4.14)$$

After analytical evaluation the integrated result, which is the one substituted in the final integrated antenna expression after IBP reduction becomes [11],

$$R_3 = 2^{-7+4\epsilon}\pi^{-3+2\epsilon} \frac{\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} (q^2)^{1-2\epsilon}. \quad (4.15)$$

The R_4 Integral

The first of the tree-level NNLO MI is R_4 , which corresponds to the direct integral over the four-final-state-particle phase-space. It is written as,

$$\begin{aligned} R_4 &= \int d\Phi_4 \\ &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \\ &\quad (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \delta(q^2 - S_4). \end{aligned} \quad (4.16)$$

To make the integral boundary limits integrable, we start by factorising the phase space into a two-particle phase space and a tripole phase space factor, $d\Phi_4 = d\Phi_2 d\Phi_T$ and then we map the six physical invariants s_{ij} into five dimensionless variables that range strictly between 0 and 1, making a five-dimensional

hypercube. We start by defining $y_{ij} = s_{ij}/q^2$ and $y_{ijk} = s_{ijk}/q^2$. Our five dimensionless variables become [11]:

- The rescaled triple-invariant of the emitter jet, $y_{134} = \frac{s_{134}}{q^2}$;
- The longitudinal momentum fraction of the unresolved parton 3, $z_1 = \frac{p_3 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled longitudinal momentum fraction of unresolved parton 4, $z_2 = (1 - z_1)t = \frac{p_4 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled invariant, $y_{14} = \frac{s_{14}}{q^2} = y_{134}(1 - z_1)v$;
- The alignment variable χ , that parametrises the physical boundaries of the Gram determinant for y_{13} as, $y_{13} = (y_{13,b} - y_{13,a})\chi + y_{13,a} = \Delta y_{13}\chi + y_{13,a}$.

After applying the mapping, the MI becomes

$$R_4 = P_2 \frac{16^{-2} \pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} (q^2)^{2-2\epsilon} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \left[y_{134}(1 - y_{134})(1 - z_1) \right]^{1-2\epsilon} \left[z_1 t(1 - t)v(1 - v) \right]^{-\epsilon} \left[\chi(1 - \chi) \right]^{-\frac{1}{2}-\epsilon}. \quad (4.17)$$

We get [4, 11],

$$\begin{aligned} R_4 &= P_2 (q^2)^{2-2\epsilon} \frac{2^{-11+8\epsilon} \pi^{-\frac{7}{2}+2\epsilon} \Gamma^3(1-\epsilon)}{\Gamma(3-3\epsilon) \Gamma\left(\frac{5}{2}-2\epsilon\right)} \\ &= S_{\Gamma,2} (q^2)^{2-2\epsilon} \frac{\Gamma^5(1-\epsilon) \Gamma(2-2\epsilon)}{\Gamma(3-3\epsilon) \Gamma(4-4\epsilon)} \\ &= S_{\Gamma} (q^2)^{2-2\epsilon} \left[\frac{1}{12} + \frac{59}{72}\epsilon + O(\epsilon^2) \right]. \end{aligned} \quad (4.18)$$

The R_6 Integral

The R_6 MI is written as [11],

$$\begin{aligned} R_6 &= \int d\Phi_4 \frac{1}{s_{134}s_{234}} \\ &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{134}s_{234}} \Theta(-\Delta_4) \delta(q^2 - S_4). \end{aligned} \quad (4.19)$$

Because the s_{234} is non-linear in the tripole parametrisation, we may not continue using direct integration [11]. We may, however, use the Optical Theorem as a cut of the three-loop vacuum polarisation diagram I_6 as,

$$2 \operatorname{Im} I_6 = -2P_2 \operatorname{Re} A_4 + 2R_6 \quad (4.20)$$

with I_6 and A_4 being known from the literature and written as,

$$\begin{aligned} I_6 &= \frac{(4\pi)^{3\epsilon}}{(16\pi^2)^3} \frac{\Gamma^6(1-\epsilon) \Gamma^3(1+\epsilon)}{3\epsilon^3 \Gamma^3(2-2\epsilon)} (-q^2)^{-3\epsilon} \\ &\quad \left[1 + \epsilon + \epsilon^2 + \epsilon^3(14\zeta_3 - 7) + \epsilon^4 \left(-67 + 14\zeta_3 + \frac{21\pi^4}{90} \right) + O(\epsilon^5) \right] \end{aligned} \quad (4.21)$$

$$A_4 = \left(\frac{(4\pi)^\epsilon \Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{16\pi^2 \Gamma(1-2\epsilon)} \right)^2 (-q^2)^{-2\epsilon} \left[-\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2} + \frac{\pi^2}{6} \right) - \epsilon \left(\frac{65}{2} + \frac{5\pi^2}{6} - 2\zeta_3 \right) - \epsilon^2 \left(\frac{211}{2} + \frac{19\pi^2}{6} - 10\zeta_3 \right) + \mathcal{O}(\epsilon^3) \right]. \quad (4.22)$$

Note that A_4 is described in Subsubsection 4.1.2.

We then apply the Minkowski boundary prescription [11] $(q^2)^{-\epsilon} = (q^2)^{-\epsilon} e^{i\pi\epsilon}$, isolate the imaginary and real parts, and get, by using $\mathcal{A}_4$ and $\mathcal{I}_6$

$$\begin{aligned} R_6 &= S_\Gamma(q^2)^{-2\epsilon} \frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{6} \left(\frac{6\cos(2\pi\epsilon)}{\Gamma^2(1-2\epsilon)} \mathcal{A}_4 + \frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)\sin(3\pi\epsilon)}{\epsilon^3\pi\Gamma^2(2-2\epsilon)} \mathcal{I}_6 \right) \\ &= S_{\Gamma,2}(q^2)^{-2\epsilon} \left[-1 + \frac{\pi^2}{6} + \epsilon \left(-12 + \frac{5\pi^2}{6} + 9\zeta_3 \right) + \mathcal{O}(\epsilon^2) \right]. \end{aligned} \quad (4.23)$$

The $R_{8,a}$ Integral

The $R_{8,a}$ is the first of two MI with four denominator invariants. Its complete integral form is written as [11],

$$\begin{aligned} R_{8,a} &= \int d\Phi_4 \frac{1}{s_{13}s_{23}s_{14}s_{24}} \\ &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{13}s_{23}s_{14}s_{24}} \Theta(-\Delta_4) \delta(q^2 - \mathcal{S}_4) \\ &= P_2(q^2)^{-2-2\epsilon} \frac{16^{-2+\epsilon} \pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \\ &\quad \left[(1-y_{134})(1-z_1) \right]^{-1-2\epsilon} \left[tvz_1^{-\epsilon} (\Delta y_{13}\chi + y_{13,a}) \right]^{-1} \left[\chi(1-\chi) \right]^{-\frac{1}{2}-\epsilon}. \end{aligned} \quad (4.24)$$

This is the most complex MI integration of the set. As such, its computation will be written in detail here as most of the methods used are transferable to other MI. We start by defining the integration order. The computation will advance as,

$$\chi \rightarrow v \rightarrow y_{134} \rightarrow t \rightarrow z_1$$

with the inclusion of a u integral, also over the $[0, 1]$ interval in between the t and z_1 integrals, which will be introduced later.

The first integral is over χ , and gives,

$$I_\chi = \int_0^1 d\chi \left[\chi(1-\chi) \right]^{-\frac{1}{2}-\epsilon} (\Delta y_{13}\chi + y_{13,a})^{-1} = \frac{\Delta y_{13}^{-2\epsilon} \Gamma^2\left(\frac{1}{2}-\epsilon\right)}{y_{13,a} \Gamma(1-2\epsilon)} {}_2F_1\left(1, \frac{1}{2}-\epsilon; 1-2\epsilon; -\frac{\Delta y_{13}}{y_{13,a}}\right) \quad (4.25)$$

where ${}_2F_1(\dots)$ is a hypergeometric function. We now consider the definition $y_{13,(a,b)} = y_{134}(A \mp B)$, $A = \sqrt{(1-t)(1-v)}$, $B = \sqrt{z_1 tv}$, and $Z = -B/A$, and get,

$$-\frac{\Delta y_{13}}{y_{13,a}} = \frac{4Z}{(1+Z)^2}, \quad (1+Z)^2 = \frac{y_{13,a}}{y_{134}A^2}. \quad (4.26)$$

Finally, by taking,

$${}_2F_1\left(1, b; 2b; \frac{4Z}{(1+Z)^2}\right) = (1+Z)^2 {}_2F_1\left(1, \frac{3}{2} - b; b + \frac{1}{2}; Z^2\right) \quad (4.27)$$

and

$$16^\epsilon \Delta y_{13}^{-2\epsilon} = \left[t(1-t)v(1-v)y_{134}^2 z_1 \right]^{-\epsilon} \quad (4.28)$$

we get,

$$I_\chi = \frac{\pi \Gamma(1-2\epsilon)}{y_{134} \Gamma^2(1-\epsilon)} \left(t v y_{134}^2 z_1 \right)^{-\epsilon} \left[(1-t)(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 t v}{(1-t)(1-v)}\right). \quad (4.29)$$

The $R_{8,a}$ expression becomes,

$$R_{8,a} = P_2(q^2)^{-2-2\epsilon} \frac{16^{-2+\epsilon} \pi^{-4+2\epsilon}}{\Gamma^2(1-\epsilon)} \int_{[0,1]^4} dy_{134} dz_1 dt dv \left[y_{134}(1-y_{134}) z_1(1-z_1) \right]^{-1-2\epsilon} \left[t(1-t)v(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 t v}{(1-t)(1-v)}\right). \quad (4.30)$$

We now continue to the v integral. To start, we restate the hypergeometric function as a series, which enables us to write,

$$\begin{aligned} I_v &= \int_0^1 dv \left[v(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 t v}{(1-t)(1-v)}\right) \\ &= \sum_{n \geq 0} \frac{(1+\epsilon)_n}{(1-\epsilon)_n} \left(\frac{z_1 t}{1-t} \right)^n \frac{\Gamma(-\epsilon-n)\Gamma(-\epsilon+n)}{\Gamma(-2\epsilon)} \\ &= -\frac{2\Gamma^2(1-\epsilon)}{\epsilon\Gamma(1-2\epsilon)} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \end{aligned} \quad (4.31)$$

Plugging this expression back in and computing the trivial y_{134} integral we get,

$$R_{8,a} = -P_2(q^2)^{-2-2\epsilon} \frac{2^{-7+8\epsilon} \pi^{-\frac{7}{2}+2\epsilon}}{\epsilon^2 \Gamma\left(\frac{1}{2}-2\epsilon\right)} \int_{[0,1]^2} dz_1 dt (1-z_1)^{-1-2\epsilon} \left[z_1 t(1-t) \right]^{-1-\epsilon} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \quad (4.32)$$

We now restate the hypergeometric function as an Euler integral as,

$${}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right) = -\epsilon \int_0^1 du u^{-1-\epsilon} \left[1 + \frac{u z_1 t}{1-t} \right]^{-1} \quad (4.33)$$

and after computing the direct t, u and z_1 integrals, in that order, we get to the final expression,

$$\begin{aligned} R_{8,a} &= P_2(q^2)^{-2-2\epsilon} 2^{-7+4\epsilon} \pi^{-3+2\epsilon} \csc(\pi\epsilon) \Gamma(-2\epsilon) \left[\frac{2^{8\epsilon} \pi}{\epsilon^2 \Gamma^2\left(\frac{1}{2}-2\epsilon\right)} {}_3F_2\left(\begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ 1-2\epsilon, -4\epsilon \end{matrix}; 1\right) - \right. \\ &\quad \left. - \frac{2\Gamma^3(-\epsilon)}{\Gamma(1-4\epsilon)\Gamma(1-\epsilon)\Gamma(1+\epsilon)\Gamma(-3\epsilon)} {}_4F_3\left(\begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix}; 1\right) \right] \end{aligned} \quad (4.34)$$

which after a final normalisation becomes [11, 27],

$$\begin{aligned}
R_{8,a} = & S_\Gamma(q^2)^{-2-2\epsilon} \left[\frac{6}{\epsilon^2} \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)}{\Gamma(1-3\epsilon)\Gamma(1-4\epsilon)} {}_4F_3 \left(\begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix}; 1 \right) - \right. \\
& \left. - \frac{1}{\epsilon^4} \frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^3(1-2\epsilon)}{\Gamma^2(1-4\epsilon)} {}_3F_2 \left(\begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ -4\epsilon, 1-2\epsilon \end{matrix}; 1 \right) \right] \\
= & S_{\Gamma,2}(q^2)^{-2-2\epsilon} \left[\frac{5}{\epsilon^4} - \frac{20\pi^2}{3\epsilon^2} - \frac{126\zeta_3}{\epsilon} + \frac{7\pi^4}{18} + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.35}$$

The $R_{8,b}$ Integral

The second of the two four-denominator-invariant MI is $R_{8,b}$. It is written as [11],

$$R_{8,b} = \int d\Phi_4 \frac{1}{s_{13}s_{134}s_{23}s_{234}}. \tag{4.36}$$

Due to the presence of s_{234} , this MI must be computed using the Optical Theorem, much like R_6 was. For this case, we are able to map $R_{8,b}$ as one of the cuts of the three-loop vacuum polarisation diagram I_8 as,

$$2 \operatorname{Im} I_8 = -2P_2 \operatorname{Re} A_6 + 4 \operatorname{Re} V_8 + R_{8,a} + 4R_{8,b}. \tag{4.37}$$

with,

$$I_8 = S + \mathcal{O}(\epsilon^0), \quad \operatorname{Im} S = -\frac{13\pi^4}{90}; \tag{4.38}$$

$$V_8 = -i \int d\Phi_3 \frac{1}{s_{12}} \operatorname{Box}(s_{13}, s_{13}, s_{12}); \tag{4.39}$$

$$A_6 = \left(\frac{(4\pi)^\epsilon}{16\pi^2} \frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \right)^2 (q^2)^{-2-2\epsilon} \left[-\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{23\zeta_3}{\epsilon} + \frac{103\pi^4}{180} + \mathcal{O}(\epsilon) \right]. \tag{4.40}$$

Note that the A_6 and V_8 integrals are described in Subsubsections 4.1.2 and 4.1.2. In those Subsubsections, the integrals are given normalised using the usual S_Γ prefactor.

Using $\mathcal{V}_8$, $\mathcal{A}_6$ and $\mathcal{R}_{8,a}$ to denote the series parts of $\operatorname{Re} V_8$, $P_2 \operatorname{Re} A_6$ and $R_{8,a}$, respectively, we get [11],

$$\begin{aligned}
R_{8,b} = & \frac{1}{2} \operatorname{Im} S + S_\Gamma(q^2)^{-2-2\epsilon} \left(\frac{1}{2} \frac{\cos(2\pi\epsilon)\Gamma^6(1-\epsilon)\Gamma^2(1-\epsilon)}{\Gamma^2(1-2\epsilon)} \mathcal{A}_6 - \mathcal{V}_8 - \frac{1}{4} \mathcal{R}_{8,a} \right) \\
= & S_\Gamma(q^2)^{-2-2\epsilon} \left[\frac{3}{4\epsilon^4} - \frac{17\pi^2}{12\epsilon^2} - \frac{44\zeta_3}{\epsilon} - \frac{61\pi^4}{60} + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.41}$$

The $V_{5,a}$ Integral

$V_{5,a}$ is a one-loop bubble integrated over a three-particle phase space, written as,

$$\begin{aligned}
V_{5,a} = & \operatorname{Re} \left[-i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p-k_1-k_2-k_3)^2} \right] \\
= & \operatorname{Re} \left[\int d\Phi_3 B_0(q^2; 0, 0) \right].
\end{aligned} \tag{4.42}$$

Since $q^2 = (p_1 + p_2 + p_3)^2$, this MI factors to the three-particle phase-space integral of the $B_0(q^2; 0, 0)$ scalar one-loop function, discussed in Subsubsection 4.1.1. After parametrising the phase space and applying the methods introduced in Subsubsection 4.1.2 we get [4],

$$\begin{aligned} V_{5,a} &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \cos(\pi\epsilon) \left(-\frac{1}{\epsilon}\right) \frac{\Gamma^6(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \left[-\frac{1}{2\epsilon} - \frac{13}{4} + O(\epsilon) \right]. \end{aligned} \quad (4.43)$$

The $V_{5,b}$ Integral

$V_{5,b}$ is very similar in structure to $V_{5,a}$, also being the three-particle phase-space integral of a B_0 scalar one-loop function. The integral is also computed in the same manner and yields [4],

$$\begin{aligned} V_{5,b} &= \int d\Phi_3 B_0(s_{13}; 0, 0) \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \cos(\pi\epsilon) \left(-\frac{1}{\epsilon}\right) \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-4\epsilon)} \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \left[-\frac{1}{2\epsilon} - 4 + O(\epsilon) \right]. \end{aligned} \quad (4.44)$$

The V_8 Integral

The V_8 MI is the most complex of the three V -type MI. It is given as a four-denominator propagator loop and three-particle phase-space integral. That makes it be restatable as a Box function which, itself, may be restated as a D_0 function, as discussed in Subsubsection 4.1.1. It is written as [4],

$$\begin{aligned} V_8 &= \text{Re} \left[-i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p-k_1)^2(p-k_1-k_3)^2(p-k_1-k_2-k_3)^2} \right] \\ &= \text{Re} \left[-i \int d\Phi_3 \frac{1}{s_{12}} \text{Box}(s_{13}, s_{23}, s_{12}) \right] \\ &= \left(\frac{(4\pi)^\epsilon \mu^{-2\epsilon}}{16\pi^2} \right) \frac{\Gamma^2(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)} \int d\Phi_3 \frac{1}{s_{12}} \text{Re} \left[D_0(0, 0, 0, q^2; s_{13}, s_{23}; 0, 0, 0, 0) \right] \end{aligned} \quad (4.45)$$

which, after normalisation and evaluation gives,

$$V_8 = S_{\Gamma,2}(q^2)^{-2-2\epsilon} \left[-\frac{5}{2\epsilon^4} + \frac{9\pi^2}{2\epsilon^2} + \frac{89\zeta_3}{\epsilon} + \frac{13\pi^4}{180} + O(\epsilon) \right]. \quad (4.46)$$

The $A_{22,LO}$ Integral

$A_{22,LO}$ is the first of the four A -type MI, and represents a double bubble product. These are loop momenta integrals, not phase-space integrals. This first MI is given as [4],

$$\begin{aligned}
A_{22,LO} &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2(p-k_1-k_2)^2 \ell^2(\ell-k_1-k_2)^2} \\
&= -\frac{\mu^{-4\epsilon}}{(16\pi^2)^2} \left[B_0(q^2; 0, 0) \right]^2 \\
&= S_\Gamma(-q^2)^{-2\epsilon} \frac{\Gamma^2(1+\epsilon)\Gamma^6(1-\epsilon)}{\Gamma^2(2-2\epsilon)} \left(-\frac{1}{\epsilon^2} \right) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \left[-\frac{1}{\epsilon^2} - \frac{4}{\epsilon} - 12 + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.47}$$

The A_3 Integral

A_3 represents a three-propagator two-loop sunset vacuum bubble, with $q^2 = s_{12}$. It is given as [4],

$$\begin{aligned}
A_3 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-\ell-k_1-k_2)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2} B_0((p-k_1-k_2)^2; 0, 0) \\
&= S_\Gamma(-q^2)^{1-2\epsilon} \frac{\Gamma(1+2\epsilon)\Gamma^5(1-\epsilon)}{\Gamma(3-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon} \right) \\
&= S_\Gamma(-q^2)^{1-2\epsilon} \left[-\frac{1}{4\epsilon} - \frac{13}{8} + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.48}$$

The A_4 Integral

A_4 is a four-propagator two-loop vacuum planar double bubble. It is given as a triangle diagram containing a bubble insertion on one of its legs. That gives [4],

$$\begin{aligned}
A_4 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-k_1-k_2)^2 (p-\ell-k_1)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p-k_1-k_2)^2} B_0((p-k_1)^2; 0, 0) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \frac{\Gamma(1-2\epsilon)\Gamma(1+\epsilon)\Gamma^4(1-\epsilon)\Gamma(1+2\epsilon)}{\Gamma(2-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon^2} \right) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \left[-\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2} + \frac{\pi^2}{6} \right) + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.49}$$

The A_6 Integral

Lastly, the A_6 MI represents the two-loop two-point vertex. It is given as [4],

$$A_6 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-k_1-k_2)^2 (p-\ell)^2 (p-\ell-k_2)^2 (\ell-k_1)^2}. \tag{4.50}$$

Unlike the other A -type antennae, this one cannot be given in terms of the scalar one-loop functions B_0 , C_0 or D_0 . To compute this integral we have to start by expressing it as a Mellin-Barnes contour integral, then evaluate the residues, and finally expand as a Laurent series around ϵ . This process gives,

$$\begin{aligned} A_6 &= \frac{S_\Gamma(-q^2)^{-2-2\epsilon}}{(2\pi i)^2} \int_{-i\infty}^{i\infty} \int_{-i\infty}^{i\infty} dz_1 dz_2 \frac{\Gamma(-z_1)\Gamma(-z_2)\Gamma(2+2\epsilon+z_1+z_2)\Gamma^2(-\epsilon-z_1)\Gamma^2(-\epsilon-z_2)}{\Gamma(-2\epsilon-z_1-z_2)\Gamma(2+\epsilon+z_1+z_2)} \\ &= S_\Gamma(-q^2)^{-2-2\epsilon} \left[-\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{27}{\epsilon}\zeta_3 + \frac{23\pi^4}{36} + O(\epsilon) \right]. \end{aligned} \quad (4.51)$$

4.1.3 Exploratory Massive Master Integrals

AntCalc features an exploratory, but complete, massive Implementation of the A_3^0 route. To integrate it, two new MI are required. These two massive MI are,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)} \\ I_2^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)} s_{13} \end{aligned} \quad (4.52)$$

where $\Phi_X^{(m_q,0,m_q)}$ is the massive tripole phase space and is defined as,

$$d\Phi_X^{(m_q,0,m_q)} = \frac{d\Phi_3^{(m_q,0,m_q)}(q; k_1, k_3, k_2)}{d\Phi_2^{(m_q,m_q)}(q; \tilde{k}_I, \tilde{k}_K)}, \quad p \rightarrow q(k_1, m_q) + g(k_3, 0) + \bar{q}(k_2, m_q) \quad (4.53)$$

and m_q is the quark mass. Note that the two- and three-particle massive phase spaces are described as in Eq. 4.9 with $k_1^2 = k_2^2 = m_q^2$, $k_3^2 = 0$, and $q^2 > 4m_q^2$.

These two integrals evaluate to [24],

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= (q^2)^{1-\epsilon} r^{2-2\epsilon} 2^{-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; r\right) \\ I_2^{(m_q,0,m_q)} &= (q^2)^{2-\epsilon} r^{3-2\epsilon} 2^{1-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; r\right) \end{aligned} \quad (4.54)$$

where,

$$r = 1 - \frac{4m_q^2}{q^2}. \quad (4.55)$$

4.2 NLO Antenna Results

The simplest non-trivial SMQCD antenna is A_3^0 . As such, it may be used as an example of how *AntCalc* handles the physics involved in building and then integrating it over the phase space. In this chapter, we are going to go over how this antenna is mathematically computed by hand, and how *AntCalc* handles those same calculations on its end.

4.2.1 The Tree-Level Three-Parton Antenna A_3^0

Analytic Construction of the Unintegrated A_3^0 Antenna

The A_3^0 is the tree-level final-final quark-antiquark-gluon antenna. It represents the normalised $\gamma^*(p) \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$ process. This process, in the current model, is able to produce two distinct topologies which differ by what quark line the gluon is emitted from. Those diagrams are shown below in Fig. 4.1.

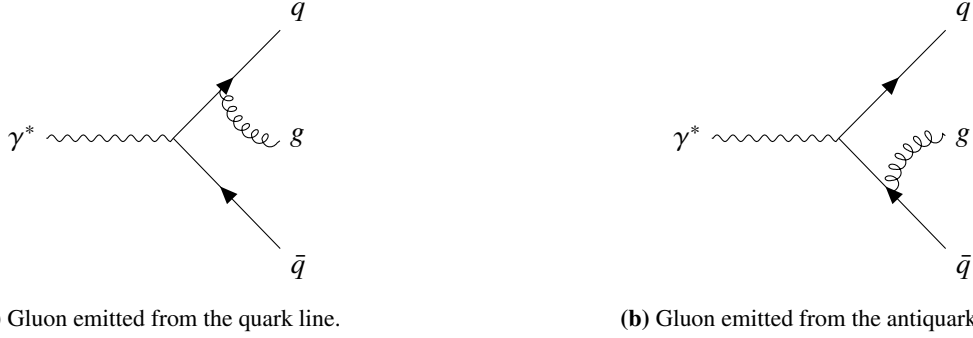


Figure 4.1: Feynman diagrams depicting the two topologies described under the A_3^0 antenna, corresponding to the process $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$.

The amplitude is given as follows directly from the diagrams,

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[\bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{k_1 + k_3}{s_{13}} \not{\epsilon}(p) v(k_2) - \bar{u}(k_1) \not{\epsilon}^*(p) \frac{k_2 + k_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) \right] \quad (4.56)$$

where $(T^{a_3})_{i_1 i_2}$ represents the colour matrix for the quark and antiquark in positions 1 and 2 and gluon in position 3.

The amplitude is now squared as follows,

$$\begin{aligned} |\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \left(M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\ \text{with } M_A &= \bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{k_1 + k_3}{s_{13}} \not{\epsilon}(p) v(k_2) = \bar{u}(k_1) \Gamma_A v(k_2), \\ M_B &= \bar{u}(k_1) \not{\epsilon}^*(p) \frac{k_2 + k_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) = \bar{u}(k_1) \Gamma_B v(k_2). \end{aligned} \quad (4.57)$$

which in total becomes

$$\begin{aligned} &= \sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \\ &\quad \sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) \left[\Gamma_A \bar{\Gamma}_A + \Gamma_B \bar{\Gamma}_B - \Gamma_A \bar{\Gamma}_B - \Gamma_B \bar{\Gamma}_A \right] v(k_2, s_2) \bar{v}(k_2, s_2) \end{aligned} \quad (4.58)$$

Each of the Γ terms becomes, using Dirac Algebra,

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_A \bar{\Gamma}_A) v(k_2, s_2) \bar{v}(k_2, s_2) &= \\
&= \frac{1}{s_{13}^2} \sum_{\lambda_g, \lambda_\gamma} \varepsilon_\mu^* \varepsilon_\nu \varepsilon_\rho \varepsilon_\sigma^* \text{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} g_{\mu\nu} g_{\rho\sigma} \text{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} \text{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma_\rho (\not{k}_1 + \not{k}_3) \gamma_\mu \right) \\
&= \left(\frac{2-d}{s_{13}} \right)^2 \text{Tr}(\not{k}_1 \not{P} \not{k}_2 \not{P}), \quad \not{P} = (\not{k}_1 + \not{k}_3) \\
&= 4 \left(\frac{2-d}{s_{13}} \right)^2 \left(2(k_1 \cdot P)(k_2 \cdot P) - (k_1 \cdot k_2) P^2 \right) \\
&= 2(d-2)^2 \frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.59}$$

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_B \bar{\Gamma}_B) v(k_2, s_2) \bar{v}(k_2, s_2) &= \\
&= 2(d-2)^2 \frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.60}$$

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_A \bar{\Gamma}_B) v(k_2, s_2) \bar{v}(k_2, s_2) &= \\
&= 2(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.61}$$

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_B \bar{\Gamma}_A) v(k_2, s_2) \bar{v}(k_2, s_2) &= \\
&= 2(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.62}$$

and computing the colour matrix sum as

$$\sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} = \frac{1}{2} (N^2 - 1) \tag{4.63}$$

we are finally able to reach the final result, as follows, using dimensional regularisation as $d \rightarrow 4 - 2\epsilon$,

$$\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (N^2 - 1) \left(2(d-2)^2 \left(\frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) - 4(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}} \right) \\
&= 2(d-2)e_q^2 g_s^2 (N^2 - 1) \left((d-2) \left(\frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) - 2 \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}} \right) \\
&= \frac{8}{s_{13}s_{23}} \left(2q^2 s_{12} + s_{13}^2 + s_{23}^2 - 2\epsilon \left(q^2 s_{12} + s_{13}^2 + s_{23}^2 + s_{13}s_{23} \right) + \epsilon^2 (s_{13} + s_{23})^2 \right)
\end{aligned} \tag{4.64}$$

The A_3^0 antenna also requires the $\mathcal{M}_2^0$ amplitude. This amplitude corresponds to the $\gamma^* \rightarrow q\bar{q}$ Born process. The amplitude is

$$\mathcal{M}_2^0 = e_q \delta_{i_1 i_2} \bar{u}(k_1) \not{\epsilon}(p) v(k_2). \tag{4.65}$$

Applying the same spin, polarisation and colour sums as above, and considering that, for a two final-state particles system $s_{12} = q^2$, we get

$$\begin{aligned}
|\mathcal{M}_2^0|^2 &= 4e_q^2 (d-2) N s_{12} = 4e_q^2 (d-2) N q^2 \\
&= 8e_q^2 (1-\epsilon) N q^2
\end{aligned} \tag{4.66}$$

The antenna is now finally ready to be built. That is accomplished using the following formula,

$$A_3^0 = \frac{1}{2g_s^2 C_F} \frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.67}$$

By taking $C_F = (N^2 - 1)/(2N)$, the formula yields, using Eqs. 4.64 and 4.66,

$$\begin{aligned}
A_3^0 &= \frac{2q^2 s_{12} + s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}} - \epsilon \frac{(s_{13} + s_{23})^2}{q^2 s_{13} s_{23}} \\
&= \frac{1}{q^2} \left[\left(\frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} + 2 \frac{s_{12} q^2}{s_{13} s_{23}} \right) - \epsilon \left(2 + \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) \right]
\end{aligned} \tag{4.68}$$

Phase-Space Integration through R_3

We now integrate the A_3^0 antenna over the appropriate (3 final-state particle, in this case) phase-space. This integral is computed in d dimensions and over the phase-space measure in terms of the Mandelstam invariants (s_{13} , q^2 , etc.). The integral takes the form,

$$\begin{aligned}
\mathcal{A}_3^0 &= C(\epsilon, 1) \int d\Phi_{ijk} A_3^0 \\
&= \mathcal{N}(\epsilon, 1) (4\pi)^{3-2d} (q^2)^{1-\frac{d}{2}} \int ds_{12} ds_{13} ds_{23} d\Omega(s_{12} s_{13} s_{23})^{\frac{d-4}{2}} \delta(q^2 - s_{12} - s_{13} - s_{23}) A_3^0.
\end{aligned} \tag{4.69}$$

with $d\Phi_{ijk} = d\Phi_3/\Phi_2$. $C(\epsilon, 1)$ is the integral normalisation which is here given as

$$C(\epsilon, 1) = 8\pi^2(4\pi)^{-\epsilon} e^{\epsilon\gamma_E}. \quad (4.70)$$

This integral is quite complex and spans across all A_3^0 terms. Accordingly, we are not going to compute this integral directly: instead we are going to use IBP reduction to reduce the expression to a linear combination of master integrals, compute those master integrals and then use the result to compute the integrated antenna. This process is explained in Subsection 2.5.1.

After applying IBP identities and reducing the integral to a linear combination of this master integral, we reach the following expression,

$$\mathcal{A}_3^0 = \frac{C(\epsilon, 1)}{\Phi_2} \frac{2(2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3)}{q^2\epsilon^2} R_3 \quad (4.71)$$

This antenna has a single master, R_3 , discussed in Section 4.1. This master integral is the integral over phase-space itself,

$$R_3 = \int d\Phi_3 = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon} \Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} (q^2)^{1-2\epsilon}. \quad (4.72)$$

IBP Reduction of an Example Term

As an example of the IBP reduction method, we will consider the eikonal term of the unintegrated antenna in Eq. 4.68,

$$\mathcal{S}_{12}^{(3)} = \frac{2s_{12}}{s_{13}s_{23}}. \quad (4.73)$$

We define,

$$y_{13} = \frac{s_{13}}{q^2}, \quad y_{23} = \frac{s_{23}}{q^2}, \quad y_{12} = \frac{s_{12}}{q^2} = 1 - y_{13} - y_{23} \quad (4.74)$$

$$J(a, b, c) = \int_{\Delta} dy_{13} dy_{23} y_{13}^{a-1} y_{23}^{b-1} y_{12}^{c-1}, \quad \Delta : y_{13}, y_{23}, y_{12} \geq 0, \quad y_{13} + y_{23} + y_{12} = 1$$

and with these,

$$\begin{aligned} \Phi_3 &\equiv \int d\Phi_3 = R_3 \\ &= \mathcal{K}(q^2, \epsilon) \int_{\Delta} (y_{13}y_{23}y_{12})^{-\epsilon} dy_{13} dy_{23} \\ &= \mathcal{K}(q^2, \epsilon) J(1-\epsilon, 1-\epsilon, 1-\epsilon), \quad \mathcal{K}(q^2, \epsilon) = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon}}{\Gamma(2-2\epsilon)} (q^2)^{1-2\epsilon}. \end{aligned} \quad (4.75)$$

We are also able to express the integral over the eikonal term using a J function as,

$$\int d\Phi_3 \mathcal{S}_{12}^{(3)} = \frac{2\mathcal{K}}{q^2} J(-\epsilon, -\epsilon, 2-\epsilon). \quad (4.76)$$

The IBP identity we use to reduce the integral is implemented as,

$$\begin{aligned} 0 &= \int_{\Delta} dy_{13} dy_{23} \frac{\partial}{\partial y_{13}} (y_{13}^a y_{23}^{b-1} y_{12}^{c-1}) \\ &= aJ(a, b, c) - (c-1)J(a+1, b, c-1) \end{aligned} \quad (4.77)$$

such that, for y_{13} ,

$$J(-\epsilon, -\epsilon, 2 - \epsilon) = -\frac{1 - \epsilon}{\epsilon} J(1 - \epsilon, -\epsilon, 1 - \epsilon) \quad (4.78)$$

and for y_{23} ,

$$J(1 - \epsilon, -\epsilon, 2 - \epsilon) = -\frac{1 - \epsilon}{\epsilon} J(1 - \epsilon, 1 - \epsilon, 1 - \epsilon). \quad (4.79)$$

By the symmetry $y_{13} \leftrightarrow y_{12}$,

$$J(2 - \epsilon, -\epsilon, 1 - \epsilon) = J(1 - \epsilon, -\epsilon, 2 - \epsilon). \quad (4.80)$$

And hence,

$$J(-\epsilon, -\epsilon, 2 - \epsilon) = \frac{(1 - \epsilon)(2 - 3\epsilon)}{\epsilon^2} J(1 - \epsilon, 1 - \epsilon, 1 - \epsilon) \quad (4.81)$$

which finally makes,

$$\int d\Phi_3 \frac{2s_{12}}{s_{13}s_{23}} = \frac{2(1 - \epsilon)(2 - 3\epsilon)}{\epsilon^2 q^2} R_3. \quad (4.82)$$

Master Integral Substitution and Laurent Expansion

Substituting Eq. 4.72 into Eq. 4.71 gives

$$\begin{aligned} \mathcal{A}_3^0 &= \frac{C(\epsilon, 1)}{\Phi_2} (q^2)^{-2\epsilon} (4\pi)^{-3+2\epsilon} (2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3) \frac{\Gamma^3(1 - \epsilon)}{\epsilon^2 \Gamma(3 - 3\epsilon) \Gamma(2 - 2\epsilon)} \\ &= (q^2)^{-\epsilon} e^{\epsilon \gamma_E} (1 - 2\epsilon)(2 + \epsilon(\epsilon + 2)) \frac{\Gamma^2(-\epsilon)}{\Gamma(3 - 3\epsilon)} \end{aligned} \quad (4.83)$$

and, after expanding the integrated expression as a Laurent series in ϵ , we finally arrive at the infrared pole structure,

$$\begin{aligned} \mathcal{A}_3^0 &= (q^2)^{-\epsilon} \left[\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{57 - 7\pi^2}{12} + \epsilon \left(\frac{84 - 21\pi^2 - 8\psi^{(2)}(1) + 108\psi^{(2)}(3)}{24} \right) \right. \\ &\quad \left. + \epsilon^2 \left(\frac{35640 - 3990\pi^2 - 71\pi^4 - 720\psi^{(2)}(1) + 9720\psi^{(2)}(3)}{1440} \right) \right] \\ &= (q^2)^{-\epsilon} \left[\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon \left(\frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right) + \epsilon^2 \left(\frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) \right] \end{aligned} \quad (4.84)$$

where the usual polygamma identities, $\psi^{(2)}(1) = -2\zeta_3$ and $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$, were employed.

Implementation in *AntCalc*

Using *AntCalc*, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. 4.68,

```
In[2] := BuildAntenna[A, 3, 0] // Collect[#, Epsilon] &
```

Out [2] :=

$$\frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{2s_{12}^2}{q^2 s_{13} s_{23}} + \frac{s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} + \epsilon \left(\frac{-2}{q^2} - \frac{s_{13}}{q^2 s_{23}} - \frac{s_{23}}{q^2 s_{13}} \right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that *AntCalc* goes through the same steps as the by-hand derivation in Subsection 4.2.1. We can see that with,

In [3] :=

`BuildAntenna[A, 3, 0, ReturnRecord->True] ["IntermediateSteps"]`

Out [3] :=

Amplitude:

$$T_{i_1 i_2}^{a_3} \left[\frac{\bar{u}(k_1) \not{\epsilon}^*(k_3) (\not{k}_1 + \not{k}_3) \not{\epsilon}(p) v(k_2)}{(k_1 + k_3)^2} - \frac{\bar{u}(k_1) \not{\epsilon}(p) (\not{k}_2 + \not{k}_3) \not{\epsilon}^*(k_3) v(k_2)}{(k_2 + k_3)^2} \right]$$

Interference:

$$\frac{4(\epsilon - 1)(N^2 - 1) \left(-2s_{12}^2 - 2s_{12}(s_{13} + s_{23}) + (\epsilon - 1)s_{13}^2 + 2\epsilon s_{13}s_{23} + (\epsilon - 1)s_{23}^2 \right)}{s_{13}s_{23}}$$

Result:

$$-\frac{2\epsilon}{q^2} + \frac{2s_{12}^2}{q^2 s_{13} s_{23}} + \frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{s_{13}}{q^2 s_{23}} - \frac{\epsilon s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} - \frac{\epsilon s_{23}}{q^2 s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. 4.56, 4.64 and 4.68.

Integrating antennae using *AntCalc* is also simple. Every step in Subsubsection 4.2.1 is handled by a single function: `IntegrateAntenna[antenna]`.

Before going over how this function is used, a small clarification needs to be made: the `IntegrateAntenna` function is not compatible with the default output of the `BuildAntenna` function described earlier. This is so that the built antenna is immediately useful and easily readable. As described in Chapter 3, however, the IBP reduction steps are handled by the `LiteRed2` package, which does not expect Mandelstam invariants, but rather, momenta. Accordingly, to use `BuildAntenna`'s output in `IntegrateAntenna` we must toggle the relevant flag: `BuildAntenna[A, 3, 0, IntegrableForm -> True]`. This will return a large association with all required information for integrating the antenna.

We may now proceed to the integration. The whole process goes as follows,

In[4] := `intA30 = BuildAntenna[A, 3, 0, IntegrableForm -> True];`

In[5] := `IntegrateAntenna[intA30, ExpansionOrder->2]`

Out [5] :=

$$\frac{19}{4} + \epsilon^{-2} + \frac{3}{2\epsilon} - \frac{7\pi^2}{12} + \epsilon^2 \left(\frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) + \epsilon \left(\frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right)$$

Much like with `BuildAntenna`, `IntegrateAntenna` also allows us to inspect its intermediate steps. Here, however, the available options show the master integral linear combination expression both with R_3 and with it substituted by the appropriate expression, that is, the equivalent to Eqs. 4.71 and 4.83 before normalisation. The code and results are,

```

In[6]:= resA30 = IntegrateAntenna[intA30,ExpansionOrder->2,
ReturnRecord->True];
In[7]:= resA30["MasterCombination"]//Simplify//Collect[#,R3]&
Out[7]:=

$$\frac{-4(d-3)(4+d(\epsilon-2)-4\epsilon)}{(d-4)^2q^2}R_3$$

In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=

$$-\left(q^2\right)^{-2\epsilon}\frac{2^{-5+4\epsilon}(d-3)(4+d(\epsilon-2)-4\epsilon)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$


```

The results here are given as a combination of d and ϵ . This is a result of the way the function processes each step. Since these are snapshots of what is happening inside the function itself, they show the expressions before dimension regularisation is applied.

Finally, *AntCalc* also enables us to forgo the work we did building and the integrating the antenna with the use of the `BuildAndIntegrateAntenna[type, numFinalParticles, numLoops]` function. It goes over the whole process, as we did, and outputs the ϵ Laurent series,

```

In[9]:= BuildAndIntegrateAntenna[A,3,0,ExpansionOrder->2]
Out[9]:=

$$\frac{19}{4} + \epsilon^{-2} + \frac{3}{2\epsilon} - \frac{7\pi^2}{12} + \epsilon^2 \left( \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) + \epsilon \left( \frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right)$$


```

The agreement between the analytic derivation and the corresponding *AntCalc* outputs validates both the construction and the integration workflow for A_3^0 . Its local soft and collinear behaviour is validated separately in the following subsection.

4.2.2 Local Validation of A_3^0

To be usable as a subtraction term, an antenna function's unintegrated form must also reproduce the singular behaviour of the corresponding matrix element locally in phase space. In particular, A_3^0 must reproduce the universal eikonal factor when the gluon becomes soft and the Altarelli-Parisi splitting function in the relevant collinear limits. These two quantities provide local validation of the results obtained analytically and using *AntCalc*.

Soft Limit and the Eikonal Factor

When a gluon becomes soft, gauge theory amplitudes factorise universally to the amplitude without the soft gluon times a universal soft current as [7],

$$|\mathcal{M}_{n+1}(\dots, i, j, k, \dots)|^2 \xrightarrow{j \rightarrow 0} -g_s^2 \sum_{i,k} T_i \cdot T_k \mathcal{S}_{ik}^{(j)} |\mathcal{M}_n(\dots, i, k, \dots)|^2 \quad (4.85)$$

where T_i are colour-charge operators and,

$$\mathcal{S}_{ik}^{(j)} = \frac{2s_{ik}}{s_{ij}s_{jk}} \quad (4.86)$$

is the eikonal factor.

Since this property is universal, we can use it to validate our results. For the A_3^0 antenna, the eikonal factor may be computed as,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3 \rightarrow 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \quad (4.87)$$

After building the antenna using *AntCalc*, the limit may be computed using *Wolfram Mathematica* as follows,

```
In[2]:= A30 = BuildAntenna[A, 3, 0];

softSubs = {s13 -> lambda * s13, s23 -> lambda * s23};
A30Soft = A30 /. softSubs;

A30SoftSeries = Series[A30Soft, {lambda, 0, 0}];

eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /. q2 -> s12
// Simplify;
Out[2]:=
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

This result agrees with the expected eikonal factor.

Collinear Limit and the Altarelli–Parisi Splitting Function

In the same manner, when two partons i and j become collinear, the amplitude factorises to [7, 28],

$$|\mathcal{M}_{n+1}(\dots, i, j, \dots)|^2 \xrightarrow{i||j} \frac{8\pi\alpha_s}{s_{ij}} P_{IJ \rightarrow K}(z) |\mathcal{M}_n(\dots, K, \dots)|^2 \quad (4.88)$$

where K is the parent parton in which i and j recombine, z is the momentum fraction carried by one of the daughter partons, and $P_{IJ \rightarrow K}$ is the splitting function. This function is universal but channel-dependent and, since it carries the spin and helicity structure of the collinear splitting, it depends on the partons involved. For A_3^0 , where a gluon may become collinear with a quark, the function appears as,

$$P_{qg \rightarrow q}(z) = C_F \frac{1+z^2}{1-z}. \quad (4.89)$$

Noting that the building process already normalises the fundamental Casimir C_F , using *AntCalc* in *Wolfram Mathematica*, the AP splitting function may be computed as,

```
In[3]:= A30 = BuildAntenna[A, 3, 0];

collinearSubs = {s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0};
A30Collinear = A30 /. collinearSubs;

A30CollinearSeries = Series[A30Collinear, {lambda, 0, 0}];

apSplittingFunc = SeriesCoefficient[A30CollinearSeries, -1]
```

```
// Simplify;
```

```
Out[3] :=
```

$$\frac{1+z^2}{1-z}$$

This result agrees with the expected colour-stripped $qg \rightarrow q$ AP kernel.

4.2.3 The Virtual A_2^1 Contribution

To compute A_2^1 , we may follow the same steps. This is a loop-antenna, however, which means that its build result has been PaVe reduced. It is given in terms of the B_0 and C_0 scalar one-loop function. The unintegrated antenna is computed, using *AntCalc*, as,

```
In[2] := BuildAntenna[A, 2, 1]
```

```
Out[2] :=
```

$$-\frac{1}{2} \left[(3+2\epsilon) B_0(q^2; 0, 0) \right] - q^2 C_0(0, 0, q^2; 0, 0, 0)$$

The integrated final result is, then [4],

```
In[3] := BuildAndIntegrateAntenna[A, 2, 1, ExpansionOrder->2]
```

```
Out[3] :=
```

$$-\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon \left(-8 + \frac{7\pi^2}{8} + \frac{7\zeta_3}{3} \right) + \epsilon^2 \left(-16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} + \frac{7\zeta_3}{2} \right)$$

Note that, exclusively for this antenna, the same PaVe scalar functions appear in both the build- and integration-side reductions. No phase-space IBP reduction is required.

4.3 NNLO Antenna Results

4.3.1 Two-Parton Contributions: the A_2^2 Set

The A_2^2 route is computed similarly to A_2^1 . This route, however, yields four different antenna components and is the longest-running calculation supported by *AntCalc* in its current version. The unintegrated antennae are computed using,

```
In[2] := BuildAntenna[A, 2, 2]
```

```
Out[2] :=
```

$$\left\{ \frac{11 \left[(3+2\epsilon) B_0(s_{12}; 0, 0) + 2s_{12} C_0(0, 0, s_{12}; 0, 0, 0) \right]}{12\epsilon} + \dots, \right. \\ 24\epsilon^{14} \left(391\pi^{12} - 1149918\pi^8 \zeta_3 - 62009136\pi^4 \zeta_3^2 - 823011840\zeta_3^3 \right) + \dots, \\ - \frac{1}{6} \frac{(3+2\epsilon) B_0(s_{12}; 0, 0) + 2s_{12} C_0(0, 0, s_{12}; 0, 0, 0)}{\epsilon} + \dots, \\ \left. 80(13-42\epsilon) \frac{\zeta_3}{1920\epsilon \Gamma^2(2-2\epsilon)} + \dots \right\}$$

The expressions have been truncated here. These four components represent, respectively, the leading, subleading, quark-loop and self-interference terms, or, A_2^2 , $\tilde{A}_2^2$, $\hat{A}_2^2$ and $\check{A}_2^2$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix A. We may then obtain the integrated expressions as,

```
In[3] := BuildAndIntegrateAntenna[A, 2, 2, ExpansionOrder->0]
```

```
Out[3] :=
```

$$\left\{ \begin{aligned} & \frac{1}{4\epsilon^4} + \frac{17}{8\epsilon^3} + \frac{433 - 72\pi^2}{144\epsilon^2} + \frac{121350 - 44820\pi^2 + 15120\zeta_3}{25920\epsilon} \\ & + \frac{-45415 - 64590\pi^2 + 4734\pi^4 + 37440\zeta_3}{25920}, \\ & - \frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} + \frac{-1476 + 312\pi^2}{576\epsilon^2} + \frac{-3978 + 864\pi^2 + 1536\zeta_3}{576\epsilon} \\ & + \frac{-10359 + 2850\pi^2 - 118\pi^4 + 4176\zeta_3}{576}, \\ & - \frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{65 + 9\pi^2}{216\epsilon} + \frac{4085 - 546\pi^2 + 72\zeta_3}{1296}, \\ & \frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1230 - 20\pi^2}{480\epsilon^2} + \frac{-60\pi^2 + 560(6 + \zeta_3)}{480\epsilon} \\ & + \frac{-205\pi^2 - 7\pi^4 - 240(-36 + 7\zeta_3)}{480} \end{aligned} \right\}$$

The integrated Laurent series agree with the established $\mathcal{A}_2^2$ set results [23].

4.3.2 Three-Parton Contributions: the A_3^1 Set

Computing the A_3^1 antenna set is very similar to the work done for the A_2^2 set, with the big differentiator being that this set does not include a self-interference term. The unintegrated antennae are computed using,

```
In[2] := BuildAntenna[A, 3, 1]
```

```
Out[2] :=
```

$$\left\{ \begin{aligned} & \epsilon(1 + 3\epsilon)s_{13}s_{23} + \dots, \\ & \frac{2s_{12} \left[(-1 + \epsilon)s_{13} + \epsilon s_{23} \right]}{2(-1 + \epsilon)q^2 s_{13}} D_0(0, 0, 0; q^2, s_{12}, s_{23}; 0, 0, 0, 0) + \dots, \\ & \frac{2}{3} \frac{s_{12}^2}{\epsilon q^2 s_{13} s_{23}} + \dots \end{aligned} \right\}$$

The expressions have been truncated here. These three components represent, respectively, the leading, subleading and quark-loop terms, or, A_3^1 , $\tilde{A}_3^1$, and $\hat{A}_3^1$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix A. We may then obtain the integrated expressions as,

```
In[3] := BuildAndIntegrateAntenna[A, 3, 1, ExpansionOrder->0]
Out[3] :=
```

$$\left\{ -\frac{1}{4\epsilon^4} - \frac{31}{12\epsilon^3} + \frac{-9540 + 660\pi^2}{1440\epsilon^2} + \frac{-38820 + 3520\pi^2 + 11040\zeta_3}{1440\epsilon} \right. \\ + \frac{-156930 + 12240\pi^2 - 123\pi^4 + 55120\zeta_3}{1440} + O(\epsilon), \\ \frac{-15 + 4\pi^2}{24\epsilon^2} + \frac{-19 + \pi^2 + 28\zeta_3}{4\epsilon} + \frac{1215\pi^2 + 56\pi^4 + 540(-35 + 18\zeta_3)}{720}, \\ \left. \frac{1}{3\epsilon^3} + \frac{1}{2\epsilon^2} + \frac{114 - 14\pi^2}{72\epsilon} + \frac{327 - 21\pi^2 - 200\zeta_3}{72} + O(\epsilon) \right\}$$

The integrated Laurent series agree with the established $\mathcal{A}_3^1$ set results [4, 23].

4.3.3 Four-Parton Contributions

The A_4^0 Antenna Set

The double real-radiation antenna set is A_4^0 , and it has two components, A_4^0 and $\tilde{A}_4^0$. It represents the $\gamma^* \rightarrow qgg\bar{q}$ process.

Building the unintegrated antenna follows the same usual process,

```
In[2] := BuildAntenna[A, 4, 0]
Out[2] :=
```

$$\left\{ -\frac{3s_{12}}{2(-1+\epsilon)(q^2)^3} + \dots, \right. \\ \left. -\frac{\epsilon}{(-1+\epsilon)q^2s_{13}} + \dots \right\}$$

and so does computing the integrated antenna functions,

```
In[3] := BuildAndIntegrateAntenna[A, 4, 0, ExpansionOrder->0]
Out[3] :=
```

$$\left\{ \frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} - \frac{1}{\epsilon^2} \left(\frac{217}{18} - \frac{13\pi^2}{12} \right) + \frac{1}{\epsilon} \left(\frac{43224}{864} - \frac{589\pi^2}{144} - \frac{71\zeta_3}{4} \right) \right. \\ + \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon), \\ \frac{1}{\epsilon^4} + \frac{3}{\epsilon^3} + \frac{1}{\epsilon^2} \left(13 - \frac{3\pi^2}{2} \right) + \frac{1}{\epsilon} \left(\frac{845}{16} - \frac{9\pi^2}{2} - \frac{80\zeta_3}{3} \right) \\ \left. + \frac{6921}{32} - \frac{473\pi^2}{24} + \frac{17\pi^4}{72} - 80\zeta_3 + O(\epsilon) \right\}$$

where for each case, the lists are $\{A_4^0, \tilde{A}_4^0\}$ and $\{\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0\}$, respectively.

The complete unintegrated expressions are collected in Appendix A. The integrated results agree with the established $\mathcal{A}_4^0$ antenna results [4].

The B_4^0 and C_4^0 Antennae

Finally, the $\gamma^* \rightarrow q\bar{q}q\bar{q}$ process is represented by the B_4^0 and C_4^0 antennae. The B_4^0 antenna describes non-identical quark flavours, while C_4^0 gives the identical-flavour contribution.

The antenna building process is completed as usual using *AntCalc*,

```
In[2] := BuildAntenna[B, 4, 0]
Out[2] :=
```

$$-\frac{\epsilon s_{12}}{(-1 + \epsilon)q^2 s_{134}^2} + \dots$$

```
In[3] := BuildAntenna[C, 4, 0]
Out[3] :=
```

$$-\frac{1}{2} \frac{\epsilon s_{12}}{(-1 + \epsilon)q^2 s_{123} s_{134}} + \dots$$

The integration process follows in the same way,

```
In[4] := BuildAndIntegrateAntenna[B, 4, 0, ExpansionOrder->0]
Out[4] :=
```

$$-\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} + \frac{1}{\epsilon} \left(-\frac{407}{216} + \frac{11\pi^2}{72} \right) - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + \mathcal{O}(\epsilon)$$

```
In[5] := BuildAndIntegrateAntenna[C, 4, 0, ExpansionOrder->0]
Out[5] :=
```

$$\frac{1}{\epsilon} \left(-\frac{13}{32} + \frac{\pi^2}{16} - \frac{\zeta_3}{4} \right) - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + \mathcal{O}(\epsilon)$$

The complete unintegrated expressions for both antennae are collected in Appendix A. The integrated results of both antennae agree with the established $\mathcal{B}_4^0$ and C_4^0 results [4].

4.4 Summary of the Massless Antenna Set

The massless results presented in this chapter demonstrate that, using *AntCalc*, we are able to build and integrate all quark-antiquark final-final standard-model (SMQCD) antenna functions through NNLO. The selected case studies cover tree-level NLO and NNLO, one-, and two-loop contributions, showcasing the programme's ability to handle these different cases. Due to the size of the unintegrated antennae, some have been truncated. The complete results are shown in Appendix A. The resulting integrated antennae agree with the established results.

The more detailed A_3^0 explanation shows a method of local validation of the unintegrated result based on computing the eikonal factor and the AP splitting function using the antenna. This method might also be used with other antenna functions whenever the corresponding unresolved limit is present. A global validation is presented in Chapter 5, where we compute the massless hadronic R-ratio using all necessary integrated antennae.

Having established the massless sector, the following section considers the exploratory massive extension of the A_3^0 route.

4.5 Exploratory Massive Extension: A_3^0

The massive A_3^0 route is currently the only massive route supported in *AntCalc*. Its inner workflow is similar to that of the massless A_3^0 route with the added complexity of the quarks being massive.

Kinematically, instead of considering only the centre of mass energy as a scale, the quark mass must also be considered. Hence, q^2 and m_q^2 are scales, and they combine to create the relevant threshold variable r ,

$$r = 1 - \frac{4m_q^2}{q^2}. \quad (4.90)$$

The phase space has $k_1^2 = k_2^2 = m_q^2$ and $k_3^2 = 0$. The cut families also shift the massless cuts by a mass term as,

$$\begin{aligned} k_{1,2}^2 &\rightarrow m_q^2 - k_{1,2}^2, \\ k_3^2 &\rightarrow k_3^2. \end{aligned} \quad (4.91)$$

This also implies using massive Mandelstam invariants which, in this case, with k_3 presenting the massless gluon's momentum, become,

$$s_{ij} = (q - k_k)^2 - m_i^2 - m_j^2, \quad \{i, j, k\} = \{1, 2, 3\}. \quad (4.92)$$

LiteRed2, then, has to be updated with these new families before recognising the topology. It will then generate two master integrals. In the current *AntCalc* implementation, the IBP reduction step produces two MI which do not exactly match the $I_1^{(m_q, 0, m_q)}$ and $I_2^{(m_q, 0, m_q)}$ integrals described in Subsection 4.1.3. Instead, it produces J_1 and J_2 , shown below using *LiteRed2*'s $j[\dots]$ notation, where the complete basis, named MX30Basis123, uses the denominators $\{m_q^2 - k_1^2, m_q^2 - k_2^2, k_3^2, -s_{13}, -s_{23}\}$ in that order,

$$\begin{aligned} J_1^{(m_q, 0, m_q)} &= j[\text{MX30Basis123}, 1, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut} \left[\frac{1}{D_1 D_2 D_3} \right], \\ J_2^{(m_q, 0, m_q)} &= j[\text{MX30Basis123}, 2, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut} \left[\frac{1}{D_1^2 D_2 D_3} \right]. \end{aligned} \quad (4.93)$$

Here, $\mathcal{N}_{\text{LiteRed2}}$ represents the normalisation implicit in the cut-integral convention used for the family in *LiteRed2*. These two MI do not directly equal the two literature masters, but they can be mapped to them. We need here to introduce a conversion factor of $-1/4$ between the literature and *AntCalc*'s conventions. Using it, we can complete the mapping as,

$$\begin{aligned} I_1^{(m_q, 0, m_q)} &= -\frac{1}{4} J_1^{(m_q, 0, m_q)}, \\ I_2^{(m_q, 0, m_q)} &= -\frac{1}{4} \left(a J_1^{(m_q, 0, m_q)} + b J_2^{(m_q, 0, m_q)} \right). \end{aligned} \quad (4.94)$$

with,

$$\begin{aligned} a &= \frac{(1 - 2\epsilon)m_q^2 + (1 - \epsilon)q^2}{3(1 - \epsilon)}, \\ b &= \frac{m_q^2(4m_q^2 - q^2)}{3(1 - \epsilon)}. \end{aligned} \quad (4.95)$$

4.5.1 Build and Integrate Results

To build and integrate the massive A_3^0 in *AntCalc*, we use the same commands as for the massless A_3^0 route, as discussed in Subsubsection 4.2.1, with the added option `quarkMass -> mQ`, where mQ is the quark mass, so far written here as m_q . The building process is completed as,

```
In[2] := BuildAntenna[A, 3, 0, quarkMass->mQ]
Out[2] :=
```

$$\left[-4(-2 + \epsilon)m_q^4(s_{13}^2 + s_{23}^2) \right] + \dots$$

The integration process follows in the same usual way, noting that much like in all other integrated cases, the default output is given with $q^2 \rightarrow 1$, making here $r \rightarrow 1 - 4m_q^2$,

```
In[3] := BuildAndIntegrateAntenna[A, 3, 0, quarkMass->mQ,
ExpansionOrder->0]
Out[3] :=
```

$$\frac{1}{2100m_q^2(1 + 2m_q^2)^2 \pi^2} \left[\frac{1}{\epsilon} \left(70m_q^2(-1 + 2m_q^2 + 8m_q^4) {}_2F_1\left(\frac{1}{2}, 2; \frac{7}{2}; 1 - 4m_q^2\right) + \dots \right) \right. \\ \left. + \left(10(-1 + 2m_q^2 + 8m_q^4) {}_2F_1\left(\frac{1}{2}, 2; \frac{7}{2}; 1 - 4m_q^2\right) + \dots \right) \right] + O(\epsilon)$$

These expressions agree with the established literature results [24]. The full unintegrated and integrated antenna expressions were truncated in this section. They are shown completely in Appendix A.

4.6 Overview of Implemented Routes

Having covered both the massless antenna set and the exploratory massive extension, this section collects every implemented route in one place. Table 4.2 summarises how each route is constructed and integrated; Table 4.3 summarises the corresponding validation evidence and status. Together, they replace the individual validation statements given throughout this chapter with a single point of reference.

Route	Process / Scope	API Call ²	Construction Check	Integration Route
A_2^0	$\gamma^* \rightarrow q\bar{q}$, LO	[A, 2, 0]	Self-interference, Born-normalised reference	Trivial (reference, $A_2^0 \equiv 1$)
A_3^0	$\gamma^* \rightarrow qg\bar{q}$, NLO real	[A, 3, 0]	Self-interference; colour-coefficient extraction; matched to by-hand derivation	IBP / <i>LiteRed2</i> , x30 basis; MI: R_3
A_2^1	$\gamma^* \rightarrow q\bar{q}$, NLO virtual	[A, 2, 1]	Tree-loop interference; PaVe-reduced scalar extraction	PaVe / <i>FeynHelpers</i> ; MI: B_0, C_0
A_3^1 set ($A_3^1, \tilde{A}_3^1, \hat{A}_3^1$)	$\gamma^* \rightarrow qg\bar{q}$, NNLO, one loop	[A, 3, 1]	Tree-loop interference; colour-coefficient extraction (leading/subleading/quark-loop)	IBP / <i>LiteRed2</i> , A31 basis; build-side MI are PaVe scalars (B_0, C_0, D_0 , Table 4.1); integration MI: $V_{5,a}, V_{5,b}, V_8$

Table 4.2 continued.

Route	Process / Scope	API Call	Construction Check	Integration Route
A_2^2 set ($A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$)	$\gamma^* \rightarrow q\bar{q}$, NNLO, two loops	[A, 2, 2]	Two-loop tree interference and one-loop self-interference (two source contributions)	IBP / <i>LiteRed2</i> , A22TwoLoopTree/A22OneLoopSelf bases; MI: $A_{22,LO}$, A_3 , A_4 (also A_6 for $\tilde{A}_2^2$)
$A_4^0, \tilde{A}_4^0$	$\gamma^* \rightarrow qgg\bar{q}$, NNLO, double-real	[A, 4, 0]	Colour-ordered amplitude; colour-coefficient extraction	IBP / <i>LiteRed2</i> , X40 basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$
B_4^0	$\gamma^* \rightarrow q\bar{q}q'\bar{q}'$, NNLO	[B, 4, 0]	Sector self-interference (direct amplitude only)	IBP / <i>LiteRed2</i> , X40 basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$
C_4^0	$\gamma^* \rightarrow q\bar{q}q\bar{q}$, NNLO	[C, 4, 0]	Sector symmetrised interference (direct and exchange)	IBP / <i>LiteRed2</i> , X40 basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$
Massive A_3^0	$\gamma^* \rightarrow q(m_q)g\bar{q}(m_q)$, exploratory	[A, 3, 0, quarkMass->mQ]	Same route as massless A_3^0 , with the quarkMass option	IBP / <i>LiteRed2</i> , massive family; MI mapped from <i>LiteRed2</i> 's J_1, J_2 to the literature $I_1^{(m_q,0,m_q)}, I_2^{(m_q,0,m_q)}$

Table 4.2: Construction and integration overview of every antenna route implemented in *AntCalc*.

Route	Literature Reference (Order Checked)	Local Validation	Global Validation	Status
A_2^0	— (reference normalisation)	Not applicable	LO normalisation of the R -ratio	By construction
A_3^0	[4] (NLO)	Eikonal factor and AP splitting function reproduced	Feeds r_{NLO}	Pass
A_2^1	[4] (NLO)	Not applicable	Feeds r_{NLO}	Pass
A_3^1 set	[4, 23] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
A_2^2 set	[23] (NNLO)	Not applicable	Feeds r_{NNLO}	Pass
$A_4^0, \tilde{A}_4^0$	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
B_4^0	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
C_4^0	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
Massive A_3^0	[24]	Not performed	Not applicable (does not feed the massless R -ratio)	Literature agreement; no global observable validation yet

Table 4.3: Validation evidence and status of every antenna route implemented in *AntCalc*. “Not applicable” marks routes with no real unresolved parton to take a soft/collinear limit of (A_2^0, A_2^1, A_2^2); “Not performed” marks routes where such a limit exists but was not checked in this work — see Section 4.4 for why A_3^0 was chosen to demonstrate the method rather than repeating it on every route. The massive A_3^0 route is deliberately marked as less mature than the massless set: it agrees with the literature locally, but has not yet been checked against any global observable.

Chapter 5

Global Validation: The Massless Hadronic R-Ratio

5.1 R-Ratio Definition

The hadronic R-ratio is usually defined as the ratio of the total cross-section for e^+e^- annihilation into hadrons to the point-like cross-section for $e^+e^- \rightarrow \mu^+\mu^-$. It is mathematically described as follows,

$$\begin{aligned} R_{\text{hadr}} &= \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_{\text{em}}^2} \sigma(e^+e^- \rightarrow \text{hadrons}). \end{aligned} \quad (5.1)$$

Looking closely, the $\sigma(e^+e^- \rightarrow \text{hadrons})$ term is what QCD predicts for $\gamma^* \rightarrow \text{hadrons}$, computed via the optical theorem [11]. What makes the R-ratio a clean QCD observable, and a good benchmark for *AntCalc* is that the ratio features terms for higher perturbative orders as a perturbative series in α_s ,

$$R_{\text{hadr}} = \left(c_{\text{LO}} + \frac{\alpha_s}{\pi} c_{\text{NLO}} + \left(\frac{\alpha_s}{\pi} \right)^2 c_{\text{NNLO}} + \mathcal{O}(\alpha_s^3) \right) \quad (5.2)$$

here the perturbative coefficients, c_{LO} , c_{NLO} and c_{NNLO} , respectively receive contributions from the multiparticle final states at the relative orders via their respective antennae, as described in Chapter 2. For example, A_3^0 as described in Chapter 4 contributes to c_{NLO} . Moreover, the leading-order term $R^{(0)}$ simply counts the number of quark colours and flavours weighted by their electric charges,

$$R^{(0)} = N_c \sum_q e_q^2 \quad (5.3)$$

since $R_{\text{hadr}} = R^{(0)}(1 + \mathcal{O}(\alpha_s)) = R^{(0)}R(\alpha_s)$.

5.2 Global Validation with *AntCalc*

As discussed, the hadronic R-ratio is a very useful quantity to compute using the integrated antenna functions obtained from *AntCalc*. As such, the programme provides an R-ratio calculator. It is the longest-running function in the whole package. Its result gives,

```
In[2] := BuildRRatio[SMQCD]
```

```
Out[2] :=
```

$$1 + \left(\frac{\alpha_s}{2\pi} \right) \frac{3}{2} C_F + \left(\frac{\alpha_s}{2\pi} \right)^2 C_F \left[N \left(\frac{243}{16} - 11\zeta_3 \right) + \frac{3}{16N} + N_f \left(-\frac{11}{4} + 2\zeta_3 \right) \right]$$

As expected, the results for each perturbative order are finite. This validates the mutual normalisation and assembly of the integrated antennae: at each perturbative order, their infrared poles cancel in the inclusive sum, leaving a finite coefficient.

5.2.1 Assembly from Integrated Antennae

Using *AntCalc*, we are also able to compute the R-ratio step-by-step, without having to resort to the automated function by just computing the intermediate steps ourselves. As described in Subsection 2.4.3, antenna functions are purely kinematic. By taking all components, we are able to construct a process-wide object carrying both kinematic and colour information. These are T -objects; their unintegrated and integrated forms are denoted T and $\mathcal{T}$, respectively. Conceptually, the unintegrated T -object is defined as ¹,

$$T_{\text{topology}}^{2(k+1)} = (\text{colour/flavour weight}) \times T_{q\bar{q}}^2 \times (\text{colour-weighted sum of antennae at } k\text{-order}). \quad (5.4)$$

This construction takes a simple form for the A-type antenna relative T -objects [4]:

$$T_{\text{topology}}^{2(k+1)} = 2C_F T_{q\bar{q}}^2 \sum_a f_a A_{n,(k),a}^l \quad (5.5)$$

where a denotes the component (leading, subleading, fermion loop, etc), f_a is the colour weight, that is N , $1/N$, N_f , etc, and $A_{n,(k),a}^l$ is the relative component antenna with n final-state particles and l loops. For this antenna type, this definition needs two extensions at NNLO that do not cleanly fit into an algorithmic formula:

- The A_3^1 antenna set represents a particularly interesting process. Unlike in lower order loop processes, here two very distinct kinds of Feynman diagram appear: those in which the loop and the emission of the final-state gluon are completely separate events, and those in which the loop involves all three external legs. The first kind of diagram maps exactly a A_2^1 antenna topology followed by a A_3^0 topology: first the two hard quarks exchange a gluon and then one of the hard quarks emits a final-state gluon². Since this behaviour is not unique to the A_3^1 antenna, it must be subtracted from the bare result to retain the antenna's singular behaviour. When we then reconstruct the T -object, the $A_2^1 A_3^0$ term has to be added back on such that the process is considered in its totality [4].
- The A_2^2 antenna set, also at NNLO, is the first time we find a double-loop topology. Here the extension required is not a subtraction of some kind of behaviour, but rather an addition of a new one. This set is, in reality, composed of two different processes: the first follows the standard rule closely, as it produces a leading, a subleading and a fermion loop component. The second, however, has a singular antenna which describes the self-interference of the one-loop correction with itself, now that the system presents with two loops. This antenna is denoted by $\check{A}_2^2$. It is also the only genuinely square (not cross) term at NNLO. This means that it requires a normalisation of $(2C_F)^2$ [4].

As mentioned before, Eq. 5.5 is only valid for T -objects constructed from A-type antenna functions. In this work we consider all quark-antiquark antennae up to NNLO. This includes the B_4^0 and C_4^0 antennae introduced in Subsection 2.4.4 of Chapter 2. These antennae populate the $T_{q\bar{q}q'\bar{q}'}^6$ and $T_{q\bar{q}q\bar{q}}^6$ objects, respectively.

$T_{q\bar{q}q'\bar{q}'}^6$ is built in the same way as other single-topology antenna T -objects, like $T_{q\bar{q}g}^4$, with the only difference being that here all quark flavours are summed over, requiring the inclusion of a $(N_f - 1)$ term, constructing,

$$T_{q\bar{q}q'\bar{q}'}^6 = 2C_F T_{q\bar{q}}^2 (N_f - 1) B_4^0. \quad (5.6)$$

The identical-flavour quark-antiquark-quark-antiquark final state is described by the sum of antennae for non-identical-flavour and for identical-flavour-only [4]. Using Eq. 2.45 in the interference term, with the identity piece vanishing by tracelessness, we can describe the identical-flavour T -object as,

$$T_{q\bar{q}q\bar{q}}^6 = 2C_F T_{q\bar{q}}^2 \left[B_4^0 - \frac{1}{N} C_4^0 \right] \quad (5.7)$$

importantly without the inclusion of the quark-flavour sum term $(N_f - 1)$.

With this definition of the integrated antenna X_n^l (Eq. 2.65 of Chapter 2), we are now also able to introduce the often used integrated T -object as,

$$\mathcal{T}_{\text{topology}}^{2(k+1)} = 2C_F T_{q\bar{q}}^2 \sum_a f_a X_{n,(k),a}^l \quad (5.8)$$

written schematically from Eq. 5.5, though the structure may change with the antenna type used to define the relevant T -object. This equation describes that the integrated T -object, $\mathcal{T}_{\text{topology}}^{2(k+1)}$ is defined using the integrated antennae X_n^l , integrated over their respective phase-spaces rather than integrating the composite unintegrated T -object directly [4, 23].

¹The T -objects are commonly notated by T^2 , T^4 and T^6 , depending on their perturbative order: LO, NLO and NNLO, respectively. That is the same as the k -index, introduced in Subsection 2.4.2 of Chapter 2, incremented by one and then doubled, $2(k + 1)$. This is the notation used in [23] and the one that will be used for the rest of the present work.

²This is merely a conceptual, diagrammatic interpretation, as the actual process is much less sequential than this description suggests.

The R-ratio is assembled using these $\mathcal{T}$ -objects. We simply sum the normalised objects order-by-order as,

$$\begin{aligned}
R &= \frac{R_{\text{hadr}}}{R^{(0)}} = R_{\text{LO}} + R_{\text{NLO}} + R_{\text{NNLO}} \\
&= r_{\text{LO}} + \frac{\alpha_s}{2\pi} r_{\text{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} \\
&= 1 + \frac{\alpha_s}{2\pi} \frac{\mathcal{T}_{q\bar{q}}^4 + \mathcal{T}_{q\bar{q}g}^4}{\mathcal{T}_{q\bar{q}}^2} + \left(\frac{\alpha_s}{2\pi}\right)^2 \frac{\mathcal{T}_{q\bar{q}}^6 + \mathcal{T}_{q\bar{q}g}^6 + \mathcal{T}_{q\bar{q}q\bar{q}}^6 + \mathcal{T}_{q\bar{q}q'q'}^6 + \mathcal{T}_{q\bar{q}gg}^6}{\mathcal{T}_{q\bar{q}}^2} + \mathcal{O}(\alpha_s^3).
\end{aligned} \tag{5.9}$$

Expanding the $\mathcal{T}$ -objects, we are able to describe the R-ratio formula using the integrated antenna functions as,

$$\begin{aligned}
r_{\text{LO}} &= 1 \\
r_{\text{NLO}} &= 2C_F \left(\mathcal{A}_2^1 + \mathcal{A}_3^0 \right) \\
r_{\text{NNLO}} &= 2C_F \left[N \left(\mathcal{A}_2^1 \mathcal{A}_3^0 + \mathcal{A}_3^1 + \mathcal{A}_4^0 + \mathcal{A}_2^2 \right) - \frac{1}{N} \left(\mathcal{A}_2^1 \mathcal{A}_3^0 + \tilde{\mathcal{A}}_3^1 + \tilde{\mathcal{A}}_4^0 + C_4^0 - \tilde{\mathcal{A}}_2^2 \right) \right. \\
&\quad \left. + N_f \left(\mathcal{B}_4^0 + \hat{\mathcal{A}}_3^1 + \hat{\mathcal{A}}_2^2 \right) + 2C_F \tilde{\mathcal{A}}_2^2 \right].
\end{aligned} \tag{5.10}$$

Assembling these pieces together, using the integrated antenna results from *AntCalc*, we finally get,

$$R = 1 + \left(\frac{\alpha_s}{2\pi}\right) \frac{3}{2} C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F \left[N \left(\frac{243}{16} - 11\zeta_3 \right) + \frac{3}{16N} + N_f \left(-\frac{11}{4} + 2\zeta_3 \right) \right] \tag{5.11}$$

which is the exact massless hadronic R-ratio coefficient through NNLO [23] and is also exactly the expression we get by running `BuildRRatio[SMQCD]`.

Chapter 6

Future Work

In its current version, *AntCalc* is implemented entirely in *Wolfram Mathematica*. This is the right language to build a trial-stage programme of its nature due to its forgiving mathematical environment and its general adoption throughout the field. But this choice of language comes at a cost: efficiency. Though convenient, *Mathematica* is slow and memory-hungry. In its current state, running the full `BuildRRatio` call takes one and a half to two hours and, on machines with less RAM, may cause the calculation to crash halfway through. It is then apparent that the next step in *AntCalc*'s development is taking it mostly out of the *Mathematica* environment and making it a truly modular, multi-language programme that exploits only the best characteristics of each tool, including *Mathematica*, keeping the current high-level workflow. The current plan for this step is to make the orchestrator a *Python* programme and to store the profiles in plain, machine-readable files, such as those in the JSON format. Then, retain *FeynArts* and *FeynCalc* for diagram generation and amplitude manipulation, respectively, but hand the algebraic manipulations required to obtain the antenna, or its components, over to *FORM*. Then, lastly, make IBP reduction faster and more stable by replacing *LiteRed2* with *Kira*, written in C++, keeping *Mathematica* as an algebraic simplifier.

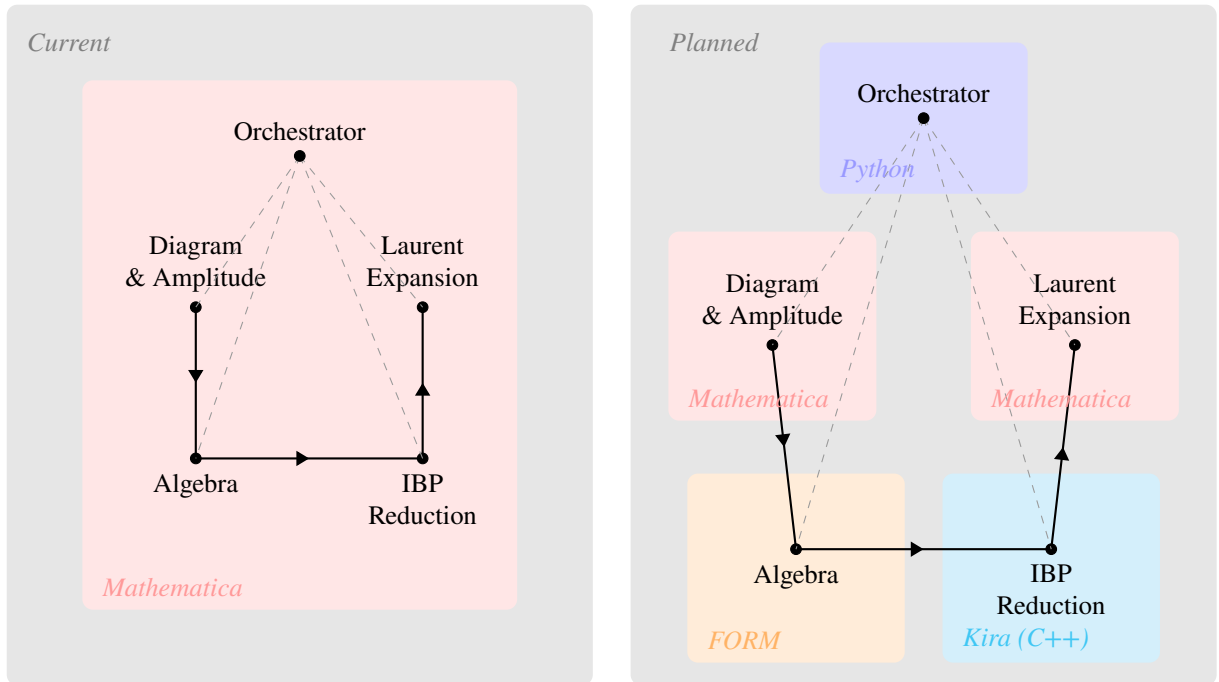


Figure 6.1: Diagram showing the current and planned workflow steps relative to their programming languages.

This step will then allow us to keep expanding *AntCalc*'s profile library without fear of hitting a computational bottleneck. The first point of expansion would be to the rest of the massive quark-antiquark final-final antenna functions, especially A_4^0 , B_4^0 and C_4^0 at first, and then A_3^1 and A_2^2 . After these are completed, expanding the scope to other relevant antenna types, particularly the quark-gluon (D and E) and the gluon-gluon (F , G and H) antennae, both in the massless and in the massive regimes becomes the logical next step. Lastly, extensions to kinematics other than final-final, namely initial-initial and initial-final configurations, and higher perturbative orders, particularly $N^3\text{LO}$, antennae would also be relevant.

A further, more structural, direction is to generalise *AntCalc* from a validated library of known antenna functions into a construction tool for antenna functions that have not yet been computed at all. The profile-driven architecture described in Chapter 3 already separates the process-specific description of a route, its profile, from the shared build and integration workflows; letting a user supply their own process description, rather than only selecting an existing route key, is a natural continuation of that design rather than a new one. Such user-defined routes would not, by definition, have a published result to validate against. This is precisely the gap that the amplitude-level Ward-identity check described in Subsection 3.7.1 is suited to: since it certifies gauge invariance directly from the generated amplitude, independently of any external reference, it, and checks like it, could be run automatically on a newly constructed antenna as a first, literature-independent certificate of correctness, before any attempt is made to compare it with an established result.

Connections to other established tools would also be pertinent. A connection to a Feynman diagram generation tool such as *MadGraph* [29] and one to a parton-level event generator such as *NNLOJet* [30] could allow us to generate more complex antennae, from more complex topologies and also enable the numerical validation of the results.

Chapter 7

Conclusion

This thesis developed the *Mathematica* package *AntCalc* to remove a friction that recurs at the start of most new antenna-subtraction calculations: before the intended physics can even be pursued, a group must first reconstruct and integrate the antenna functions it needs from scratch, much as Feynman-diagram algebra once had to be carried out by hand before tools such as *FeynCalc* became standard. *AntCalc* removes this initial barrier, providing a validated, ready-to-use library of antenna functions that can be loaded directly into a *Mathematica* notebook, making their construction and everyday use easier, more standardised, and streamlined.

The resulting framework was implemented completely within *Mathematica*, using the *FeynCalc* environment (including *FeynArts* and *FeynHelpers*) to handle the building stage and *LiteRed2* for the IBP reduction in the integration stage. That framework was shown to produce all massless quark-antiquark final-final antennae through NNLO. It was also shown to be able to be extended to the massive regime by implementing the massive A_3^0 route in its entirety. It begins by using *FeynArts* to produce one or more Feynman diagrams relevant for the process under study and generates their amplitudes. It then develops the amplitude into one antenna or multiple antenna components using *FeynCalc*. The unintegrated antenna is then handed to *LiteRed2* for IBP reduction and, finally, after the pre-computed MI are identified and substituted, the integrated antenna is Laurent-expanded around the dimensional regularisation parameter ϵ .

These unintegrated and integrated antennae were shown to be in agreement with established results in the literature. Furthermore, the massless A_3^0 antenna, whose workflow was shown in great detail, was locally validated by computing the eikonal factor and the Altarelli–Parisi splitting function, using a method that may be used to validate other appropriate antennae. All massless integrated antenna results were also validated when we computed the massless hadronic R-ratio.

The package is beginning to reach the computational limits of its current workflow, running completely within *Mathematica*. As such, the most pressing piece of future work is the overhaul to a new, truly modular and multi-language, workflow, that is able to harness the best characteristics of different tools, like Kira for IBP reduction and FORM for algebraic manipulation, without compromising efficiency. Once this overhaul is done, *AntCalc* could be expanded to handle much more complex cases, particularly those in the massive regime, other antenna types, such as quark-gluon, other kinematic configurations, such as initial-final antennae, and those in higher perturbative orders, such as N³LO.

Appendix A

Complete Antenna Expressions

This appendix includes the complete expressions of the antennae computed using *AntCalc*. It shows the unintegrated and integrated antennae, along with the commands used to compute them.

A note on conventions: the unintegrated antenna functions below are given exactly as *AntCalc* returns them, in terms of s_{ij} and q^2 . The integrated antennae, however, are quoted at $q^2 = 1$, matching *IntegrateAntenna*'s default convention. This does not break generality as established in Subsection 2.5.2 and Eq. 2.87, a quantity of this kind scales with q^2 as a single power fixed by its mass dimension and ϵ -scaling, so the complete q^2 -dependent result is recovered by multiplying the expression given here by $(q^2)^{\lambda+\kappa\epsilon}$ for the appropriate λ and κ . The explicit case of $\mathcal{A}_3^0$, which scales as $(q^2)^{-\epsilon}$, is worked out in full in Eq. 4.83. We must also here introduce the shorthand,

$$C_\epsilon \equiv \frac{(4\pi)^\epsilon e^{-\gamma_E \epsilon}}{8\pi^2} = \frac{1}{C(\epsilon, 1)} \quad (\text{A.1})$$

as the one-loop normalisation factor.

A.1 LO Antenna

A.1.1 A_2^0

Running `BuildAntenna[A, 2, 0]`, we get the built result,

$$A_2^0 = 1. \quad (\text{A.2})$$

By running `BuildAndIntegrateAntenna[A, 2, 0]` we get the integrated result,

$$\mathcal{A}_2^0 = \frac{1}{4} + \epsilon \frac{9}{8} - \epsilon^2 \left(\frac{9}{16} - \frac{\pi^2}{48} \right). \quad (\text{A.3})$$

A.2 NLO Antennae

A.2.1 A_2^1

Running `BuildAntenna[A, 2, 0]`, we get the built result,

$$A_2^1 = -\frac{1}{2}(3 + 2\epsilon)B_0(q^2; 0, 0) - q^2 C_0(0, 0, q^2; 0, 0, 0). \quad (\text{A.4})$$

By running `BuildAndIntegrateAntenna[A, 2, 1]` we get the integrated result,

$$\mathcal{A}_2^1 = -\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon \left(\frac{7\zeta_3}{3} - 8 + \frac{7\pi^2}{8} \right) + \epsilon^2 \left(\frac{7\zeta_3}{2} - 16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} \right) + \mathcal{O}(\epsilon^3). \quad (\text{A.5})$$

A.2.2 A_3^0

Running `BuildAntenna[A, 3, 0]`, we get the built result,

$$A_3^0 = \frac{2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{q^2 s_{13} s_{23}} + \epsilon \frac{2}{q^2} + (\epsilon - 1) \frac{s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}}. \quad (\text{A.6})$$

By running `BuildAndIntegrateAntenna[A, 3, 0]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_3^0 = & \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon \left(-\frac{25\zeta_3}{3} + \frac{109}{8} - \frac{7\pi^2}{8} \right) \\ & + \epsilon^2 \left(-\frac{25\zeta_3}{2} + \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} \right) + \mathcal{O}(\epsilon^3). \end{aligned} \quad (\text{A.7})$$

A.3 NNLO Antennae

A.3.1 A_2^2

Running `BuildAntenna[A, 2, 2, Component->Leading]`, we get the built result,

$$A_2^2 = A_2^{2,(0)} + (-s_{12})^{-\epsilon} C_\epsilon A_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon} A_2^{2,(-2\epsilon)}. \quad (\text{A.8})$$

where,

$$A_2^{2,(0)} = \frac{1}{16\epsilon^4} - \frac{1}{\epsilon^2} \left(\frac{9}{32} - \frac{11\pi^2}{96} \right) - \frac{1}{\epsilon} \left(\frac{377}{64} - \frac{55\pi^2}{96} \right) + \frac{863}{384} - \frac{319\pi^2}{576} - \frac{13\zeta_3}{36} \quad (\text{A.9})$$

$$A_2^{2,(-\epsilon)} = -\frac{11}{6\epsilon^3} + \frac{11}{4\epsilon^2} + \frac{22}{3\epsilon} + \frac{44}{3} + \mathcal{O}(\epsilon) \quad (\text{A.10})$$

$$\begin{aligned} A_2^{2,(-2\epsilon)} = & \frac{3}{16\epsilon^4} + \frac{7}{24\epsilon^3} + \frac{1}{\epsilon^2} \left(\frac{155}{288} - \frac{37\pi^2}{96} \right) + \frac{1}{\epsilon} \left(\frac{5597}{1728} - \frac{355\pi^2}{288} + \frac{7\zeta_3}{12} \right) \\ & - \frac{193531}{10368} - \frac{577\pi^2}{1728} + \frac{263\pi^4}{1440} + \frac{73\zeta_3}{12}. \end{aligned} \quad (\text{A.11})$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Leading]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_2^2 = & \frac{1}{4\epsilon^2} + \frac{17}{8\epsilon^3} + \frac{1}{\epsilon^2} \left(\frac{433}{144} - \frac{\pi^2}{2} \right) + \frac{1}{\epsilon} \left(\frac{4045}{864} - \frac{83\pi^2}{48} + \frac{7\zeta_3}{12} \right) \\ & - \frac{9083}{5184} - \frac{2153\pi^2}{864} + \frac{263\pi^4}{1440} + \frac{13\zeta_3}{9} + \mathcal{O}(\epsilon). \end{aligned} \quad (\text{A.12})$$

A.3.2 $\tilde{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Subleading]`, we get the built result,

$$\tilde{A}_2^2 = \tilde{A}_2^{2,(0)} + s_{12}^{-2\epsilon} \tilde{A}_2^{2,(-2\epsilon)}. \quad (\text{A.13})$$

where,

$$\begin{aligned} \tilde{A}_2^{2,(0)} = & -\frac{3}{16\epsilon^4} - \frac{9}{16\epsilon^3} - \frac{1}{\epsilon^2} \left(-\frac{89}{32} + \frac{11\pi^2}{32} \right) + \frac{1}{\epsilon} \left(\frac{533}{64} - \frac{11\pi^2}{6} - \frac{13\zeta_3}{12} \right) \\ & - \frac{1803}{128} - \frac{341\pi^2}{192} + \frac{79\pi^4}{2880} - \frac{39\zeta_3}{8}. \end{aligned} \quad (\text{A.14})$$

$$\begin{aligned} \tilde{A}_2^{2,(-2\epsilon)} = & -\frac{1}{16\epsilon^4} - \frac{3}{16\epsilon^3} + \frac{1}{\epsilon^2} \left(\frac{7}{32} + \frac{19\pi^2}{96} \right) + \frac{1}{\epsilon} \left(-\frac{975}{64} + \frac{10\pi^2}{3} + \frac{15\zeta_3}{4} \right) \\ & - \frac{499}{128} + \frac{1291\pi^2}{192} - \frac{233\pi^4}{960} + \frac{97\zeta_3}{8}. \end{aligned} \quad (\text{A.15})$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Subleading]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_2^2 = & -\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} - \frac{1}{\epsilon^2} \left(\frac{41}{16} - \frac{13\pi^2}{24} \right) + \frac{1}{\epsilon} \left(-\frac{221}{32} + \frac{3\pi^2}{2} + \frac{8\zeta_3}{3} \right) \\ & - \frac{1151}{64} + \frac{475\pi^2}{96} - \frac{59\pi^4}{288} + \frac{29\zeta_3}{4} + \mathcal{O}(\epsilon).\end{aligned}\tag{A.16}$$

A.3.3 $\hat{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Nf]`, we get the built result,

$$\hat{A}_2^2 = (-s_{12})^{-\epsilon} C_\epsilon \hat{A}_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon} \hat{A}_2^{2,(-2\epsilon)}.\tag{A.17}$$

where,

$$\hat{A}_2^{2,(-\epsilon)} = \frac{1}{3\epsilon^3} - \frac{1}{2\epsilon^2} - \frac{4}{3\epsilon} - \frac{8}{3}\tag{A.18}$$

$$\hat{A}_2^{2,(-2\epsilon)} = \frac{1}{12\epsilon^3} + \frac{7}{18\epsilon^2} + \frac{1}{\epsilon} \left(\frac{353}{216} - \frac{11\pi^2}{72} \right) + \frac{7541}{1296} - \frac{77\pi^2}{108} - \frac{13\zeta_3}{18}.\tag{A.19}$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Nf]` we get the integrated result,

$$\mathcal{A}_2^2 = -\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{1}{\epsilon} \left(\frac{65}{216} + \frac{\pi^2}{24} \right) + \frac{4085}{1296} - \frac{91\pi^2}{216} + \frac{\zeta_3}{18} + \mathcal{O}(\epsilon).\tag{A.20}$$

A.3.4 $\check{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Breve]`, we get the built result,

$$\check{A}_2^2 = \frac{(-2 + \epsilon - 2\epsilon^2)\Gamma^6(1 - \epsilon)\Gamma^2(1 + \epsilon)(120 - 20\pi^2\epsilon^2 + 80(13 - 42\epsilon)\zeta_3\epsilon^3 + \pi^4\epsilon^4)}{1920\epsilon^4\Gamma^2(2 - 2\epsilon)}.\tag{A.21}$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Breve]` we get the integrated result,

$$\begin{aligned}\check{\mathcal{A}}_2^2 = & \frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2} \left(\frac{41}{16} - \frac{\pi^2}{24} \right) + \frac{1}{\epsilon} \left(7 - \frac{\pi^2}{8} + \frac{7\zeta_3}{6} \right) \\ & + 18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} + \mathcal{O}(\epsilon).\end{aligned}\tag{A.22}$$

A.3.5 A_3^1

Running `BuildAntenna[A, 3, 1, Component->Leading]`, we get the built result,

$$\begin{aligned}A_3^1 = & \frac{1}{2(\epsilon - 1)q^2} \left((-s_{13})^{-\epsilon} C_\epsilon A_3^{1,(s_{13})} + (-s_{23})^{-\epsilon} C_\epsilon A_3^{1,(s_{23})} + (-q^2)^{-\epsilon} C_\epsilon A_3^{1,(q^2)} \right. \\ & \left. + \left[\frac{\pi^2 q^2}{4} + \frac{1}{4} \log^2 \left(\frac{s_{13}}{s_{23}} \right) + \text{Li}_2 \left(1 - \frac{q^2}{s_{13}} \right) + \text{Li}_2 \left(1 - \frac{q^2}{s_{23}} \right) \right] A_3^{1,(\text{Trans.})} \right).\end{aligned}\tag{A.23}$$

where,

$$A_3^{1,(s_{13})} = \mathcal{F}_1(s_{12}; s_{13}, s_{23}), \quad A_3^{1,(s_{23})} = \mathcal{F}_1(s_{12}; s_{23}, s_{13})\tag{A.24}$$

$$\begin{aligned}\mathcal{F}_1(s_{12}; x, y) = & \frac{1}{4(q^2)^2 (s_{12} + y)^2} \left[\frac{1}{\epsilon} \left(10s_{12}^2 + 4s_{12}(x + 4y) + 3y(x + 2y) \right) \right. \\ & \left. - \frac{1}{x} \left(s_{12}^3 - s_{12}^2(23x - 3y) - s_{12} \left(13x^2 + 36xy - 3y^2 \right) - 10x^2y - 13xy^2 + y^3 \right) \right]\end{aligned}\tag{A.25}$$

$$A_3^{1,(q^2)} = \frac{\mathcal{P} \left(20s_{12}^2(q^2)^2 + 28s_{12}q^2s_{13}s_{23} + 11s_{13}^2s_{23}^2 \right)}{12\epsilon(q^2)^2 s_{13}s_{23} (s_{12}q^2 + 13s_{23})^2} + \mathcal{O}(\epsilon), \quad \mathcal{P} = (q^2)^2 + s_{12}^2 - 2s_{13}s_{23} \quad (\text{A.26})$$

$$A_3^{1,(\text{Trans.})} = -\frac{1}{2q^2} A_3^0. \quad (\text{A.27})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Leading]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_3^1 = & -\frac{1}{4\epsilon^4} - \frac{31}{12\epsilon^3} - \frac{1}{\epsilon^2} \left(\frac{53}{8} - \frac{11\pi^2}{24} \right) - \frac{1}{\epsilon} \left(\frac{647}{24} - \frac{22\pi^2}{9} - \frac{23\zeta_3}{3} \right) \\ & - \frac{5231}{48} + \frac{17\pi^2}{2} - \frac{41\pi^4}{480} + \frac{689\zeta_3}{18} + \mathcal{O}(\epsilon). \end{aligned} \quad (\text{A.28})$$

A.3.6 $\tilde{A}_3^1$

Running `BuildAntenna[A, 3, 1, Component->Subleading]`, we get the built result,

$$\begin{aligned} \tilde{A}_3^1 = & \frac{\mathcal{R}(\epsilon, s_{13}, s_{23})}{2(\epsilon-1)q^2} \left[(-s_{12}^{-\epsilon}) \tilde{A}_3^{1,(s_{12})} + (-s_{13})^{-\epsilon} \tilde{A}_3^{1,(s_{13})} + (-s_{23})^{-\epsilon} \tilde{A}_3^{1,(s_{23})} + \right. \\ & + \frac{(-q^2)^{-\epsilon}}{(s_{12}+s_{13})^3(s_{12}+s_{23})^3} \tilde{A}_3^{1,(q^2)} + \epsilon^2(2\epsilon-1) \left(\frac{\pi^2}{4} + \text{Li}_2 \left(1 - \frac{q^2}{s_{12}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{12})} \\ & \left. + \left(\text{Li}_2 \left(1 - \frac{q^2}{s_{13}} \right) + \frac{1}{4} \log^2 \left(\frac{s_{12}}{s_{13}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{13})} + \left(\text{Li}_2 \left(1 - \frac{q^2}{s_{23}} \right) + \frac{1}{4} \log^2 \left(\frac{s_{12}}{s_{23}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{23})} \right] \quad (\text{A.29}) \\ \mathcal{R}(\epsilon, s_{13}, s_{23}) = & \frac{C\epsilon}{\epsilon^2(2\epsilon-1)s_{13}s_{23}} \end{aligned}$$

where, using $u = s_{13} + s_{23}$ and $v = s_{13}s_{23}$,

$$\begin{aligned} \tilde{A}_3^{1,(s_{12})} = & -4s_{12}^2u^2 + 4\epsilon \left(3s_{12}^2u^2 - s_{12}^2v + s_{12}u^3 - 2s_{12}uv + u^4 - u^2v \right) \\ & - 2\epsilon^2u \left(4s_{12}^2u + 4s_{12}u^2 - 4s_{12}v + 7u^3 - 6uv \right) + 2\epsilon^3u^2 \left(5u^2 - 4v \right) + 4\epsilon^4u^4 \end{aligned} \quad (\text{A.30})$$

$$\tilde{A}_3^{1,(s_{13})} = \mathcal{G}(s_{12}, s_{13}, s_{23}), \quad \tilde{A}_3^{1,(s_{23})} = \mathcal{G}(s_{12}, s_{23}, s_{13}) \quad (\text{A.31})$$

$$\begin{aligned} \mathcal{G}(s_{12}, x, y) = & -2(s_{12}+y)^2 [2(s_{12}+y)^2 - 2y(s_{12}+y) + y^2] + \epsilon \left(12(s_{12}+y)^4 \right. \\ & + 4(s_{12}+y)^3(x-3y) + 4(s_{12}+y)^2(x^2-xy+2y^2) - 2xy(s_{12}+y)(x-2y) + x^2y^2 \Big) \\ & - \epsilon^2 \left(8(s_{12}+y)^4 + (s_{12}+y)^3(8x-7y) + (s_{12}+y)^2(14x^2+xy+10y^2) + xy(s_{12}+y)(x+6y) \right. \\ & + x^2y^2 \Big) + \epsilon^3(s_{12}+y) \left(y(s_{12}+y)^2 + (s_{12}+y)^2 + (s_{12}+y)(10x^2+14xy+5y^2) \right. \\ & \left. + 5x^2y \right) + \epsilon^4(s_{12}+y)(x+y) \left((s_{12}+y)(4x-y) + 2xy \right) \end{aligned} \quad (\text{A.32})$$

$$\begin{aligned} \tilde{A}_3^{1,(q^2)} = & 2s_{12}^5(3u^2-2v) + 12s_{12}^4u(u^2-v) + s_{12}^3(9u^4-10u^3v-8v^2) + s_{12}^2u(3u^4-16v^2) \\ & + 4s_{12}v(u^4-2u^2v-v^2) + 2uv^2(u^2-2v^2) - \epsilon u \left(2s_{12}^5u + 4s_{12}^4(u^2-v) + 2s_{12}^3u(3u^2-5v) \right. \\ & \left. + 4s_{12}^2(u^4-v^2) + s_{12}uv(9u^2-8v) + 2v^2(3u^2-2v) \right) \end{aligned} \quad (\text{A.33})$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{12})} = 4s_{12}^2 + 2s_{12}(x+y) + x^2 + y^2 - 4\epsilon \left(s_{12}^2 + s_{12}(x+y) + x^2 + xy + y^2 \right)$$

$$+\epsilon^2\left(4(x+y)^2-2xy\right)+\epsilon^3(x+y)^2 \quad (\text{A.34})$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{13})} = \mathcal{H}(s_{12}, s_{13}, s_{23}), \quad \tilde{A}_3^{1,(\text{Trans. } s_{23})} = \mathcal{H}(s_{12}, s_{23}, s_{13}) \quad (\text{A.35})$$

$$\begin{aligned} \mathcal{H}(s_{12}, x, y) &= s_{12}^2 + (s_{12} + y)^2 - 2\epsilon\left(s_{12}^2 + s_{12}(x + y) + x^2 + xy + y^2\right) \\ &\quad + \epsilon^2\left(3x(x + y) + y^2\right) + \epsilon^3x(x + y). \end{aligned} \quad (\text{A.36})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Subleading]` we get the integrated result,

$$\tilde{\mathcal{A}}_3^1 = -\frac{1}{\epsilon^2}\left(\frac{5}{8} - \frac{\pi^2}{6}\right) - \frac{1}{\epsilon}\left(\frac{19}{4} - \frac{\pi^2}{4} - 7\zeta_3\right) - \frac{105}{4} + \frac{27\pi^2}{16} + \frac{7\pi^4}{90} + \frac{27\zeta_3}{2} + \mathcal{O}(\epsilon). \quad (\text{A.37})$$

A.3.7 $\hat{A}_3^1$

Running `BuildAntenna[A, 3, 1, Component->Nf]`, we get the built result,

$$\begin{aligned} \hat{A}_3^1 &= \frac{4s_{12}^2}{3s_{13}s_{23}} + \frac{4s_{12}}{3s_{13}} + \frac{4s_{12}}{3s_{23}} + \frac{4s_{23}}{3s_{13}} + \frac{4s_{13}}{3s_{23}} + \frac{4}{3} \\ &\quad - \frac{1}{\epsilon}\left(\frac{4s_{12}^2}{3s_{13}s_{23}} + \frac{4s_{12}}{3s_{23}} + \frac{2s_{23}}{3s_{13}} + \frac{2s_{13}}{3s_{23}}\right) \\ &\quad - \epsilon\left(\frac{2s_{13}}{3s_{23}} + \frac{2s_{23}}{3s_{13}} + \frac{4}{3}\right). \end{aligned} \quad (\text{A.38})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Nf]` we get the integrated result,

$$\tilde{\mathcal{A}}_3^1 = \frac{1}{3\epsilon^3} + \frac{1}{2\epsilon^2} + \frac{1}{\epsilon}\left(\frac{19}{12} - \frac{7\pi^2}{36}\right) + \frac{109}{24} - \frac{7\pi^2}{24} - \frac{25\zeta_3}{9} + \mathcal{O}(\epsilon). \quad (\text{A.39})$$

A.3.8 A_4^0

Running `BuildAntenna[A, 4, 0, Component->Leading]`, we get the built result,

$$A_4^0 = \frac{1}{2(\epsilon - 1)q^2} \left(A_4^{0,(0)} + \epsilon A_4^{0,(1)} + \epsilon^2 A_4^{0,(2)} + \epsilon^3 A_4^{0,(3)} \right) \quad (\text{A.40})$$

where,

$$\begin{aligned} A_4^{0,(0)} &= \frac{2s_{12}s_{14}}{s_{13}s_{134}^2} + \frac{s_{14}s_{23}}{s_{13}s_{134}^2} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}^2} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{s_{24}}{s_{134}s_{234}} - \frac{5s_{24}}{s_{134}^2} \\ &\quad - \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} - \frac{2s_{34}}{s_{13}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{4s_{12}^2s_{34}}{s_{13}s_{134}s_{234}s_{24}} \\ &\quad - \frac{s_{12}s_{34}}{s_{234}^2s_{24}} - \frac{2s_{13}s_{34}}{s_{234}^2s_{24}} - \frac{s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{2s_{12}}{s_{13}s_{134}} + \frac{2}{s_{134}} - \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} \\ &\quad - \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{s_{13}}{s_{134}s_{234}} - \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{2}{s_{234}} - \frac{4s_{14}}{s_{13}s_{24}} - \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\ &\quad + \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{3s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{13}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{234}s_{24}} - \frac{3s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\ &\quad - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} + \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{6s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\ &\quad - \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} - \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}} \\ &\quad - \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}}{s_{234}^2s_{24}} - \frac{4s_{13}s_{23}}{s_{234}^2s_{24}} + \frac{s_{14}s_{23}}{s_{234}^2s_{24}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} \\ &\quad - \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} - \frac{s_{14}}{s_{134}s_{34}} - \frac{6s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} - \frac{3s_{23}}{s_{134}s_{34}} \end{aligned}$$

$$\begin{aligned}
& -\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} - \frac{6s_{14}s_{24}}{s_{134}^2s_{34}} + \frac{2s_{24}}{s_{134}s_{34}} - \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& - \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{3s_{12}s_{24}}{s_{234}^2s_{34}} - \frac{4s_{13}s_{24}}{s_{234}^2s_{34}} \\
& + \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{234}s_{34}} + \frac{2s_{13}}{s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} \\
& - \frac{3s_{14}}{s_{234}s_{34}} + \frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{6s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{23}}{s_{234}s_{34}} - \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} + \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} - \frac{s_{13}^2}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{3s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} \\
& - \frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{2s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} \\
& - \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{3s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}} \\
& + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} - \frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}} \\
& - \frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{3s_{12}s_{13}}{s_{134}^2s_{34}} \\
& - \frac{6s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{6s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2}{s_{34}} - \frac{3s_{12}}{s_{134}^2} - \frac{3s_{12}}{s_{234}^2} - \frac{5s_{13}}{s_{234}^2} - \frac{3s_{14}^2}{s_{134}s_{34}^2} \\
& - \frac{5s_{12}s_{14}^2}{s_{134}^2s_{34}^2} - \frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{3s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{6s_{12}s_{14}}{s_{134}s_{34}^2} \\
& - \frac{s_{13}s_{14}}{s_{134}s_{34}^2} - \frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{4s_{14}}{s_{34}^2} + \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{34}^2} - \frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} - \frac{3s_{14}s_{23}}{s_{134}s_{34}^2} + \frac{s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& - \frac{s_{13}s_{23}}{s_{134}s_{34}^2} + \frac{4s_{23}}{s_{34}^2} - \frac{3s_{14}^2s_{24}}{s_{134}s_{34}^2} + \frac{4s_{14}s_{24}}{s_{134}s_{34}^2} + \frac{2s_{12}s_{24}}{s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{14}s_{24}}{s_{234}s_{34}^2} + \frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{23}s_{24}}{s_{234}s_{34}^2} \\
& - \frac{3s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{6s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{12}s_{13}}{s_{134}s_{34}^2} \\
& + \frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2} + \frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2} + \frac{3s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} - \frac{3s_{23}^2}{s_{234}s_{34}^2} + \frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2} + \frac{3s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{6s_{12}s_{23}}{s_{234}s_{34}^2} + \frac{4s_{13}s_{23}}{s_{234}s_{34}^2} + \frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{3s_{14}s_{23}}{s_{234}s_{34}^2} \\
& - \frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2} - \frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2} + \frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2} + \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} \\
& - \frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2} + \frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2} - \frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2} + \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2} \\
& - \frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{3s_{14}s_{34}^3}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{2s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
& + \frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} - \frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{3s_{14}s_{34}^3}{s_{234}^2s_{24}s_{34}^2} \\
& - \frac{s_{12}s_{13}}{s_{134}^2s_{34}^2} - \frac{5s_{12}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned}$$

(A.41)

$$\begin{aligned}
A_4^{0,(1)} = & \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} + \frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} \\
& + \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} - \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{24}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{14}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}}{s_{13}s_{134}} + \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{6s_{12}}{s_{134}s_{234}} + \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} + \frac{5s_{12}s_{23}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}}{s_{13}s_{24}} + \frac{5s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\
& + \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} + \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}}{s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{234}^2s_{24}} + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} \\
& + \frac{4s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{5s_{12}s_{24}}{s_{234}^2s_{34}} - \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} \\
& - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} - \frac{4s_{12}}{s_{234}s_{34}} \\
& + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{5s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{4s_{12}s_{23}}{s_{234}^2s_{34}} + \frac{5s_{12}}{s_{134}^2} \\
& + \frac{5s_{12}}{s_{234}^2} + \frac{6s_{12}s_{14}^2}{s_{134}^2s_{34}^2} + \frac{2s_{12}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{6s_{12}s_{14}}{s_{134}s_{34}^2} + \frac{8s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{24}}{s_{234}s_{34}^2} \\
& - \frac{2s_{12}s_{13}}{s_{134}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{6s_{12}s_{23}}{s_{234}s_{34}^2} - \frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2} \\
& - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{12}s_{13}^2}{s_{134}^2s_{34}^2} + \frac{6s_{12}s_{23}^2}{s_{234}^2s_{34}^2} \\
& + \frac{8s_{14}}{s_{13}s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{8s_{23}}{s_{13}s_{134}} + \frac{8s_{14}s_{24}}{s_{13}s_{134}^2} + \frac{4s_{24}}{s_{134}s_{234}} + \frac{14s_{24}}{s_{134}^2} \\
& + \frac{6s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} + \frac{6s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{6s_{13}s_{34}}{s_{234}^2s_{24}} + \frac{4}{s_{134}} + \frac{2s_{14}}{s_{134}s_{234}} \\
& + \frac{2s_{23}}{s_{134}s_{234}} + \frac{4s_{13}}{s_{134}s_{234}} + \frac{4}{s_{234}} + \frac{8s_{14}}{s_{13}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} + \frac{8s_{23}}{s_{13}s_{24}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{8s_{14}}{s_{234}s_{24}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{8s_{23}}{s_{234}s_{24}} + \frac{8s_{13}s_{23}}{s_{234}^2s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{24}} \\
& + \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} + \frac{2s_{24}^2}{s_{134}s_{234}s_{34}} + \frac{7s_{14}}{s_{134}s_{34}} + \frac{s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} + \frac{s_{14}s_{23}}{s_{134}^2s_{34}} \\
& + \frac{5s_{23}}{s_{134}s_{34}} + \frac{8s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} + \frac{11s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{24}}{s_{134}s_{34}} + \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{6s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{24}}{s_{234}s_{34}} + \frac{11s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{11s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{13}}{s_{134}s_{34}} - \frac{2s_{13}}{s_{234}s_{34}} \\
& + \frac{5s_{14}}{s_{234}s_{34}} - \frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} + \frac{7s_{23}}{s_{234}s_{34}} + \frac{2s_{13}^2}{s_{134}s_{234}s_{34}} \\
& + \frac{8s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} \\
& - \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}} + \frac{8s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{11s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2}{s_{34}} \\
& + \frac{14s_{13}}{s_{234}^2} + \frac{3s_{14}^2}{s_{134}s_{34}^2} + \frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{5s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{3s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{s_{13}s_{14}}{s_{134}s_{34}^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{4s_{14}}{s_{34}^2} - \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{3s_{14}s_{23}}{s_{134}s_{34}^2} - \frac{3s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& + \frac{s_{13}s_{23}}{s_{134}s_{34}^2} - \frac{4s_{23}}{s_{34}^2} + \frac{3s_{14}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{4s_{14}s_{24}}{s_{134}s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{s_{14}s_{24}}{s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{4s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{s_{23}s_{24}}{s_{234}s_{34}^2} + \frac{5s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{3s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{3s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
& + \frac{3s_{23}^2}{s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{4s_{13}s_{23}}{s_{234}s_{34}^2} - \frac{4s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{3s_{14}s_{23}}{s_{234}s_{34}^2} \\
& + \frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2} - \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2} \\
& - \frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{2s_{14}^2s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} \\
& - \frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} - \frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2} + \frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned} \tag{A.42}$$

$$\begin{aligned}
A_4^{0,(2)} = & -\frac{s_{13}^2}{s_{134}s_{234}s_{34}} - \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{s_{12}s_{13}^2}{s_{134}^2s_{34}^2} - \frac{6s_{13}s_{34}}{s_{234}^2s_{24}} - \frac{7s_{13}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{13}s_{23}}{s_{234}^2s_{24}} - \frac{12s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{7s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{7s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{s_{13}}{s_{134}s_{34}} + \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} - \frac{5s_{13}s_{23}}{s_{234}^2s_{34}} - \frac{11s_{13}}{s_{234}^2} \\
& - \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{7s_{24}}{s_{134}s_{234}} \\
& - \frac{11s_{24}}{s_{134}^2} - \frac{2s_{34}}{s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{8s_{34}}{s_{234}s_{24}} - \frac{3s_{12}s_{34}}{s_{234}^2s_{24}} - \frac{11}{s_{134}} - \frac{4s_{14}}{s_{134}s_{234}} \\
& - \frac{4s_{23}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{11}{s_{234}} - \frac{s_{12}}{s_{134}s_{24}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\
& - \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{8s_{23}}{s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{234}^2s_{24}} + \frac{3s_{14}s_{23}}{s_{234}^2s_{24}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} - \frac{5s_{14}}{s_{134}s_{34}} + \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} \\
& - \frac{s_{14}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{23}}{s_{134}s_{34}} - \frac{5s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{24}}{s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}}{s_{234}s_{34}} + \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{5s_{23}}{s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} - \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{3s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} \\
& + \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{4s_{12}}{s_{134}} - \frac{4s_{12}}{s_{234}} - \frac{s_{12}s_{14}^2}{s_{134}^2s_{34}^2} - \frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2} \\
& - \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{s_{12}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{2s_{12}s_{24}^2}{s_{13}s_{134}s_{234}s_{24}} - \frac{8s_{14}}{s_{13}s_{134}} \\
& + \frac{3s_{12}s_{14}}{s_{13}s_{134}} + \frac{3s_{14}s_{23}}{s_{13}s_{134}} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{6s_{24}s_{34}}{s_{13}s_{134}} \\
& - \frac{8s_{34}}{s_{13}s_{134}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{14}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{6s_{34}}{s_{13}s_{24}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{3s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}}{s_{13}s_{234}} - \frac{2s_{12}s_{14}}{s_{13}s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{3s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\
& -\frac{4s_{14}}{s_{13}s_{24}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{2s_{12}s_{14}}{s_{13}s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{13}s_{234}s_{24}} \\
& -\frac{s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} - \frac{3s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} - \frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{13}s_{234}s_{34}} - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} \\
& - \frac{3s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned} \tag{A.43}$$

$$\begin{aligned}
A_4^{0,(3)} &= \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{12}}{s_{134}s_{234}} \\
&+ \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{2s_{24}}{s_{134}s_{234}} + \frac{2s_{24}}{s_{134}^2} + \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} \\
&+ \frac{4s_{34}}{s_{134}s_{234}} + \frac{2s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{2s_{13}s_{34}}{s_{234}^2s_{24}} + \frac{2}{s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} \\
&+ \frac{2}{s_{234}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{24}} + \frac{2s_{13}}{s_{234}^2}.
\end{aligned} \tag{A.44}$$

By running `BuildAndIntegrateAntenna[A, 4, 0, Component->Leading]` we get the integrated result,

$$\begin{aligned}
\mathcal{A}_4^0 &= \frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} + \frac{434 - 39\pi^2}{36\epsilon^2} - \frac{1}{\epsilon} \left(-\frac{43223}{864} + \frac{589\pi^2}{144} + \frac{71\zeta_3}{4} \right) \\
&+ \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon).
\end{aligned} \tag{A.45}$$

A.3.9 $\tilde{A}_4^0$

Running `BuildAntenna[A, 4, 0, Component->Subleading]`, we get the built result,

$$\tilde{A}_4^0 = \frac{1}{2(\epsilon - 1)q^2} \left(\tilde{A}_4^{0,(0)} + \epsilon \tilde{A}_4^{0,(1)} + \epsilon^2 \tilde{A}_4^{0,(2)} + \epsilon^3 \tilde{A}_4^{0,(3)} \right) \tag{A.46}$$

where,

$$\begin{aligned}
\tilde{A}_4^{0,(0)} &= -\frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} + \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} \right. \\
&- \frac{2s_{14}^2(2s_{12} - 3s_{23} + 2s_{24})}{s_{13}s_{134}^2s_{34}} - \frac{2(3s_{12}s_{14} + 3s_{12}s_{34} - 3s_{14}s_{23} + 2s_{14}s_{24} - s_{23}s_{34} + s_{24}s_{34})}{s_{13}s_{134}^2} \\
&- \frac{2s_{12}s_{34}(s_{12} + s_{24})}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}(s_{12} + s_{23})}{s_{13}s_{134}s_{14}s_{24}} \\
&- \frac{2s_{12}^2(2s_{12} + s_{14} + s_{24} + 2s_{34})}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}(2s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} \\
&- \frac{2s_{14}^2s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{2s_{14}s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} - s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
&- \frac{2s_{12}(2s_{12}^2 + s_{12}s_{14} + s_{12}s_{23} + 2s_{12}s_{34} - 3s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{14}^2s_{23}(2s_{12} + s_{14} - s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
&+ \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{23}^2 + s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} - \frac{2(2s_{12}^2 - 2s_{12}s_{14} + s_{12}s_{23} + s_{12}s_{24} + s_{12}s_{34} - s_{14}s_{23})}{s_{13}s_{134}s_{234}} \\
&+ \frac{4s_{14}^2s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} - \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} - 2s_{12}s_{23} - 3s_{14}s_{23} - 3s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
&- \frac{2(4s_{12}^2 + 3s_{12}s_{14} + 2s_{12}s_{23} + s_{12}s_{34} - s_{14}s_{23} - s_{23}^2)}{s_{13}s_{134}s_{24}} - \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{4s_{14}(s_{12} - 2s_{23} + s_{24})}{s_{13}s_{134}s_{34}} + \frac{2(s_{12} + s_{14} - 2s_{23} + s_{24})}{s_{13}s_{134}} \\
& - \frac{4s_{12}^3}{s_{13}s_{14}s_{23}s_{24}} - \frac{2s_{12}^2}{s_{13}s_{14}s_{23}} - \frac{2s_{12}^2}{s_{13}s_{14}s_{24}} - \frac{2s_{12}}{s_{13}s_{14}} \\
& - \frac{2s_{12}s_{34}(s_{12} + s_{14})}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}(2s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{13}s_{23}s_{24}} \\
& + \frac{4s_{14}s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{2s_{23}(4s_{12}^2 - 2s_{12}s_{14} + 2s_{12}s_{23} - 3s_{14}^2 - 3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}} \\
& - \frac{2(4s_{12}^2 + 2s_{12}s_{14} + 3s_{12}s_{23} + s_{12}s_{34} - s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{24}} + \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{34}^2} \\
& - \frac{4(2s_{12}^2 + s_{12}s_{23} - s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{34}} - \frac{2(2s_{12} - s_{14})}{s_{13}s_{234}} \\
& - \frac{6s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{24}s_{34}^2} + \frac{2(4s_{12}^2 - 2s_{14}^2 - 3s_{14}s_{23} - 2s_{23}^2)}{s_{13}s_{24}s_{34}} \\
& - \frac{2(s_{12} + 2s_{14} + 2s_{23} + s_{34})}{s_{13}s_{24}} + \frac{2s_{14}s_{23}}{s_{13}s_{34}^2} + \frac{2s_{23}}{s_{13}s_{34}} \\
& + \frac{4s_{13}^3s_{24}}{s_{134}s_{14}s_{34}^2} - \frac{2s_{13}^2(2s_{12} + 2s_{23} - 3s_{24})}{s_{134}^2s_{14}s_{34}} \\
& - \frac{2(3s_{12}s_{13} + 3s_{12}s_{34} + 2s_{13}s_{23} - 3s_{13}s_{24} + s_{23}s_{34} - s_{24}s_{34})}{s_{134}^2s_{14}} \\
& - \frac{4(2s_{12}s_{13}^2 + 4s_{12}s_{13}s_{14} + 2s_{12}s_{14}^2 + s_{13}^2s_{23} - s_{13}^2s_{24} + s_{13}s_{14}s_{23} + s_{13}s_{14}s_{24} - s_{14}^2s_{23} + s_{14}^2s_{24})}{s_{134}^2s_{34}^2} \\
& - \frac{10(2s_{12}s_{13} + 2s_{12}s_{14} + s_{13}s_{23} + s_{14}s_{24})}{s_{134}^2s_{34}} - \frac{2(8s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& - \frac{2s_{13}^2s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2} + \frac{2s_{13}s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} - s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}} \\
& - \frac{2s_{12}(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{24} + 2s_{12}s_{34} - 3s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}} + \frac{4s_{13}^2s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{34}^2} \\
& - \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} - 2s_{12}s_{24} - 3s_{13}s_{24} - 3s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}} \\
& - \frac{2(4s_{12}^2 + 3s_{12}s_{13} + 2s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24} - s_{24}^2)}{s_{134}s_{14}s_{23}} \\
& - \frac{2s_{12}^2(2s_{12} + s_{13} + s_{23} + 2s_{34})}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{13}^2s_{24}(2s_{12} + s_{13} - s_{23})}{s_{134}s_{14}s_{234}s_{34}^2} \\
& + \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{23}s_{24} + s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}} \\
& - \frac{2(2s_{12}^2 - 2s_{12}s_{13} + s_{12}s_{23} + s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24})}{s_{134}s_{14}s_{234}} \\
& - \frac{2s_{12}(2s_{12} + 2s_{23} + s_{34})}{s_{134}s_{14}s_{24}} - \frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2} \\
& + \frac{4s_{13}(s_{12} + s_{23} - 2s_{24})}{s_{134}s_{14}s_{34}} + \frac{2(s_{12} + s_{13} + s_{23} - 2s_{24})}{s_{134}s_{14}} \\
& - \frac{2s_{13}s_{24}^2(2s_{12} - s_{14} + s_{24})}{s_{134}s_{23}s_{234}s_{34}^2} + \frac{2s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{13}^2 + s_{13}s_{14})}{s_{134}s_{23}s_{234}s_{34}} \\
& - \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} - 2s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24})}{s_{134}s_{23}s_{234}} \\
& + \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{23}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{13} - s_{13}s_{24} - s_{24}^2)}{s_{134}s_{23}s_{34}} - \frac{2(2s_{12} - s_{24})}{s_{134}s_{23}}
\end{aligned}$$

$$\begin{aligned}
& - \frac{2s_{14}s_{23}^2(2s_{12} - s_{13} + s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{2s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{13}s_{14} + s_{14}^2)}{s_{134}s_{234}s_{24}s_{34}} \\
& - \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} - 2s_{12}s_{23} + s_{12}s_{34} - s_{14}s_{23})}{s_{134}s_{234}s_{24}} \\
& - \frac{2(4s_{12}s_{13}s_{23} + 6s_{12}s_{13}s_{24} + 6s_{12}s_{14}s_{23} + 4s_{12}s_{14}s_{24} - s_{13}s_{14}s_{23} - s_{13}s_{14}s_{24} - s_{13}s_{23}s_{24} - s_{14}s_{23}s_{24})}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2}{s_{134}s_{234}s_{34}} \left(8s_{12}^2 - 2s_{12}s_{13} - 2s_{12}s_{14} - 2s_{12}s_{23} - 2s_{12}s_{24} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{24} \right. \\
& \quad \left. - s_{14}^2 + s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{24} - s_{24}^2 \right) \\
& - \frac{2(2s_{12} + s_{13} + s_{14} + s_{23} + s_{24})}{s_{134}s_{234}} \\
& + \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{134}s_{24}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{14} - s_{14}s_{23} - s_{23}^2)}{s_{134}s_{24}s_{34}} - \frac{2(2s_{12} - s_{23})}{s_{134}s_{24}} \\
& + \frac{4(4s_{12}s_{13} + 4s_{12}s_{14} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{23} - s_{13}s_{24} - s_{14}^2 - s_{14}s_{23} + s_{14}s_{24})}{s_{134}s_{34}^2} \\
& + \frac{2(8s_{12} - s_{13} - s_{14} + s_{23} + s_{24})}{s_{134}s_{34}} + \frac{4}{s_{134}} \\
& - \frac{2s_{12}s_{34}(s_{12} + s_{13})}{s_{14}s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}(4s_{12}^2 - 2s_{12}s_{13} + 2s_{12}s_{24} - 3s_{13}^2 - 3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}} \\
& - \frac{2(4s_{12}^2 + 2s_{12}s_{13} + 3s_{12}s_{24} + s_{12}s_{34} - s_{13}^2 - s_{13}s_{24})}{s_{14}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{14}s_{23}s_{24}} \\
& - \frac{6s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{34}^2} + \frac{2(4s_{12}^2 - 2s_{13}^2 - 3s_{13}s_{24} - 2s_{24}^2)}{s_{14}s_{23}s_{34}} \\
& - \frac{2(s_{12} + 2s_{13} + 2s_{24} + s_{34})}{s_{14}s_{23}} - \frac{2s_{12}(2s_{12} + 2s_{13} + s_{34})}{s_{14}s_{234}s_{24}} \\
& + \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{234}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{24} - s_{13}^2 - s_{13}s_{24})}{s_{14}s_{234}s_{34}} \\
& - \frac{2(2s_{12} - s_{13})}{s_{14}s_{234}} + \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} + \frac{2s_{24}}{s_{14}s_{34}} \\
& - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^3}{s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}^2(2s_{12} - 3s_{13} + 2s_{14})}{s_{23}s_{234}s_{34}} - \frac{2(3s_{12}s_{24} + 3s_{12}s_{34} - 3s_{13}s_{24} - s_{13}s_{34} + 2s_{14}s_{24} + s_{14}s_{34})}{s_{23}s_{234}^2} \\
& - \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(s_{12} - 2s_{13} + s_{14})}{s_{23}s_{234}s_{34}} \\
& + \frac{2(s_{12} - 2s_{13} + s_{14} + s_{24})}{s_{23}s_{234}} - \frac{2s_{12}}{s_{23}s_{24}} + \frac{2s_{13}s_{24}}{s_{23}s_{34}^2} + \frac{2s_{13}}{s_{23}s_{34}} \\
& + \frac{4s_{14}s_{23}^3}{s_{234}s_{24}s_{34}^2} - \frac{2s_{23}^2(2s_{12} + 2s_{13} - 3s_{14})}{s_{234}s_{24}s_{34}} \\
& - \frac{2(3s_{12}s_{23} + 3s_{12}s_{34} + 2s_{13}s_{23} + s_{13}s_{34} - 3s_{14}s_{23} - s_{14}s_{34})}{s_{234}^2s_{24}} \\
& - \frac{4(2s_{12}s_{23}^2 + 4s_{12}s_{23}s_{24} + 2s_{12}s_{24}^2 + s_{13}s_{23}^2 + s_{13}s_{23}s_{24} - s_{13}s_{24}^2 - s_{14}s_{23}^2 + s_{14}s_{23}s_{24} + s_{14}s_{24}^2)}{s_{234}^2s_{34}^2} \\
& - \frac{10(2s_{12}s_{23} + 2s_{12}s_{24} + s_{13}s_{23} + s_{14}s_{24})}{s_{234}^2s_{34}} - \frac{2(8s_{12} + s_{13} + s_{14})}{s_{234}^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{4s_{23}(s_{12} + s_{13} - 2s_{14})}{s_{234}s_{24}s_{34}} \\
& + \frac{2(s_{12} + s_{13} - 2s_{14} + s_{23})}{s_{234}s_{24}} \\
& + \frac{4\left(4s_{12}s_{23} + 4s_{12}s_{24} + s_{13}s_{23} - s_{13}s_{24} - s_{14}s_{23} + s_{14}s_{24} - s_{23}^2 - 2s_{23}s_{24} - s_{24}^2\right)}{s_{234}s_{34}^2} \\
& + \frac{2(8s_{12} + s_{13} + s_{14} - s_{23} - s_{24})}{s_{234}s_{34}} + \frac{4}{s_{234}} \\
& + \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} + \frac{2s_{14}}{s_{24}s_{34}} - \frac{4(2s_{12} - s_{13} - s_{14} - s_{23} - s_{24})}{s_{34}^2} - \frac{4}{s_{34}} \Big] \tag{A.47}
\end{aligned}$$

$$\begin{aligned}
\tilde{A}_4^{0,(1)} = & \frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[\frac{4s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} + \frac{2s_{14}^2(2s_{12} - 3s_{23} + 4s_{24})}{s_{13}s_{134}^2s_{34}} \right. \\
& + \frac{2(3s_{12}s_{14} + 6s_{12}s_{34} - 6s_{14}s_{23} + 4s_{14}s_{24} - 2s_{23}s_{34} + 3s_{24}s_{34})}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{14}s_{23}} + \frac{2s_{12}s_{34}(s_{12} + 2s_{23} + s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
& + \frac{2s_{12}(2s_{12}^2 + s_{12}s_{14} + s_{12}s_{24} + 2s_{14}s_{24} + s_{14}s_{34} + s_{24}s_{34} + 2s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} \\
& + \frac{4s_{12}(s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} + \frac{2s_{14}^2s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
& - \frac{4s_{12}s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{2(2s_{12}^3 + s_{12}^2s_{14} + s_{12}^2s_{23} - s_{12}s_{14}s_{34} - s_{12}s_{23}s_{34} + s_{14}^2s_{23} + s_{14}s_{23}^2)}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{2s_{14}^2s_{23}(2s_{12} + s_{14} - s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + 3s_{14}s_{23} + 2s_{14}s_{24} + s_{23}^2 + s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + 3s_{12}s_{14} + s_{12}s_{23} + s_{12}s_{24} + 2s_{12}s_{34} - s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{24} - 2s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
& - \frac{4s_{14}^2s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} \\
& + \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} - 2s_{12}s_{23} - 3s_{14}s_{23} - 5s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
& + \frac{2(4s_{12}^2 + 3s_{12}s_{14} + 4s_{12}s_{23} + 2s_{12}s_{34} - 2s_{14}s_{23} - 2s_{23}^2 - s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
& + \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}(-s_{12} + 2s_{14} + 2s_{23} - 2s_{24})}{s_{13}s_{134}s_{34}} \\
& + \frac{2(-s_{12} + 2s_{14} + 4s_{23} - 2s_{24} + s_{34})}{s_{13}s_{134}} + \frac{4s_{12}(s_{12}^2 + s_{12}s_{34} + s_{34}^2)}{s_{13}s_{14}s_{23}s_{24}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{14}s_{23}} + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{14}s_{24}} + \frac{2s_{12}}{s_{13}s_{14}} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}s_{24}} + \frac{4s_{12}(s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{23}s_{24}} + \frac{4s_{12}}{s_{13}s_{23}} \\
& - \frac{4s_{14}s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} \\
& + \frac{2s_{23}(4s_{12}^2 - 2s_{12}s_{14} + 2s_{12}s_{23} - 5s_{14}^2 - 3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2(4s_{12}^2 + 4s_{12}s_{14} + 3s_{12}s_{23} + 2s_{12}s_{34} - 2s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{34})}{s_{13}s_{234}s_{24}} \\
& - \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{23} - 2s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{34}} \\
& + \frac{4(s_{12} - s_{14})}{s_{13}s_{234}} + \frac{6s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{24}s_{34}^2} \\
& + \frac{8(-s_{12}^2 + s_{14}^2 + s_{14}s_{23} + s_{23}^2)}{s_{13}s_{24}s_{34}} + \frac{2(-2s_{12} + 3s_{14} + 3s_{23} + s_{34})}{s_{13}s_{24}} \\
& - \frac{2s_{14}s_{23}}{s_{13}s_{34}^2} - \frac{2(2s_{14} + s_{23})}{s_{13}s_{34}} - \frac{2}{s_{13}} \\
& - \frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2} + \frac{2s_{13}^2(2s_{12} + 4s_{23} - 3s_{24})}{s_{134}^2s_{14}s_{34}} \\
& + \frac{2(3s_{12}s_{13} + 6s_{12}s_{34} + 4s_{13}s_{23} - 6s_{13}s_{24} + 3s_{23}s_{34} - 2s_{24}s_{34})}{s_{134}^2s_{14}} \\
& + \frac{4(2s_{12}s_{13}^2 + 4s_{12}s_{13}s_{14} + 2s_{12}s_{14}^2 + s_{13}^2s_{23} - s_{13}^2s_{24} + s_{13}s_{14}s_{23} + s_{13}s_{14}s_{24} - s_{14}^2s_{23} + s_{14}^2s_{24})}{s_{134}^2s_{34}^2} \\
& + \frac{2(10s_{12}s_{13} + 10s_{12}s_{14} + 9s_{13}s_{23} + 2s_{13}s_{24} + 2s_{14}s_{23} + 9s_{14}s_{24})}{s_{134}^2s_{34}} \\
& + \frac{4(5s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& + \frac{2s_{13}^2s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2} \\
& - \frac{4s_{12}s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^3 + s_{12}^2s_{13} + s_{12}^2s_{24} - s_{12}s_{13}s_{34} - s_{12}s_{24}s_{34} + s_{13}^2s_{24} + s_{13}s_{24}^2)}{s_{134}s_{14}s_{23}s_{234}} \\
& - \frac{4s_{13}^2s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{34}^2} \\
& + \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} - 2s_{12}s_{24} - 3s_{13}s_{24} - 5s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}} \\
& + \frac{2(4s_{12}^2 + 3s_{12}s_{13} + 4s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}s_{24} - 2s_{24}^2 - s_{24}s_{34})}{s_{134}s_{14}s_{23}} \\
& + \frac{2s_{12}(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{23} + 2s_{13}s_{23} + s_{13}s_{34} + s_{23}s_{34} + 2s_{34}^2)}{s_{134}s_{14}s_{234}s_{24}} \\
& + \frac{2s_{13}^2s_{24}(2s_{12} + s_{13} - s_{23})}{s_{134}s_{14}s_{234}s_{34}^2} \\
& - \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + 2s_{13}s_{23} + 3s_{13}s_{24} + s_{23}s_{24} + s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + 3s_{12}s_{13} + s_{12}s_{23} + s_{12}s_{24} + 2s_{12}s_{34} - s_{13}^2 - s_{13}s_{23} - 2s_{13}s_{24} - 2s_{13}s_{34})}{s_{134}s_{14}s_{234}} \\
& + \frac{4s_{12}(s_{12} + 2s_{23} + s_{34})}{s_{134}s_{14}s_{24}} + \frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2} \\
& + \frac{4s_{13}(-s_{12} + 2s_{13} - 2s_{23} + 2s_{24})}{s_{134}s_{14}s_{34}} + \frac{2(-s_{12} + 2s_{13} - 2s_{23} + 4s_{24} + s_{34})}{s_{134}s_{14}} \\
& + \frac{2s_{13}s_{24}^2(2s_{12} - s_{14} + s_{24})}{s_{134}s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{13}^2 + s_{13}s_{14} + 3s_{13}s_{24} + 2s_{14}s_{24})}{s_{134}s_{23}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} + 3s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}s_{24} - s_{14}s_{24} - s_{24}^2 - 2s_{24}s_{34})}{s_{134}s_{23}s_{234}}
\end{aligned}$$

$$\begin{aligned}
& - \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{23}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{13} - s_{13}s_{24} - 2s_{24}^2)}{s_{134}s_{23}s_{34}} \\
& + \frac{4(s_{12} - s_{24})}{s_{134}s_{23}} + \frac{2s_{14}s_{23}^2(2s_{12} - s_{13} + s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} \\
& - \frac{2s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{13}s_{14} + 2s_{13}s_{23} + s_{14}^2 + 3s_{14}s_{23})}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} + 3s_{12}s_{23} + 2s_{12}s_{34} - s_{13}s_{23} - 2s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{34})}{s_{134}s_{234}s_{24}} \\
& + \frac{2(4s_{12}s_{13}s_{23} + 6s_{12}s_{13}s_{24} + 6s_{12}s_{14}s_{23} + 4s_{12}s_{14}s_{24} - s_{13}s_{14}s_{23} - s_{13}s_{14}s_{24} - s_{13}s_{23}s_{24} - s_{14}s_{23}s_{24})}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2}{s_{134}s_{234}s_{34}} \left((-8s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{14} + 2s_{12}s_{23} + 2s_{12}s_{24} + s_{13}^2 + 2s_{13}s_{14} + 2s_{13}s_{23} + s_{14}^2 \right. \\
& \quad \left. + 2s_{14}s_{24} + s_{23}^2 + 2s_{23}s_{24} + s_{24}^2) \right) \\
& + \frac{2(16s_{12} + 3s_{13} + 3s_{14} + 3s_{23} + 3s_{24})}{s_{134}s_{234}} \\
& - \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{134}s_{24}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{14} - s_{14}s_{23} - 2s_{23}^2)}{s_{134}s_{24}s_{34}} \\
& + \frac{4(s_{12} - s_{23})}{s_{134}s_{24}} \\
& + \frac{4(-4s_{12}s_{13} - 4s_{12}s_{14} + s_{13}^2 + 2s_{13}s_{14} - s_{13}s_{23} + s_{13}s_{24} + s_{14}^2 + s_{14}s_{23} - s_{14}s_{24})}{s_{134}s_{34}^2} \\
& + \frac{2(-8s_{12} + 5s_{13} + 5s_{14} - 3s_{23} - 3s_{24})}{s_{134}s_{34}} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{13} + s_{34})}{s_{14}s_{23}s_{234}s_{24}} \\
& - \frac{4s_{13}s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
& + \frac{2s_{24}(4s_{12}^2 - 2s_{12}s_{13} + 2s_{12}s_{24} - 5s_{13}^2 - 3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}} \\
& + \frac{2(4s_{12}^2 + 4s_{12}s_{13} + 3s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}^2 - 2s_{13}s_{24} - s_{13}s_{34})}{s_{14}s_{23}s_{234}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{14}s_{23}s_{24}} + \frac{6s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{34}^2} \\
& + \frac{8(-s_{12}^2 + s_{13}^2 + s_{13}s_{24} + s_{24}^2)}{s_{14}s_{23}s_{34}} + \frac{2(-2s_{12} + 3s_{13} + 3s_{24} + s_{34})}{s_{14}s_{23}} \\
& + \frac{4s_{12}(s_{12} + 2s_{13} + s_{34})}{s_{14}s_{234}s_{24}} - \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{234}s_{34}^2} \\
& + \frac{4(2s_{12}^2 + s_{12}s_{24} - 2s_{13}^2 - s_{13}s_{24})}{s_{14}s_{234}s_{34}} + \frac{4(s_{12} - s_{13})}{s_{14}s_{234}} \\
& + \frac{4s_{12}}{s_{14}s_{24}} - \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} - \frac{2(2s_{13} + s_{24})}{s_{14}s_{34}} - \frac{2}{s_{14}} \\
& + \frac{4s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}} - \frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2} \\
& + \frac{2s_{24}^2(2s_{12} - 3s_{13} + 4s_{14})}{s_{23}s_{234}^2s_{34}} \\
& + \frac{2(3s_{12}s_{24} + 6s_{12}s_{34} - 6s_{13}s_{24} - 2s_{13}s_{34} + 4s_{14}s_{24} + 3s_{14}s_{34})}{s_{23}s_{234}^2} \\
& + \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(-s_{12} + 2s_{13} - 2s_{14} + 2s_{24})}{s_{23}s_{234}s_{34}} \\
& + \frac{2(-s_{12} + 4s_{13} - 2s_{14} + 2s_{24} + s_{34})}{s_{23}s_{234}} + \frac{2s_{12}}{s_{23}s_{24}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{2s_{13}s_{24}}{s_{23}s_{34}^2} - \frac{2(s_{13} + 2s_{24})}{s_{23}s_{34}} - \frac{2}{s_{23}} \\
& -\frac{4s_{14}s_{23}^3}{s_{23}^2s_{24}s_{34}^2} + \frac{2s_{23}^2(2s_{12} + 4s_{13} - 3s_{14})}{s_{23}^2s_{24}s_{34}} \\
& + \frac{2(3s_{12}s_{23} + 6s_{12}s_{34} + 4s_{13}s_{23} + 3s_{13}s_{34} - 6s_{14}s_{23} - 2s_{14}s_{34})}{s_{23}^2s_{24}} \\
& + \frac{4(2s_{12}s_{23}^2 + 4s_{12}s_{23}s_{24} + 2s_{12}s_{24}^2 + s_{13}s_{23}^2 + s_{13}s_{23}s_{24} - s_{13}s_{24}^2 - s_{14}s_{23}^2 + s_{14}s_{23}s_{24} + s_{14}s_{24}^2)}{s_{23}^2s_{34}^2} \\
& + \frac{2(10s_{12}s_{23} + 10s_{12}s_{24} + 9s_{13}s_{23} + 2s_{13}s_{24} + 2s_{14}s_{23} + 9s_{14}s_{24})}{s_{23}^2s_{34}} \\
& + \frac{4(5s_{12} + s_{13} + s_{14})}{s_{23}^2} + \frac{8s_{14}s_{23}^2}{s_{23}s_{24}s_{34}^2} \\
& + \frac{4s_{23}(-s_{12} - 2s_{13} + 2s_{14} + 2s_{23})}{s_{23}s_{24}s_{34}} \\
& + \frac{2(-s_{12} - 2s_{13} + 4s_{14} + 2s_{23} + s_{34})}{s_{23}s_{24}} \\
& + \frac{4(-4s_{12}s_{23} - 4s_{12}s_{24} - s_{13}s_{23} + s_{13}s_{24} + s_{14}s_{23} - s_{14}s_{24} + s_{23}^2 + 2s_{23}s_{24} + s_{24}^2)}{s_{23}s_{34}^2} \\
& + \frac{2(-8s_{12} - 3s_{13} - 3s_{14} + 5s_{23} + 5s_{24})}{s_{23}s_{34}} \\
& - \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} - \frac{2(s_{14} + 2s_{23})}{s_{24}s_{34}} - \frac{2}{s_{24}} \\
& + \left[\frac{4(2s_{12} - s_{13} - s_{14} - s_{23} - s_{24})}{s_{34}^2} - \frac{8}{s_{34}} \right] \tag{A.48}
\end{aligned}$$

$$\begin{aligned}
\tilde{A}_4^{0,(2)} = & \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[-\frac{4s_{14}^2s_{24}}{s_{13}s_{134}s_{34}} - \frac{2(3s_{12}s_{34} - 3s_{14}s_{23} + 2s_{14}s_{24} + 3s_{24}s_{34})}{s_{13}s_{134}^2} - \frac{2s_{12}s_{34}(s_{24} + 2s_{34})}{s_{13}s_{134}s_{14}s_{23}} \right. \\
& - \frac{2s_{12}s_{34}(s_{23} + 2s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
& - \frac{2s_{12}(2s_{14}s_{24} + s_{14}s_{34} + s_{24}s_{34} + 3s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}(2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} \\
& - \frac{2s_{34}(s_{12}s_{34} - s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{14}^2(s_{23} + s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
& - \frac{2(2s_{12}s_{14} + 2s_{12}s_{34} - s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{24} - 3s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
& + \frac{4s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{2(2s_{12}s_{23} + 3s_{12}s_{34} - 3s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
& - \frac{4s_{14}(2s_{14} - s_{24})}{s_{13}s_{134}s_{34}} - \frac{2(3s_{14} + 2s_{23} - s_{24} + 2s_{34})}{s_{13}s_{134}} \\
& - \frac{6s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{23}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{24}} \\
& - \frac{2s_{12}s_{34}(s_{14} + 2s_{34})}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}(2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} \\
& - \frac{2s_{12}s_{34}}{s_{13}s_{23}s_{24}} - \frac{4s_{12}}{s_{13}s_{23}} + \frac{4s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} \\
& - \frac{2(2s_{12}s_{14} + 3s_{12}s_{34} - s_{14}^2 - 3s_{14}s_{23} - 2s_{14}s_{34})}{s_{13}s_{234}s_{24}} \\
& + \frac{4s_{14}^2}{s_{13}s_{234}s_{34}} + \frac{2s_{14}}{s_{13}s_{234}} - \frac{4(s_{14}^2 + s_{23}^2)}{s_{13}s_{24}s_{34}} \\
& - \frac{2(s_{14} + s_{23})}{s_{13}s_{24}} + \frac{4s_{14}}{s_{13}s_{34}} + \frac{2}{s_{13}} \tag{A.48}
\end{aligned}$$

$$\begin{aligned}
& - \frac{4s_{13}^2 s_{23}}{s_{134}^2 s_{14} s_{34}} - \frac{2(3s_{12} s_{34} + 2s_{13} s_{23} - 3s_{13} s_{24} + 3s_{23} s_{34})}{s_{134}^2 s_{14}} \\
& - \frac{4(2s_{13} s_{23} + s_{13} s_{24} + s_{14} s_{23} + 2s_{14} s_{24})}{s_{134}^2 s_{34}} - \frac{2(2s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& - \frac{2s_{34}(s_{12} s_{34} - s_{13} s_{24})}{s_{134} s_{14} s_{23} s_{24}} + \frac{4s_{13} s_{24}^2}{s_{134} s_{14} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{24} + 3s_{12} s_{34} - 3s_{13} s_{24} - s_{24}^2 - 2s_{24} s_{34})}{s_{134} s_{14} s_{23}} \\
& - \frac{2s_{12}(2s_{13} s_{23} + s_{13} s_{34} + s_{23} s_{34} + 3s_{34}^2)}{s_{134} s_{14} s_{23} s_{24}} \\
& + \frac{4s_{13}^2 (s_{23} + s_{24})}{s_{134} s_{14} s_{23} s_{34}} - \frac{2(2s_{12} s_{13} + 2s_{12} s_{34} - s_{13}^2 - s_{13} s_{23} - 2s_{13} s_{24} - 3s_{13} s_{34})}{s_{134} s_{14} s_{23} s_{34}} \\
& - \frac{2s_{12}(2s_{23} + s_{34})}{s_{134} s_{14} s_{24}} - \frac{4s_{13}(2s_{13} - s_{23})}{s_{134} s_{14} s_{34}} \\
& - \frac{2(3s_{13} - s_{23} + 2s_{24} + 2s_{34})}{s_{134} s_{14}} + \frac{4s_{24}^2 (s_{13} + s_{14})}{s_{134} s_{23} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{24} + 2s_{12} s_{34} - 2s_{13} s_{24} - s_{14} s_{24} - s_{24}^2 - 3s_{24} s_{34})}{s_{134} s_{23} s_{23} s_{34}} \\
& + \frac{4s_{24}^2}{s_{134} s_{23} s_{34}} + \frac{2s_{24}}{s_{134} s_{23}} + \frac{4s_{23}^2 (s_{13} + s_{14})}{s_{134} s_{23} s_{24} s_{34}} \\
& - \frac{2(2s_{12} s_{23} + 2s_{12} s_{34} - s_{13} s_{23} - 2s_{14} s_{23} - s_{23}^2 - 3s_{23} s_{34})}{s_{134} s_{23} s_{24}} \\
& + \frac{8(s_{13} + s_{14})(s_{23} + s_{24})}{s_{134} s_{23} s_{34}} - \frac{4(4s_{12} + 2s_{13} + 2s_{14} + 2s_{23} + 2s_{24} + s_{34})}{s_{134} s_{23} s_{34}} \\
& + \frac{4s_{23}^2}{s_{134} s_{24} s_{34}} + \frac{2s_{23}}{s_{134} s_{24}} - \frac{4(2s_{13} + 2s_{14} - s_{23} - s_{24})}{s_{134} s_{34}} - \frac{4}{s_{134}} \\
& - \frac{2s_{12} s_{34}(s_{13} + 2s_{34})}{s_{14} s_{23} s_{23} s_{24}} + \frac{4s_{13}^2 s_{24}}{s_{14} s_{23} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{13} + 3s_{12} s_{34} - s_{13}^2 - 3s_{13} s_{24} - 2s_{13} s_{34})}{s_{14} s_{23} s_{23} s_{34}} \\
& - \frac{2s_{12} s_{34}}{s_{14} s_{23} s_{24}} - \frac{4(s_{13}^2 + s_{24}^2)}{s_{14} s_{23} s_{34}} - \frac{2(s_{13} + s_{24})}{s_{14} s_{23}} \\
& - \frac{2s_{12}(2s_{13} + s_{34})}{s_{14} s_{23} s_{24}} + \frac{4s_{13}^2}{s_{14} s_{23} s_{34}} + \frac{2s_{13}}{s_{14} s_{23} s_{34}} \\
& - \frac{4s_{12}}{s_{14} s_{24}} + \frac{4s_{13}}{s_{14} s_{34}} + \frac{2}{s_{14}} \\
& - \frac{4s_{14} s_{24}^2}{s_{23} s_{23}^2 s_{34}} - \frac{2(3s_{12} s_{34} - 3s_{13} s_{24} + 2s_{14} s_{24} + 3s_{14} s_{34})}{s_{23} s_{23}^2} \\
& + \frac{4s_{24}(s_{14} - 2s_{24})}{s_{23} s_{23} s_{34}} - \frac{2(2s_{13} - s_{14} + 3s_{24} + 2s_{34})}{s_{23} s_{23} s_{34}} \\
& + \frac{4s_{24}}{s_{23} s_{34}} + \frac{2}{s_{23}} - \frac{4s_{13} s_{23}^2}{s_{23}^2 s_{24} s_{34}} \\
& - \frac{2(3s_{12} s_{34} + 2s_{13} s_{23} + 3s_{13} s_{34} - 3s_{14} s_{23})}{s_{23}^2 s_{24}} \\
& - \frac{4(2s_{13} s_{23} + s_{13} s_{24} + s_{14} s_{23} + 2s_{14} s_{24})}{s_{23}^2 s_{34}} - \frac{2(2s_{12} + s_{13} + s_{14})}{s_{23}^2} \\
& + \frac{4s_{23}(s_{13} - 2s_{23})}{s_{23} s_{24} s_{34}} + \frac{2(s_{13} - 2s_{14} - 3s_{23} - 2s_{34})}{s_{23} s_{24}} \\
& + \frac{4(s_{13} + s_{14} - 2s_{23} - 2s_{24})}{s_{23} s_{34}} - \frac{4}{s_{23} s_{34}} + \frac{4s_{23}}{s_{24} s_{34}} + \frac{2}{s_{24}} \Big]
\end{aligned}
\tag{A.49}$$

$$\begin{aligned}
\tilde{A}_4^{0,(3)} = & \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{23}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} \right. \\
& - \frac{2s_{12}s_{34}^2}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{14}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}s_{24}} \\
& + \frac{2s_{34}(s_{23} + s_{24})}{s_{13}s_{134}^2} + \frac{2s_{14}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{23}s_{34}}{s_{13}s_{134}s_{24}} \\
& + \frac{2s_{14}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{34}(s_{23} + s_{24})}{s_{134}^2s_{14}} + \frac{2s_{24}s_{34}}{s_{134}s_{14}s_{23}} \\
& + \frac{2s_{13}s_{34}}{s_{134}s_{14}s_{234}} + \frac{2s_{24}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{23}s_{34}}{s_{134}s_{234}s_{24}} \\
& + \frac{2s_{13}s_{34}}{s_{14}s_{23}s_{234}} + \frac{2s_{34}(s_{13} + s_{14})}{s_{23}s_{234}^2} + \frac{2s_{34}(s_{13} + s_{14})}{s_{234}^2s_{24}} \\
& \left. + \frac{6s_{34}}{s_{13}s_{134}} + \frac{4s_{34}}{s_{13}s_{24}} + \frac{6s_{34}}{s_{134}s_{14}} + \frac{8s_{34}}{s_{134}s_{234}} + \frac{4s_{34}}{s_{14}s_{23}} + \frac{6s_{34}}{s_{23}s_{234}} + \frac{6s_{34}}{s_{234}s_{24}} \right] \quad (A.50)
\end{aligned}$$

using,

$$\mathcal{D}_{4,gg} = s_{13}s_{24}s_{34}^2s_{134}^2s_{234}^2$$

By running `BuildAndIntegrateAntenna[A, 4, 0, Component->Subleading]` we get the integrated result,

$$\begin{aligned}
\tilde{\mathcal{A}}_4^0 = & \frac{1}{\epsilon^4} + \frac{3}{\epsilon^3} + \frac{1}{\epsilon^2} \left(13 - \frac{3\pi^2}{2} \right) + \frac{1}{\epsilon} \left(\frac{845}{16} - \frac{9\pi^2}{2} - \frac{80\zeta_3}{3} \right) \\
& + \frac{6921}{32} - \frac{473\pi^2}{24} + \frac{17\pi^4}{72} - 80\zeta_3 + \mathcal{O}(\epsilon). \quad (A.51)
\end{aligned}$$

A.3.10 B_4^0

Running `BuildAntenna[B, 4, 0]`, we get the built result,

$$\begin{aligned}
B_4^0 = & \epsilon^2 \left(\frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}} \right. \\
& - \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{234}^2s_{34}} \\
& \left. - \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} \right) \\
& + \epsilon \left(\frac{4s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} \right. \\
& + \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{12}}{s_{134}^2} - \frac{2s_{12}}{s_{234}^2} \\
& + \frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} - \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}^2s_{34}} \\
& + \frac{4s_{14}s_{24}}{s_{134}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}} \\
& + \frac{4s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& + \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} \\
& \left. + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2} \right) \\
& - \frac{4s_{12}}{s_{134}s_{234}} - \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{2s_{14}s_{24}}{s_{234}^2s_{34}} - \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} - \frac{4s_{12}^2}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} \\
& - \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} \\
& - \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} \\
& + \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}.
\end{aligned} \tag{A.52}$$

By running `BuildAndIntegrateAntenna[B, 4, 0]` we get the integrated result,

$$\mathcal{B}_4^0 = -\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} - \frac{1}{\epsilon} \frac{11(37 - 3\pi^2)}{216} - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + \mathcal{O}(\epsilon). \tag{A.53}$$

A.3.11 C_4^0

Running `BuildAntenna[C, 4, 0]`, we get the built result,

$$\begin{aligned}
C_4^0 = & \epsilon^3 \left(\frac{s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}} \right. \\
& - \frac{s_{12}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}} + \frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}} \\
& - \frac{s_{14}s_{24}}{s_{14}s_{24}} + \frac{s_{13}s_{24}}{s_{13}s_{24}} + \frac{s_{13}s_{24}}{s_{13}s_{24}} - \frac{s_{14}s_{24}}{s_{14}s_{24}} - \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{s_{14}s_{24}}{s_{234}^2s_{34}} - \frac{s_{12}s_{24}}{s_{23}s_{234}^2} + \frac{s_{13}s_{24}}{s_{23}s_{234}^2} \\
& + \frac{s_{14}}{s_{123}s_{134}} - \frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{23}s_{234}^2} \\
& + \frac{2s_{13}}{s_{123}s_{134}} + \frac{s_{12}}{s_{123}s_{134}} + \frac{2s_{13}}{s_{123}s_{234}} + \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}}{s_{134}s_{234}} \\
& + \frac{s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} \\
& - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} + \frac{s_{12}}{s_{234}^2} + \frac{2s_{13}}{s_{234}^2} + \frac{s_{14}}{s_{234}^2} \Big) \\
& + \epsilon^2 \left(-\frac{s_{12}^2}{s_{123}s_{134}s_{23}} - \frac{s_{12}^2}{s_{134}s_{23}s_{234}} - \frac{3s_{12}s_{24}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{12}s_{24}}{s_{134}s_{23}s_{234}} \right. \\
& - \frac{2s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} + \frac{2s_{12}s_{34}}{s_{23}s_{234}^2} + \frac{2s_{12}}{s_{123}s_{134}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}} \\
& - \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}} - \frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} \\
& - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}}{s_{234}^2} + \frac{2s_{14}}{s_{123}s_{134}} + \frac{2s_{13}s_{24}}{s_{123}s_{134}s_{23}} \\
& + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}} - \frac{2s_{13}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{14}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{13}}{s_{123}s_{234}} \\
& + \frac{s_{14}}{s_{123}s_{234}} - \frac{2s_{14}}{s_{134}s_{234}} - \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}^2}{s_{123}s_{134}s_{34}} + \frac{3s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} \\
& + \frac{2s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} + \frac{2s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} - \frac{2s_{14}s_{23}}{s_{123}s_{134}s_{34}} - \frac{3s_{14}s_{24}}{s_{123}s_{134}s_{34}} \\
& + \frac{2s_{13}s_{24}}{s_{123}s_{134}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{2s_{13}^2s_{24}}{s_{123}s_{134}s_{23}s_{34}} - \frac{2s_{14}s_{24}}{s_{123}s_{234}s_{34}} - \frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}} \Big)
\end{aligned}$$

$$\begin{aligned}
& + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{s_{14}^2}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} \\
& - \frac{4s_{13}}{s_{234}^2} - \frac{2s_{14}}{s_{234}^2} \Big) \\
& + \epsilon \left(\frac{s_{12}^2}{s_{123}s_{134}s_{23}} - \frac{s_{12}^2}{s_{123}s_{23}s_{234}} + \frac{2s_{12}^2}{s_{134}s_{23}s_{234}} + \frac{2s_{12}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}} \right. \\
& + \frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}} + \frac{3s_{12}s_{24}}{s_{23}s_{234}^2} + \frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} + \frac{2s_{12}s_{34}}{s_{134}s_{23}s_{234}} \\
& - \frac{s_{12}s_{34}}{s_{23}s_{234}^2} - \frac{s_{12}}{s_{123}s_{134}} - \frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} + \frac{s_{12}}{s_{134}s_{234}} \\
& + \frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}} - \frac{2s_{12}s_{14}}{s_{123}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}} + \frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}} - \frac{s_{12}s_{14}}{s_{123}s_{134}s_{24}} \\
& - \frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{2s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{4s_{12}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}} - \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} + \frac{s_{12}}{s_{234}^2} - \frac{s_{14}}{s_{123}s_{134}} \\
& - \frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} + \frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}} + \frac{s_{13}s_{24}}{s_{23}s_{234}^2} + \frac{4s_{14}s_{24}}{s_{23}s_{234}^2} \\
& - \frac{4s_{13}}{s_{123}s_{134}} + \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}^2}{s_{123}s_{134}s_{34}} - \frac{2s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} - \frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} \\
& - \frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} - \frac{3s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}} - \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} + \frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}} + \frac{2s_{14}s_{24}}{s_{123}s_{134}s_{34}} \\
& - \frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}} - \frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{234}^2s_{34}} \\
& + \frac{3s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{14}^2}{s_{123}s_{234}s_{34}} - \frac{s_{14}^2}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}} + \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} \\
& + \frac{2s_{14}s_{23}}{s_{123}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{13}}{s_{234}^2} + \frac{s_{14}}{s_{234}^2} \Big) \\
& - \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}} - \frac{2s_{12}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{14}s_{24}}{s_{23}s_{234}^2} + \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{13}}{s_{123}s_{134}} + \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} - \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}}{s_{134}s_{234}} \\
& + \frac{s_{12}}{s_{123}s_{234}} - \frac{s_{12}}{s_{134}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}} + \frac{s_{12}^2}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}^2}{s_{134}s_{23}s_{234}} + \frac{2s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}} + \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} - \frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{2s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}} - \frac{s_{14}^2}{s_{123}s_{234}s_{34}} + \frac{s_{14}^2}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}} \\
& + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}.
\end{aligned} \tag{A.54}$$

By running `BuildAndIntegrateAntenna[C, 4, 0]` we get the integrated result,

$$C_4^0 = \frac{1}{\epsilon} \frac{(-13 + 2\pi^2 - 8\zeta_3)}{32} - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + \mathcal{O}(\epsilon). \tag{A.55}$$

A.4 Exploratory Massive A_3^0

Running `BuildAntenna[A, 3, 0, quarkMass->mQ]`, we get the built result,

$$\begin{aligned}
A_3^{0,(m_q)} = & \frac{1}{s_{13}s_{23}(2m_q^2 + q^2)} \left[\frac{1}{\epsilon^2} \frac{m_q^2}{(2m_q^2 + q^2)^2} \left(m_q^2 (s_{13}^2 + 4s_{13}s_{23} + s_{23}^2) + s_{13}s_{23}s_{123} \right) \right. \\
& - \frac{1}{\epsilon} \frac{1}{4(2m_q^2 + q^2)} \left(2m_q^2 (3s_{13}^2 + 4s_{13}s_{23} + 3s_{23}^2) + s_{12} (s_{13} + s_{23})^2 + (s_{13} + s_{23})^3 \right) \\
& - \frac{1}{4s_{13}s_{23}} \left(-8m_q^4 (s_{13}^2 + s_{23}^2) - 2m_q^2 (s_{12} (s_{13}^2 - s_{13}s_{23} + s_{23}^2) + (s_{13} + s_{23})^3) \right. \\
& \left. \left. + 2s_{12}s_{13}s_{23}s_{123} + s_{13}s_{23} (s_{13}^2 + s_{23}^2) \right) \right] + \mathcal{O}(\epsilon^3). \tag{A.56}
\end{aligned}$$

By running `BuildAndIntegrateAntenna[A, 3, 0, quarkMass->mQ]` we get the integrated result,

$$\begin{aligned}
\mathcal{A}_3^{0,(m_q)} = & \frac{2^{-2(1+\epsilon)} (1-\epsilon) \pi^{\epsilon-2} (1-4m_q^2)^{1-2\epsilon}}{\epsilon(2\epsilon-1)m_q^2(\epsilon-2m_q^2-1)} \left[-\frac{24\Gamma(4-3\epsilon)\Gamma(3-2\epsilon)}{\Gamma(8-6\epsilon)} (\epsilon-1) (4m_q^2-1) \right. \\
& \cdot \left(\epsilon^2 (8m_q^2-2) + \epsilon (3-4m_q^2) - 4m_q^2-2 \right) {}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; 1-4m_q^2\right) \\
& - \frac{4\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}{\Gamma(6-6\epsilon)} \left(4m_q^4 + \epsilon^3 (8m_q^2-2) + 6m_q^2 + \epsilon^2 (16m_q^4-12m_q^2+5) \right. \\
& \left. \left. - \epsilon (16m_q^4+4m_q^2+5) + 2 \right) {}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; 1-4m_q^2\right) \right]. \tag{A.57}
\end{aligned}$$

Appendix B

Supporting Derivations

B.1 Feynman Rules

For the massless final-final quark-antiquark antenna functions considered throughout this work, in CDR, with $d = 4 - 2\epsilon$, and using the Feynman gauge $\lambda = 1$, the required Feynman rules go as follows.

An application of these rules is shown in full in Subsection 4.2.1.

B.1.1 Propagators

Each internal line in a Feynman diagram represents a propagator: the way a quantum field carries an excitation from one spacetime point to another. These propagators differ particle-by-particle both in mathematical structure and in the way they are graphically represented in the diagrams. A description of propagators encountered throughout this work is present in Table B.1 [1, 8].

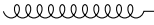
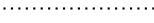
Particle-type	Mathematical Representation	Graphical Representation
Quark (q)	$\frac{i\delta_{ij}\not{p}}{p^2 + i0}$ $\left(m_q \neq 0 : \frac{i(\not{p} + m_q)}{p^2 - m_q^2 + i0}\right)$	
Gluon (g)	$\frac{-i\delta^{ab}g_{\mu\nu}}{p^2 + i0}$	
Ghost (c)	$\frac{i\delta^{ab}}{p^2 + i0}$	

Table B.1: Overview of the propagators found throughout the thesis, alongside their mathematical and graphical representations.

For an external massless quark,

$$\sum_s u_s(p)\bar{u}_s(p) = \not{p}, \quad \sum_s v_s(p)\bar{v}_s(p) = \not{p}. \quad (\text{B.1})$$

For external gluons,

$$\sum_\lambda \varepsilon_\mu^{(\lambda)}(p)\varepsilon_\nu^{(\lambda)*}(p) = -g_{\mu\nu} + t_g \quad (\text{B.2})$$

where t_g are gauge terms that vanish after summing a gauge-invariant amplitude or colour-ordered gauge-invariant partial amplitude.

B.1.2 Vertices

These lines meet in vertices, which represent the way in which quantum fields interact. The four vertices encountered throughout this work are shown in Table B.2 [1, 8].

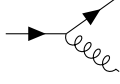
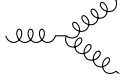
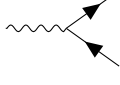
Interaction	Mathematical Representation	Graphical Representation
$q\bar{q}g$	$ig_s\gamma^\mu (T^a)_{ij}$	
ggg	$g_s f^{abc} (g^{\mu\nu}(p-q)^\rho + g^{\nu\rho}(q-r)^\mu + g^{\rho\mu}(r-p)^\nu)$ $p+q+r=0$	
$gggg$	$-ig_s^2 \left(f^{abe} f^{cde} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}) \right.$ $\quad + f^{ace} f^{bde} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho})$ $\quad \left. + f^{ade} f^{bce} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma}) \right)$	
$c\bar{c}g$	$-g_s f^{abc} p^\mu$ p is the outgoing ghost momentum	
$\gamma^* q\bar{q}$	$ieQ_q\gamma^\mu\delta_{ij}$	

Table B.2: Overview of the vertices encountered throughout this thesis, alongside their mathematical and graphical representations.

B.2 PaVe Reduction

At one-loop, amplitudes produce tensor integrals similar to those in Eq. 2.70 [13]. With the general massive¹ denominators defined as,

$$D_0 = \ell^2 - m_0^2 + i0, \quad D_i = (\ell + r_i)^2 - m_i^2 + i0 \quad (\text{B.3})$$

the tensor integral takes the form,

$$I_N^{\mu_1 \dots \mu_R} = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{D_0 D_1 \dots D_{N-1}} \quad (\text{B.4})$$

where R is the tensor rank, r_i are the combinations of external momenta and m_i is the mass associated with the i -th propagator. The reduction is carried out in $d = 4 - 2\epsilon$ dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \quad (\text{B.5})$$

PaVe reduces the Lorentz indices in the numerator $\ell^\mu \ell^\nu \ell^\rho \dots$ terms and expresses them in terms of external momenta p_i^μ , the metric tensor $g_{\mu\nu}$ and scalar one-loop integrals, through Lorentz covariance [13].

¹Throughout this section, the general and massive case will be considered for completeness. The rest of the present work, however, heavily leans towards the massless regime. In that regime the internal masses vanish and the physical external partons satisfy $p_a^2 = 0$. The shifted momenta r_i , introduced in Eq. B.3, entering propagators are combinations of external momenta and need not be lightlike.

For a rank-one N -point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \quad (\text{B.6})$$

where C_i are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \quad (\text{B.7})$$

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} (g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu) C_{00i} \quad (\text{B.8})$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 + m_i^2 - m_0^2 \quad (\text{B.9})$$

which lets us cancel denominator terms, through a method like the following example set at $i = 1$,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2} \left(\frac{1}{D_0} - \frac{1}{D_1} + \frac{1}{D_0 D_1} (-m_0^2 + m_1^2 - r_1^2) \right) \quad (\text{B.10})$$

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2p_j \cdot p_i. \quad (\text{B.11})$$

Contracting a general rank-one integral, as described in Eq. B.6, with p_j gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2} \sum_i G_{ji} C_i. \quad (\text{B.12})$$

If we now define $R_j \equiv p_j \cdot I_N$, we are then able to compute the C_i coefficients in the Lorentz decomposition by taking,

$$G_{ji} C_i = 2R_j \Rightarrow C_i = 2 \left(G^{-1} \right)_{ij} R_j. \quad (\text{B.13})$$

B.2.1 Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^\mu(p; m_0^2, m_1^2) = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^\mu}{D_0 D_1} \quad (\text{B.14})$$

where the denominators follow Eq. B.3. Lorentz covariance gives,

$$B^\mu = p^\mu B_1 \quad (\text{B.15})$$

Contracting with p_μ yields,

$$p^2 B_1 = p_\mu B^\mu = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \quad (\text{B.16})$$

Using the relation in Eq. B.9, setting $r_1 = p$, we get,

$$p^2 B_1 = \frac{1}{2} \left[A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2) B_0(p^2; m_0^2, m_1^2) \right] \quad (\text{B.17})$$

and therefore²,

$$B^\mu = p^\mu \frac{A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2) B_0}{2p^2}. \quad (\text{B.18})$$

²The result shown in Eq. B.18 assumes generic kinematics with $p^2 \neq 0$. Exceptional cases are obtained through an appropriate limit or alternative reduction.

Here, A_0 and B_0 are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$\begin{aligned} A_0(m^2) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{\ell^2 - m^2 + i0} \\ B_0(p^2; m_0^2, m_1^2) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \end{aligned} \quad (\text{B.19})$$

B.3 IBP Reduction Example

The method behind IBP reduction, as discussed in Subsection 2.5.1 may be made easier to visualise with an example. Consider a one-loop bubble, akin to that studied via PaVe reduction in Section B.2. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d \ell \frac{1}{D_0^{a_0} D_1^{a_1}} \quad (\text{B.20})$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \quad (\text{B.21})$$

We now choose $v^\mu = \ell^\mu$, and build and expand the relation from Eq. 2.71 as,

$$\begin{aligned} 0 &= \int d^d \ell \frac{\partial}{\partial \ell^\mu} \left(\frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}} \right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d \ell \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d \ell \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \quad (\text{B.22})$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \quad (\text{B.23})$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \quad (\text{B.24})$$

and finally get the explicit shifted-index identity,

$$\begin{aligned} 0 &= (d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\ &\quad - 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\ &\quad - a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\ &\quad - a_1 (m_0^2 + m_1^2 - p^2) \mathcal{I}(a_0, a_1 + 1). \end{aligned} \quad (\text{B.25})$$

B.4 Three-Particle Phase-Space to Cut Propagators

Let us now consider the massless three-particle phase-space Φ_3 . It may be described as,

$$d\Phi_3(q; p_1, p_2, p_3) \propto d^d p_1 d^d p_2 d^d p_3 \delta^+(p_1^2) \delta^+(p_2^2) \delta^+(p_3^2) \delta^{(d)}(q - p_1 - p_2 - p_3). \quad (\text{B.26})$$

We start by using momentum conservation to eliminate p_3 ,

$$p_3 = q - p_1 - p_2. \quad (\text{B.27})$$

This first step eliminates the rightmost δ -function. Then, using reverse unitarity, each of the remaining δ -functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^d p_1 d^d p_2 \frac{1}{[p_1^2]_c [p_2^2]_c [(q - p_1 - p_2)^2]_c} \quad (\text{B.28})$$

with

$$\left[\frac{1}{p^2} \right]_c \equiv \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right]_+. \quad (\text{B.29})$$

B.5 Laporta Algorithm Implementation Example

Consider an arbitrary family defined as $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$, with the denominators chosen as $D_1 = s_{12}$, $D_2 = s_{13}$ and $D_3 = s_{23}$. We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \quad (\text{B.30})$$

$$I_2 = \int d\Phi \frac{2}{s_{12}s_{13}s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \quad (\text{B.31})$$

$$I_3 = \int d\Phi \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \quad (\text{B.32})$$

where $d\Phi$ is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals I_1 and I_3 in the example above are members of the same sector, $I_1, I_3 \in (1, 0, 0)$. In our example, only two sectors appear: $(1, 0, 0)$ and $(1, 1, 1)$. These are our two target sectors. The three integrals are organised in terms of the t , r and s as follows,

	t	r	s
$I(1, -1, 0)$	1	0	1
$I(1, 1, 1)$	3	0	0
$I(3, 0, 0)$	1	2	0

(B.33)

then, by order of complexity, the algorithm orders the integrals as,

$$I(1, 1, 1) \succ I(3, 0, 0) \succ I(1, -1, 0). \quad (\text{B.34})$$

B.6 The Massive Extension

The main body of this thesis focused primarily on the massless regime, since the vast majority of the present work's chapters focus on that regime. This appendix collects the supporting extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero, they are set to their squared mass.

$$p_q^2 = p_{\bar{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\bar{q}}^2 = m_q^2. \quad (\text{B.35})$$

where p_i represents the momentum and m_i represents the mass of particle i .

The antenna subtraction formalism and the reduction architecture that was developed and described throughout Chapter 2 are retained, however. What changes is the intermediate steps and kinematics: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different, as the characteristic phase-space scale in the massive regime is not equal to its massless counterpart. A new scale of m_q^2/q^2 is also introduced and leads to distinct cut-integral families and master integrals [24].

B.6.1 Massive Kinematics and Infrared Structure

In Subsection 2.2.1, the Mandelstam invariants were introduced for the massless regime. The general definition is,

$$\begin{aligned} s_{ij} &= (p_i + p_j)^2 = p_i^2 + 2p_i \cdot p_j + p_j^2 \\ &= m_i^2 + m_j^2 + 2p_i \cdot p_j. \end{aligned} \quad (\text{B.36})$$

Through momentum conservation, which states that $q = \sum_i p_i$, we also have to reconsider the q^2 expression, as,

$$q^2 = \sum_i m_i^2 + 2 \sum_{i < j} p_i \cdot p_j. \quad (\text{B.37})$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless

collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive, consider that particle i of mass m_i and energy E_i , and the other is massless, consider that particle j , such that $m_j = 0$, then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos \theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \quad (\text{B.38})$$

As the angle between the partons θ tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta \rightarrow 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \quad (\text{B.39})$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region, $E_i \gg m_i$ and $\theta \ll 1$,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2} \left(\theta^2 + \frac{m_i^2}{E_i^2} \right) \quad (\text{B.40})$$

and consequently since,

$$\int_0^{\theta_{\max}^2} d\theta^2 \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta_{\max}^2} d\theta^2 \left(\theta^2 + \frac{m_i^2}{E_i^2} \right)^{-1} \sim \ln \left(\frac{E_i^2}{m_i^2} \right) \quad (\text{B.41})$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass m_i and the opening angle θ become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as $m_i \rightarrow 0$ [24].

Mass does not regulate UV divergences, as $\ell^2 \gg m_i^2, p_i^2$. It is noteworthy, however, that in a theory with massive quarks, the quark mass and quark field must be renormalised as,

$$g_{s,0} = \mu^\epsilon Z_g g_s, \quad m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q} \psi_q. \quad (\text{B.42})$$

B.6.2 Massive Phase-Space Integration and Reduction Methods

For final-state particles of momenta p_i and masses m_i , the phase-space becomes,

$$d\Phi_n(q; p_1, \dots, p_n) = (2\pi)^d \delta^{(d)} \left(q - \sum_{i=1}^n p_i \right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+(p_i^2 - m_i^2). \quad (\text{B.43})$$

At the level of the phase-space measure, the formal change from the massless case, as described in Subsubsection 2.5.1, is,

$$\delta^+(p_i^2) \rightarrow \delta^+(p_i^2 - m_i^2) \quad (\text{B.44})$$

which physically imposes that $p_i^2 = m_i^2$ and $p_i^0 > 0$.

Phase-space factorisation still holds for the massive regime, but the momentum mapping must preserve the relevant mass-shell conditions. So,

$$d\Phi_{m+r}(\{p\}; q) = d\Phi_m(\{\tilde{p}\}; q) d\Phi_X. \quad (\text{B.45})$$

For a final-final antenna with hard radiator i, k and unresolved particles between them, this is schematically,

$$d\Phi_{m+r} = d\Phi_m(\dots, \tilde{p}_I, \tilde{p}_K, \dots; q) d\Phi_{X_{ijk}} \quad (\text{B.46})$$

while $\tilde{p}_I^2 = m_i^2$, $\tilde{p}_K^2 = m_k^2$ and momentum conservation is satisfied,

$$\tilde{p}_I + \tilde{p}_K = p_i + \left(\sum_{\text{unresolved } j} p_j \right) + p_k. \quad (\text{B.47})$$

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left(\prod_{r=1}^l d^d \ell_r \right) X_n^l(\{p\}, \{\ell\}) \sim (q^2)^{\lambda+\kappa\epsilon} f\left(\left\{\frac{m^2}{q^2}\right\}, \epsilon\right). \quad (\text{B.48})$$

PaVe and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space is non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless

limit at $m \rightarrow 0$ and the threshold limit at $\beta \rightarrow 0$ [24].

Bibliography

- [1] R. Keith Ellis, W. James Stirling, and B. R. Webber. *QCD and collider physics*. Vol. 8. Cambridge University Press, Feb. 2011. ISBN: 978-0-511-82328-2, 978-0-521-54589-1. DOI: 10.1017/CBO9780511628788.
- [2] Thomas Hahn. “Generating Feynman diagrams and amplitudes with FeynArts 3”. In: *Comput. Phys. Commun.* 140 (2001), pp. 418–431. DOI: 10.1016/S0010-4655(01)00290-9. arXiv: hep-ph/0012260.
- [3] Matthew D. Schwartz. *Quantum Field Theory and the Standard Model*. Cambridge University Press, Mar. 2014. ISBN: 978-1-107-03473-0, 978-1-107-03473-0. DOI: 10.1017/9781139540940.
- [4] A. Gehrmann-De Ridder, T. Gehrmann, and E. W. Nigel Glover. “Antenna subtraction at NNLO”. In: *JHEP* 09 (2005), p. 056. DOI: 10.1088/1126-6708/2005/09/056. arXiv: hep-ph/0505111.
- [5] Toichiro Kinoshita. “Mass Singularities of Feynman Amplitudes”. In: *Journal of Mathematical Physics* 3.4 (1962), pp. 650–677. ISSN: 1089-7658. DOI: 10.1063/1.1724268. URL: <http://dx.doi.org/10.1063/1.1724268>.
- [6] T. D. Lee and M. Nauenberg. “Degenerate Systems and Mass Singularities”. In: *Phys. Rev.* 133 (1964). Ed. by G. Feinberg, B1549–B1562. DOI: 10.1103/PhysRev.133.B1549.
- [7] S. Catani and M. H. Seymour. “A General algorithm for calculating jet cross-sections in NLO QCD”. In: *Nucl. Phys. B* 485 (1997). [Erratum: *Nucl.Phys.B* 510, 503–504 (1998)], pp. 291–419. DOI: 10.1016/S0550-3213(96)00589-5. arXiv: hep-ph/9605323.
- [8] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to quantum field theory*. Reading, USA: Addison-Wesley, 1995. ISBN: 978-0-201-50397-5, 978-0-429-50355-9, 978-0-429-49417-8. DOI: 10.1201/9780429503559.
- [9] Gerard 't Hooft and M. J. G. Veltman. “Regularization and Renormalization of Gauge Fields”. In: *Nucl. Phys. B* 44 (1972), pp. 189–213. DOI: 10.1016/0550-3213(72)90279-9.
- [10] Zvi Bern and David A. Kosower. “The Computation of loop amplitudes in gauge theory”. In: *Nucl. Phys. B* 379 (1992), pp. 451–461. DOI: 10.1016/0550-3213(92)90240-Y.
- [11] A. Gehrmann-De Ridder, T. Gehrmann, and G. Heinrich. “Four-Particle Phase Space Integrals in Massless QCD”. In: (2003). DOI: 10.48550/ARXIV.HEP-PH/0311276. URL: <https://arxiv.org/abs/hep-ph/0311276>.
- [12] T. Gehrmann, T. Huber, and D. Maitre. “Two-Loop Quark and Gluon Form Factors in Dimensional Regularisation”. In: (2005). DOI: 10.48550/ARXIV.HEP-PH/0507061. URL: <https://arxiv.org/abs/hep-ph/0507061>.

- [13] G. Passarino and M. J. G. Veltman. “One-Loop Corrections for e^+e^- Annihilation into $\mu^+\mu^-$ in the Weinberg Model”. In: *Nucl. Phys. B* 160 (1979), pp. 151–207. doi: 10.1016/0550-3213(79)90234-7.
- [14] K. G. Chetyrkin and F. V. Tkachov. “Integration by Parts: The Algorithm to Calculate β -Functions in 4 Loops”. In: *Nucl. Phys. B* 192 (1981), pp. 159–204. doi: 10.1016/0550-3213(81)90199-1.
- [15] Charalampos Anastasiou and Kirill Melnikov. “Higgs Boson Production at Hadron Colliders in NNLO QCD”. In: *Nucl. Phys. B* 646 (2002), pp. 220–256. doi: 10.1016/S0550-3213(02)00837-4. arXiv: hep-ph/0207004.
- [16] Stefano Laporta. “High-Precision Calculation of Multi-Loop Feynman Integrals by Difference Equations”. In: *Int. J. Mod. Phys. A* 15 (2000), pp. 5087–5159. doi: 10.1142/S0217751X00002159. arXiv: hep-ph/0102033.
- [17] Ettore Remiddi. “Differential Equations for Feynman Graph Amplitudes”. In: *Nuovo Cim. A* 110 (1997), pp. 1435–1452. doi: 10.1007/BF03185566. arXiv: hep-th/9711188.
- [18] Vladyslav Shtabovenko, Rolf Mertig, and Frederik Orellana. “FeynCalc 10: Do multiloop integrals dream of computer codes?” In: *Computer Physics Communications* 306 (Jan. 2025), p. 109357. issn: 0010-4655. doi: 10.1016/j.cpc.2024.109357. URL: <http://dx.doi.org/10.1016/j.cpc.2024.109357>.
- [19] Vladyslav Shtabovenko, Rolf Mertig, and Frederik Orellana. “FeynCalc 9.3: New features and improvements”. In: *Computer Physics Communications* 256 (Nov. 2020), p. 107478. issn: 0010-4655. doi: 10.1016/j.cpc.2020.107478. URL: <http://dx.doi.org/10.1016/j.cpc.2020.107478>.
- [20] Vladyslav Shtabovenko, Rolf Mertig, and Frederik Orellana. “New developments in FeynCalc 9.0”. In: *Computer Physics Communications* 207 (Oct. 2016), pp. 432–444. issn: 0010-4655. doi: 10.1016/j.cpc.2016.06.008. URL: <http://dx.doi.org/10.1016/j.cpc.2016.06.008>.
- [21] R. Mertig, M. Böhm, and A. Denner. “Feyn Calc - Computer-algebraic calculation of Feynman amplitudes”. In: *Computer Physics Communications* 64.3 (1991), pp. 345–359. issn: 0010-4655. doi: [https://doi.org/10.1016/0010-4655\(91\)90130-D](https://doi.org/10.1016/0010-4655(91)90130-D). URL: <https://www.sciencedirect.com/science/article/pii/001046559190130D>.
- [22] R. N. Lee. *Presenting LiteRed: a tool for the Loop InTEgrals REDuction*. 2012. arXiv: 1212.2685 [hep-ph]. URL: <https://arxiv.org/abs/1212.2685>.
- [23] A. Gehrmann-De Ridder, T. Gehrmann, and E. W. Nigel Glover. “Infrared structure of $e^+e^- \rightarrow 2$ jets at NNLO”. In: *Nucl. Phys. B* 691 (2004), pp. 195–222. doi: 10.1016/j.nuclphysb.2004.05.017. arXiv: hep-ph/0403057.
- [24] A. Gehrmann-De Ridder and M. Ritzmann. “NLO Antenna Subtraction with Massive Fermions”. In: *JHEP* 07 (2009), p. 041. doi: 10.1088/1126-6708/2009/07/041. arXiv: 0904.3297 [hep-ph].
- [25] Hiren H. Patel. “Package-X: A Mathematica package for the analytic calculation of one-loop integrals”. In: *Comput. Phys. Commun.* 197 (2015), pp. 276–290. doi: 10.1016/j.cpc.2015.08.017. arXiv: 1503.01469 [hep-ph].

- [26] Vladyslav Shtabovenko. “FeynHelpers: Connecting FeynCalc to FIRE and Package-X”. In: *Comput. Phys. Commun.* 218 (2017), pp. 48–65. doi: 10.1016/j.cpc.2017.04.014. arXiv: 1611.06793 [hep-ph].
- [27] Joana Maria Rodrigues Reis. “Implementation of Quark Mass Effects in QCD Three-Jet Production Observables Produced by Hadronic Decays of the Z-Boson at the FCC-ee Collider”. Master’s thesis. Lisbon, Portugal: Faculdade de Ciências, Universidade de Lisboa, 2024. URL: <https://repositorio.ulisboa.pt/bitstreams/998ca0e7-1783-41c1-b5eb-d6625da33324/download> (visited on 08/12/2026).
- [28] Guido Altarelli and G. Parisi. “Asymptotic Freedom in Parton Language”. In: *Nucl. Phys. B* 126 (1977), pp. 298–318. doi: 10.1016/0550-3213(77)90384-4.
- [29] J. Alwall et al. “The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations”. In: *JHEP* 07 (2014), p. 079. doi: 10.1007/JHEP07(2014)079. arXiv: 1405.0301 [hep-ph].
- [30] Alexander Huss et al. “NNLOJET: A parton-level event generator for jet cross sections at NNLO QCD accuracy”. In: *SciPost Physics Codebases* (Apr. 2026). issn: 2949-804X. doi: 10.21468/scipostphyscodeb.69. URL: <http://dx.doi.org/10.21468/SciPostPhysCodeb.69>.